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Librairies Manual

# SCADE Suite Libraries Manual

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The Legal Notice (PDF) about third-party software can be found under the **help** folder of the SCADE installation.

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# Libraries Manual Overview

This manual provides basic descriptions of all libraries delivered with your SCADE Suite installation. The content covers all libraries compatible with SCADE Suite projects.


- "Fundamentals about Libraries"
- Chapter 2: "Library libdigital"
- Chapter 3: "Library libimpl"
- Chapter 4: "Library liblinear"
- Chapter 5: "Library libmath"
- Chapter 6: "Library libmathext"
- Chapter 7: "Library libpwlinear"
- Chapter 8: "Library libsmlk"
- Chapter 9: "Library libverif"

"Index"


## RELATED DOCUMENTS

- *SCADE Suite User Manual*

## TYPOGRAPHICAL CONVENTIONS

| | |
|---|---|
| **Bold** | GUI buttons, options, tabs, radio buttons, drop-down lists, window titles, panels. |
| *Italics* | Path for menu selection from main menu bar or in contextual menus, new technical concepts or words, references to product manuals. |
| `Courier New` | Code extracts, constant or variable names, file names, directory path, or typed text. |
| Title Case | Names of application components, menus. |
| UPPERCASE | File formats |

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# 1 / Fundamentals about Libraries

This manual provides an exhaustive description of each SCADE Suite library with detailed operator description and important information. The most widely used functions or filters necessary when modeling applications are provided in SCADE Suite libraries.

- "Libraries and SCADE Suite Design Activities"
- "Libraries and Model Implementation"
- "Libraries and Simulink Modeling Activities"
- "Libraries and Formal Verification Activities"

The content of this manual is divided into chapters describing each library:

- libdigital
- liblinear
- libmathext
- libsmlk
- libimpl
- libmath
- libpwlinear
- libverif

For each library operator, any differences related to the library version in use in SCADE Suite projects are reported. For basic instructions or tips about SCADE Suite library management, please consult "Managing Libraries" on page 114 in *SCADE Suite User Manual*.

# Libraries and SCADE Suite Design Activities

The libraries listed below are intended to facilitate design work. All the libraries distributed with SCADE Suite provide a set of operators which are commonly used in model design.

### WHAT IS THE GENERIC FUNCTION OF OPERATORS IN EACH LIBRARY?

- **libdigital** contains digital operators that handle Boolean data (*e.g.,* falling and rising edges, flip-flops, truth tables). Read details in "Library libdigital".
- **liblinear** contains operators that compute linear functions or filters like derivative, gain, memories, or integrators. Read details in "Library liblinear".
- **libmath** contains basic mathematical operators (*e.g.,* absolute value, rounding, mean, maximum) and operators for vectors and matrices. Read details in "Library libmath".
- **libmathext** contains operators that compute trigonometric functions and power-based functions. Read details in "Library libmathext".
- **libpwlinear** contains operators that compute piecewise linear functions or filters (clock counter, counter, dead bands, hysteresis, rate limiter, etc.) and look-up tables. Read details in "Library libpwlinear".

### WHAT IS THE GENERAL PURPOSE OF SCADE SUITE LIBRARIES?

SCADE Suite includes an extensive set of libraries with functions commonly used in software development. All SCADE Suite libraries can be modified and enriched. Users can also create, share, and reuse their own libraries between projects.

---

**Note**

Library functions are expressed in Scade language when it makes sense. Some are in C code, meaning they may be platform-dependent as libimpl and libmathext.

---

### WHAT CAN I USE THEM FOR?

Most libraries contain mathematical functions and are highly generic. They can be useful in any application developed with SCADE Suite. Other libraries support more specific activities in SCADE Suite model-based design environment. For instance, `libverif` is needed to perform formal verification on SCADE Suite models.

### WHERE CAN I FIND THEM IN MY DISTRIBUTION?

The set of libraries distributed with SCADE Suite is located in the following directory: `<SCADESuiteInstallDirectory>\libraries`

### DISCLAIMER

Although every precaution has been taken when developing and testing the libraries, Ansys assumes no responsibility or liability for any errors that may be contained in the models or any damages resulting from the use of the libraries. If the libraries are used in a qualified / certified context, all documentation, verification, and validation activities requested by the Safety standard must be performed by the end user.

# Libraries and Model Implementation

SCADE Suite models are independent from the target. Indeed, the implementation choice for Scade data types is made only when code is being built. If data types are different and known at model level, the implementation size of integers is needed; the libimpl library provides many possible implementations for integer types and functions.

### WHAT IS THE GENERIC FUNCTION OF OPERATORS IN THIS LIBRARY?

- **libimpl** contains all standard integer operators (arithmetic, comparison, logical (bitwise, left/right shifting), time operators, type conversion, etc.) and types (unsigned and signed 8, 16, and 32 bits integers). For details, see "Library libimpl".

---

**Note**

This library can easily be customized for specific target: The types are defined in the following header files: `libimpl.h` and `macro_libimpl.h`. The definitions of macro functions are contained in the `macro_libimpl.h` file. For SCADE Suite Simulator, conversion functions are defined in the `libimpl.c` file.

---

# Libraries and Simulink Modeling Activities

When importing Simulink models into SCADE Suite, Simulink blocks are mapped, depending on their purpose, to Scade nodes using libsmlk, libmath, libdigital, liblinear, libpwlinear or libmathext libraries.

The *libsmlk* library is dedicated to support the correct translation of Simulink models into SCADE Suite models.

---

**Note**

The libsmlk library is available only if SCADE Suite Simulink® Importer is installed.

---

### WHAT IS THE GENERIC FUNCTION OF OPERATORS IN THIS LIBRARY?

- **libsmlk** contains operators designed to reproduce the behavior of Simulink blocks with respect to SCADE Suite formalism. For details, see "Library libsmlk".

# Libraries and Formal Verification Activities

Formal verification allows designers to check the correctness of their design. In SCADE Suite, Design Verifier enables extended debugging, non-regression proof and property verification without having to write verification tests.

The *libverif* library is dedicated to the formal verification of SCADE Suite models. It provides you with a set of library operators that are useful for expressing the properties to be verified in your models.

---

**Note**

The libverif library is available only if Design Verifier is installed.

---

### WHAT IS THE GENERIC FUNCTION OF OPERATORS IN THIS LIBRARY?

- **libverif** contains verification operators designed to help you express properties (implies, etc.). Such operators are expressed as regular Scade nodes. For details, see "Library libverif".

# 2 / Library libdigital

This library contains digital operators that handle Boolean data and a count-down counter. You can read the technical description of the library components by package:

- "Digital Operators" in **`digital`** package
- "Truth Table Operators" in **`truthtables`** package

Access the description of library operators in alphabetical order:


"Boolean Vector to Int8"
"Boolean Vector to Int16"
"Boolean Vector to Int32"
"Boolean Vector to Int64"
"Boolean Vector to Uint8"
"Boolean Vector to Uint16"
"Boolean Vector to Uint32"
"Boolean Vector to Uint64"
"Count-Down"
"Either Edge"
"EntryDetect"
"Falling Edge"
"Falling Edge (Re-Trigger)"
"Falling Edge (without Re-Trigger)"
"Flip-Flop JK"
"Flip-Flop Priority to Reset"
"Flip-Flop Priority to Set"
"Inactive Cycles"
"Inactive Time"
"Integer to Boolean Vector"
"Relay"
"Rising Edge"
"Rising Edge (Re-Trigger)"
"Rising Edge (without Re-Trigger)"
"Toggle"
"Trigger (Either)"
"Trigger (Fall)"
"Trigger (Rise)"
"Truth Table"
"Truth Table (Exhaustive)"
"Truth Table (Index)"

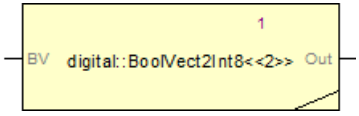
# Digital Operators

Read the description of the following operators:


"Boolean Vector to Int8"
"Boolean Vector to Int16"
"Boolean Vector to Int32"
"Boolean Vector to Int64"
"Boolean Vector to Uint8"
"Boolean Vector to Uint16"
"Boolean Vector to Uint32"
"Boolean Vector to Uint64"
"Count-Down"
"Either Edge"
"EntryDetect"
"Falling Edge"
"Falling Edge (Re-Trigger)"
"Falling Edge (without Re-Trigger)"
"Flip-Flop JK"
"Flip-Flop Priority to Reset"
"Flip-Flop Priority to Set"
"Inactive Cycles"
"Inactive Time"
"Integer to Boolean Vector"
"Relay"
"Rising Edge"
"Rising Edge (Re-Trigger)"
"Rising Edge (without Re-Trigger)"
"Toggle"
"Trigger (Either)"
"Trigger (Fall)"
"Trigger (Rise)"


## Boolean Vector to Int8

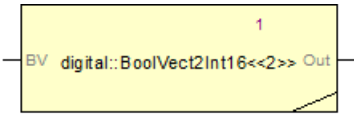
**Symbol 2.1:**




| | |
|---|---|
| SCADE name | **BoolVect2Int8** |
| Package | digital |
| Function | Outputs the integer value computed from a vector of booleans BV interpreted as its Boolean representation. |
| Size parameters | Nbool |
| Inputs | BV: bool ^Nbool |
| Outputs | `Out`: int8 |
| Comment | BV[0] is the highest bit, *i.e.*, (BoolVect2Int([true,false])=2. |

# Boolean Vector to Int16

**Symbol 2.2:**




| | |
|---|---|
| SCADE name | **BoolVect2Int16** |
| Package | digital |
| Function | Outputs the integer value computed from a vector of booleans BV interpreted as its Boolean representation. |
| Size parameters | Nbool |
| Inputs | BV: bool ^Nbool |
| Outputs | `Out`: int16 |
| Comment | BV[0] is the highest bit, *i.e.*, (BoolVect2Int([true,false])=2. |

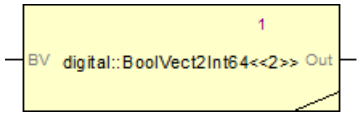
# Boolean Vector to Int32

**Symbol 2.3:**

| | |
|---|---|
| SCADE name | **BoolVect2Int32** |
| Package | digital |
| Function | Outputs the integer value computed from a vector of booleans BV interpreted as its Boolean representation. |
| Size parameters | Nbool |
| Inputs | BV: bool ^Nbool |
| Outputs | `Out`: int32 |
| Comment | BV[0] is the highest bit, *i.e.*, (BoolVect2Int([true,false])=2. |

# Boolean Vector to Int64

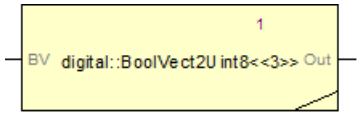
**Symbol 2.4:**




| | |
|---|---|
| SCADE name | **BoolVect2Int64** |
| Package | digital |
| Function | Outputs the integer value computed from a vector of booleans BV interpreted as its Boolean representation. |
| Size parameters | Nbool |
| Inputs | BV: bool ^Nbool |
| Outputs | `Out`: int64 |
| Comment | BV[0] is the highest bit, *i.e.*, (BoolVect2Int([true,false])=2. |

# Boolean Vector to Uint8

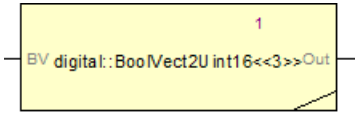
**Symbol 2.5:**




| | |
|---|---|
| SCADE name | **BoolVect2Uint8** |
| Package | digital |
| Function | Outputs the integer value computed from a vector of booleans BV interpreted as its Boolean representation. |
| Size parameters | Nbool |
| Inputs | BV: bool ^Nbool |
| Outputs | `Out`: uint8 |
| Comment | BV[0] is the highest bit, *i.e.*, (BoolVect2Int([true,false])=2. |

# Boolean Vector to Uint16

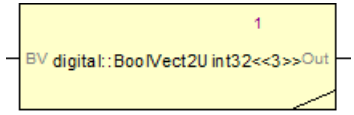
**Symbol 2.6:**




| | |
|---|---|
| SCADE name | **BoolVect2Uint16** |
| Package | digital |
| Function | Outputs the integer value computed from a vector of booleans BV interpreted as its Boolean representation. |
| Size parameters | Nbool |
| Inputs | BV: bool ^Nbool |
| Outputs | `Out`: uint16 |
| Comment | BV[0] is the highest bit, *i.e.*, (BoolVect2Int([true,false])=2. |

# Boolean Vector to Uint32

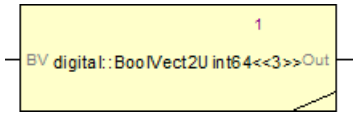
**Symbol 2.7:**




| | |
|---|---|
| SCADE name | **BoolVect2Uint32** |
| Package | digital |
| Function | Outputs the integer value computed from a vector of booleans BV interpreted as its Boolean representation. |
| Size parameters | Nbool |
| Inputs | BV: bool ^Nbool |
| Outputs | `Out`: uint32 |
| Comment | BV[0] is the highest bit, *i.e.*, (BoolVect2Int([true,false])=2. |

# Boolean Vector to Uint64

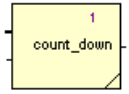
**Symbol 2.8:**




| | |
|---|---|
| SCADE name | **BoolVect2Uint64** |
| Package | digital |
| Function | Outputs the integer value computed from a vector of booleans BV interpreted as its Boolean representation. |
| Size parameters | Nbool |
| Inputs | BV: bool ^Nbool |
| Outputs | `Out`: uint64 |
| Comment | BV[0] is the highest bit, *i.e.*, (BoolVect2Int([true,false])=2. |

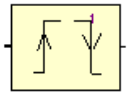
# Count-Down

**Symbol 2.9:**




| | |
|---|---|
| SCADE name | **count_down** |
| Package | digital |
| Function | Returns input N when Reset is `true`, otherwise, decrements the output by one at each cycle. |
| Inputs | `Reset`: bool and `N`: 'T |
| Outputs | `cpt`: 'T where 'T integer |
| Comment | The count_down function does not "stop" at value 0; it decrements the count at each step, until Reset is `true`, even if the count is negative. |

# Either Edge

**Symbol 2.10:** 

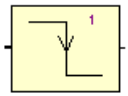
| | |
|---|---|
| SCADE name | **EitherEdge** |
| Package | digital |
| Function | Detects all edges, rising or falling (i.e., `true` to `false`, and `false` to `true` transitions). |
| Inputs | `EE_Input`: bool |
| Outputs | `EE_Output`: bool |
| Comment | Output is true during transition clock cycle. Output is initialized to `false`. |

# EntryDetect

**Symbol 2.11:**

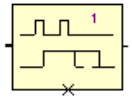

| | |
|---|---|
| SCADE name | **EntryDetect** |
| Package | digital |
| Function | Detects entry into a range based on two Boolean inputs: one input indicates whether the signal is outside the border and the other one whether the signal is at the border. |
| Inputs | `Outside`: bool and `OnTheBorder`: bool |
| Outputs | `Entry`: bool |

# Falling Edge

**Symbol 2.12:** 

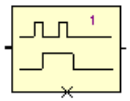
| | |
|---|---|
| SCADE name | **FallingEdge** |
| Package | digital |
| Function | Detects a falling edge (i.e., `true` to `false` transition of Boolean input). |
| Inputs | `FE_Input`: bool |
| Outputs | `FE_Output`: bool |
| Comment | Output is true during transition clock cycle. Output is initialized to `false`. |

# Falling Edge (Re-Trigger)

**Symbol 2.13:** 

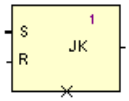
| | |
|---|---|
| SCADE name | **FallingEdgeRetrigger** |
| Package | digital |
| Function | Detects a falling edge (i.e., `true` to `false` transition of Boolean input). |
| Inputs | `FER_Input`: bool |
| Hidden inputs | `NumberOfCycle`: int |
| Outputs | `FER_Output`: bool |
| Comment | The output becomes `true` as soon as the transition is detected and remains unchanged during NumberOfCycle. Output is initialized to `false`. If another falling edge occurs while the output is `true`, it is detected. |

# Falling Edge (without Re-Trigger)

**Symbol 2.14:** 


| | |
|---|---|
| SCADE name | **FallingEdgeNoRetrigger** |
| Package | digital |
| Function | Detects a falling edge (i.e., `true` to `false` transition of Boolean input). |
| Inputs | `FENR_Input`: bool |
| Hidden inputs | `NumberOfCycle`: 'T |
| Outputs | `FENR_Output`: bool |
| Comment | The output becomes `true` as soon as the transition is detected and remains unchanged during NumberOfCycle. Output is initialized to `false`. If another falling edge occurs while the output is `true`, it is not detected. |

# Flip-Flop JK

**Symbol 2.15:** 

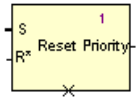

| | |
|---|---|
| SCADE name | **FlipFlopJK** |
| Package | digital |
| Function | Basic JK flip-flop of Boolean values. See Table 2.1 below. |
| Inputs | `Set` and `Reset`: bool |
| Hidden inputs | `Init`: bool |
| Outputs | `FFJK_Output`: bool |
| Comment | At the first step, the Output takes the same value as Init, regardless of the values of Set and Reset. |

**Table 2.1:** FlipFlopJK functional description

| Set | Reset | FFJK_Output |
|---|---|---|
| false | false | FFJK_Output -1 (memorization) |
| false | true | false |
| true | false | true |
| true | true | not FFJK_Output -1 (memorization) |

# Flip-Flop Priority to Reset

**Symbol 2.16:**



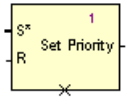

| | |
|---|---|
| SCADE name | **FlipFlopReset** |
| Package | digital |
| Function | Basic flip-flop of Boolean values with priority for reset. See Table 2.2 below. |
| Inputs | `Set` and `Reset`: bool |
| Hidden inputs | `Init`: bool |
| Outputs | `FFR_Output`: bool |
| KCG pragma | Operator expansion |
| Comment | At the first step, the Output takes the same value as Init, regardless of the values of Set and Reset. |

**Table 2.2:** FlipFlopReset functional description

| Set | Reset | FFR_Output |
|---|---|---|
| false | false | FFR_Output -1 (memorization) |
| false | true | false |
| true | false | true |
| true | true | false |

# Flip-Flop Priority to Set

**Symbol 2.17:**



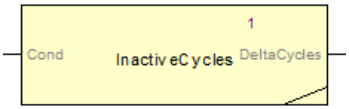

| | |
|---|---|
| SCADE name | **FlipFlopSet** |
| Package | digital |
| Function | Basic flip-flop of Boolean values with priority for set. See Table 2.3 below. |
| Inputs | `Set` and `Reset`: bool |
| Hidden inputs | `Init`: bool |
| Outputs | `FFS_Output`: bool |
| KCG pragma | Operator expansion |
| Comment | At the first step, the Output takes the same value as Init, regardless of the values of Set and Reset. |

**Table 2.3:** FlipFlopSet functional description

| Set | Reset | FFS_Output |
|---|---|---|
| false | false | FFS_Output -1 (memorization) |
| false | true | false |
| true | false | true |
| true | true | true |

# Inactive Cycles

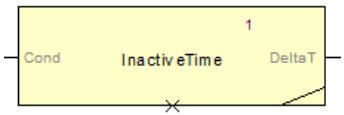
**Symbol 2.18:**




| | |
|---|---|
| SCADE name | **InactiveCycles** |
| Package | digital |
| Function | Gives the number of cycles since last time `Cond` was true.<br>Each time `Cond` is true, the counter is reset to 0. |
| Inputs | `Cond:` bool |
| Outputs | `DeltaCycles:` int32 |

# Inactive Time

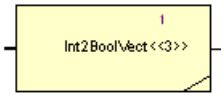
**Symbol 2.19:**




| | |
|---|---|
| SCADE name | **InactiveTime** |
| Package | digital |
| Function | Gives the time since last time `Cond` was true.<br>Each time `Cond` is true, the counter is reset to 0.0.<br>The `TimeCycle` input gives the length of a cycle. |
| Inputs | `Cond:` bool |
| Hidden inputs | `TimeCycle:` 'T where 'T is numeric |
| Outputs | `DeltaT:` 'T where 'T is numeric |

# Integer to Boolean Vector

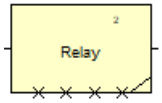
**Symbol 2.20:**




| | |
|---|---|
| SCADE name | **Int2BoolVect** |
| Package | digital |
| Function | Outputs a vector of booleans BV that is the boolean representation of the integer input, modulo 2^Nbool. |
| Size parameters | Nbool |
| Inputs | `In:` 'T where 'T is integer |
| Outputs | `BV:` bool ^Nbool |
| Comment | BV[0] is the highest bit, e.g., (Int2BoolVect<<2>>)([true,false])=2. |

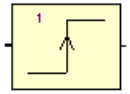
# Relay

**Symbol 2.21:**



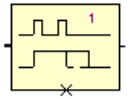

| | |
|---|---|
| SCADE name | **Relay** |
| Package | digital |
| Function | Basic relay that generates discontinuous output values based on input. |
| Inputs | `Input1:` 'T where 'T is numeric |
| Hidden inputs | `UpperLimit_On`: 'T and `LowerLimit_Off`: 'T where 'T is numeric<br>`OutputWhenOn`: 'T and `OutputWhenOff`: 'T where 'T is numeric |
| Outputs | `Output1:` 'T where 'T is numeric |
| Comment | Initially, the relay is off. `UpperLimit_On` >= `LowerLimit_Off`.<br>• when off (*i.e.*, output = `OutputWhenOff`), relay remains off unless the input reaches or exceeds the `UpperLimit_On` (*i.e.*, input >= `UpperLimit_On`)<br>• when on (*i.e.*, output = `OutputWhenOn`), relay remains on unless the input reaches or drops below `LowerLimit_Off` (*i.e.*, input <= `LowerLimit_Off`). |

# Rising Edge

**Symbol 2.22:** 

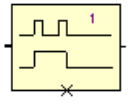

| | |
|---|---|
| SCADE name | **RisingEdge** |
| Package | digital |
| Function | Detects a rising edge (i.e., `false` to `true` transition of Boolean input). |
| Inputs | `RE_Input`: bool |
| Outputs | `RE_Output`: bool |
| Comment | The output is true during the transition clock cycle. Output is initialized to `false`. |

# Rising Edge (Re-Trigger)

**Symbol 2.23:** 

| | |
|---|---|
| SCADE name | **RisingEdgeRetrigger** |
| Package | digital |
| Function | Detects a rising edge (i.e., `false` to `true` transition of Boolean input). |
| Inputs | `RER_Input`: bool |
| Hidden inputs | `NumberOfCycle`: 'T where 'T is integer |
| Outputs | `RER_Output`: bool |
| Comment | The output becomes `true` as soon as transition is detected and remains unchanged during NumberOfCycle. If another rising edge occurs while the output is `true`, it is detected. Output is initialized to `false`. |

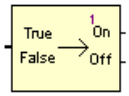
# Rising Edge (without Re-Trigger)

**Symbol 2.24:** 

| | |
|---|---|
| SCADE name | **RisingEdgeNoRetrigger** |
| Package | digital |
| Function | Detects a rising edge (i.e., `false` to `true` transition of Boolean input). |
| Inputs | `RENR_Input`: bool |
| Hidden inputs | `NumberOfCycle`: 'T where 'T is integer |
| Outputs | `RENR_Output`: bool |
| Comment | The output is set to `true` as soon as transition is detected and remains unchanged during NumberOfCycle. If another rising edge occurs while the output is `true`, it is not detected. Output is initialized to `false`. |

# Toggle

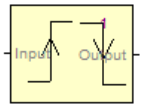
**Symbol 2.25:**




| | |
|---|---|
| SCADE name | **Toggle** |
| Package | digital |
| Function | Stores the input value into On and Off states. When the input is `true`, On is `true` and Off is `false`. When the input is `false`, On is `false` and Off is `true`. |
| Inputs | `T_Input`: bool |
| Outputs | `T_On` and `T_Off`: bool |
| KCG pragma | Operator expansion |
| Comment | `T_On` always equals `NOT T_Off`. |

# Trigger (Either)

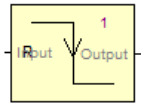
**Symbol 2.26:**




| | |
|---|---|
| SCADE name | **TriggerEither** |
| Package | digital |
| Function | Output is true if any of the following conditions is true:<br>• Input is 0 and was greater than 0 at previous cycle<br>• Input lower than 0 and was greater than 0 at previous cycle<br>• Input is lower than 0 and was equal to 0 on previous cycle, and Output was not already true at previous cycle<br>• Input is 0 and was lower than 0 at previous cycle<br>• Input greater than 0 and was lower than 0 at previous cycle<br>• Input is greater than 0 and was equal to 0 on previous cycle, and Output was not already true at previous cycle |
| Inputs | `Input`: 'T where 'T is numeric |
| Outputs | `Output`: bool |

# Trigger (Fall)

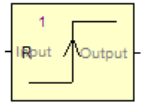
**Symbol 2.27:**




| | |
|---|---|
| SCADE name | **TriggerFall** |
| Package | digital |
| Function | Output is true if any of the following conditions is true:<br>• Input is 0 and was greater than 0 at previous cycle<br>• Input lower than 0 and was greater than 0 at previous cycle<br>• Input is lower than 0 and was equal to 0 on previous cycle, and Output was not already true at previous cycle |
| Inputs | `Input`: 'T where 'T is numeric |
| Outputs | `Output`: bool |

# Trigger (Rise)

**Symbol 2.28:**




| | |
|---|---|
| SCADE name | **TriggerRise** |
| Package | digital |
| Function | Output is true if any of the following conditions is true:<br>• Input is 0 and was lower than 0 at previous cycle<br>• Input greater than 0 and was lower than 0 at previous cycle<br>• Input is greater than 0 and was equal to 0 on previous cycle, and Output was not already true at previous cycle |
| Inputs | `Input`: 'T where 'T is numeric |
| Outputs | `Output`: bool |

# Truth Table Operators

Read the description of the following operators:

"Truth Table"
"Truth Table (Exhaustive)"
"Truth Table (Index)"

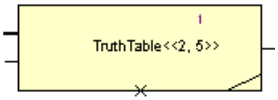
### TYPES

This package of library operators includes the following type definition:

```
ThruthTableValues = enum {T, F, X}
```

## Truth Table

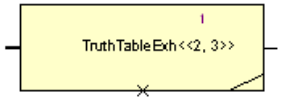
**Symbol 2.29:**




| | |
|---|---|
| SCADE name | **TruthTable** |
| Package | truthtables |
| Function | Implements a truth table. Outputs the element from `ResultValues` that corresponds to the first line of `Ttable` that matches `Conditions`. |
| Size parameters | Ncond and Nlines |
| Inputs | `Conditions`: bool ^Ncond<br>`ResultValues`: 'T ^(Nlines + 1)<br>`Ttable`: TruthTableValues ^Ncond ^Nlines |
| Outputs | `Result`: 'T |
| Comment | `ResultValues`[Nlines] is produced when no line matches.<br>Element 'T' from `Ttable` matches `true`, 'F' matches `false`, 'X' matches both. |

# Truth Table (Exhaustive)

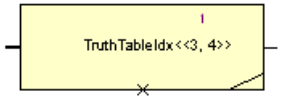
**Symbol 2.30:**




| | |
|---|---|
| SCADE name | **TruthTableExh** |
| Package | truthtables |
| Function | Outputs the element fom ResultValues indexed exhaustively by the integer value computed from a vector of booleans BV interpreted as its boolean representation. |
| Size parameters | Nbool and Nres |
| Inputs | BV: bool ^Nbool<br>`ResultValues`: 'T ^Nres |
| Outputs | `Result`: 'T |
| Comment | BV[0] is the highest bit. It is assumed that Nres = 2^Nbool. An "assume" in the operator checks that constraint. |

# Truth Table (Index)

**Symbol 2.31:**




| | |
|---|---|
| SCADE name | **TruthTableIdx** |
| Package | truthtables |
| Function | Implements a truth table. Outputs the index of the first line of `Ttable` of `Nlines` lines that matches `Conditions`. |
| Size parameters | Ncond and Nlines |
| Inputs | `Conditions`: bool ^Ncond<br>`Ttable`: TruthTableValues ^Ncond ^Nlines |
| Outputs | `LineIdx`: int16 |
| Comment | Index `Nlines` is produced when no line matches.<br>Element 'T' from `Ttable` matches `true`, 'F' matches `false`, 'X' matches both. |

# 3 / Library libimpl

This library contains types and operators that support sized integers.

The libimpl library contains the following packages:

- **`impl`**: to handle operators that support all sized integer types defined in libimpl. These are called polymorphic operators.
- **`impls`**: to handle operators that are the monomorphic specializations of the polymorphic operators from package impl.

As there is no additional cost in memory or execution time using polymorphic operators, it is simpler to use the operators from the **`impl`** package. In case, users need monomorphic operators from **`impls`**, it is possible.

The **`impls_arithmetic`**, **`impls_comp`**, and **`impls_bitwise`** packages contain specializations for types int8, uint8, int16, uint16, int32, and uint32 of the polymorphic operators. These operators (more than 150) are named from the name of the polymorphic operator suffixed by 1 or 2 type identifier corresponding to the specialization.

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**Design Verifier and Implementation**

The current version of Design Verifier for SCADE 6 does not support operators from libimpl (considered as any imported operators).

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- "Arithmetic Operators"
- "Comparison Operators"
- "Bitshift and Bitwise Operators"

Access the description of library operators in alphabetical order:


"Bit Clear"
"Bit Set"
"Decrease LSB"
"Different"
"Equal"
"Extract Bit Range"
"Greater Than or Equal"
"Increase LSB to Ceiling"
"Increase LSB to Fix"
"Increase LSB to Floor"
"Integer Division (Ceiling Rounding)"
"Integer Division (Fix Rounding)"
"Integer Division (Floor Rounding)"
"Integer Division (Saturation)"
"Integer Multiplication"
"Integer Multiplication (Int16)"
"Integer Multiplication (Uint16)"
"Integer Multiplication (Int32)"
"Integer Multiplication (Uint32)"
"Integer Multiplication (Saturation)"
"Integer Negation (Saturation)"
"Integer Subtraction (Saturation)"
"Integer Sum (Saturation)"
"Less Than or Equal"
"Strictly Greater Than"
"Strictly Less Than"
"Shift Left"
"Shift Right"


## OVERFLOW

Overflow means that the result of an operation is too large in absolute value to be represented in the type of the result. The result is wrapped with respect to the type, meaning that the result of the operation is computed modulo the largest possible value for the type plus 1, namely, mod(max_type + 1). For example, the subtraction operation defined as `iSub(<value=2>, <value=3>)` overflows if the type of the result is uint8. Saturation is one method for preventing overflows, but saturation operators are less efficient. Saturating an operation may also lead to unexpected results such as instability of the model, etc. Any libimpl operator that can overflow has a saturated version to allow you to choose between versions.

# Arithmetic Operators

The **`libimpl`** library contains all basic mathematical operators such as summation, subtraction, multiplication, division, modulo or unary minus, that allow handling sized integer data.

---

**Note**

For backward compatibility, all arithmetic operators defined in this library have as argument the following types: `'T1` and `'T2`. Mixing signed and unsigned types is not supported. `'T1` and `'T2` must be of the same type.

For all operators, they can be int8, uint8, int16, uint16, int32, uint32.

---

Read the description of the following operators:


"Integer Division (Ceiling Rounding)"
"Integer Division (Fix Rounding)"
"Integer Division (Floor Rounding)"
"Integer Division (Saturation)"
"Integer Multiplication"
"Integer Multiplication (Int16)"
"Integer Multiplication (Uint16)"
"Integer Multiplication (Int32)"
"Integer Multiplication (Uint32)"
"Integer Multiplication (Saturation)"
"Integer Negation (Saturation)"
"Integer Subtraction (Saturation)"
"Integer Sum (Saturation)"

# Integer Division (Ceiling Rounding)

**Symbol 3.1:**


| | |
|---|---|
| SCADE name | **iDivCeil** |
| Function | Computes the division of integer inputs with ceiling rounding (Out1 = RoundCeil (In1/In2). Rounding always to upper integer value. |
| Inputs | In1: 'T1 and In2: 'T2 , with 'T1 = 'T2 and 'T1, 'T2 as int8, int16, int32, uint8, uint16, or uint32 type |
| Outputs | `Out1`: 'T1 |
| Caution | Overflow if second operand is 0. |

# Integer Division (Fix Rounding)

**Symbol 3.2:**


| | |
|---|---|
| SCADE name | **iDivFix** |
| Function | Computes the division of integer inputs with rounding toward zero (Out1 = (In1/In2). |
| Inputs | In1: 'T1 and In2: 'T2 , with 'T1 = 'T2 and 'T1, 'T2 as int8, int16, int32, uint8, uint16, or uint32 type |
| Outputs | `Out1`: 'T1 |
| Caution | Overflow if second operand is 0. |

# Integer Division (Floor Rounding)

**Symbol 3.3:**


| | |
|---|---|
| SCADE name | **iDivFloor** |
| Function | Computes the division of integer inputs with floor rounding (Out1 = RoundFloor (In1/In2). Rounding always to lower integer value. |
| Inputs | In1: 'T1 and In2: 'T2 , with 'T1 = 'T2 and 'T1, 'T2 as int8, int16, int32, uint8, uint16, or uint32 type |
| Outputs | `Out1`: 'T1 |
| Caution | Overflow if second operand is 0. |

# Integer Division (Saturation)

**Symbol 3.4:**


| | |
|---|---|
| SCADE name | **iDivSat** |
| Function | Computes the division of integer inputs with saturation to avoid overflow. Implicit rounding is rounding toward zero (usual C integer division operation). (Out1 = (In1/In2) unless Out1 > max_int, then Out1 = max_int). |
| Inputs | In1: 'T1 and In2: 'T2 , with 'T1 = 'T2 and 'T1, 'T2 as int8, int16, int32, uint8, uint16, or uint32 type |
| Outputs | `Out1`: 'T1 |

# Integer Multiplication

**Symbol 3.5:**


| | |
|---|---|
| SCADE name | **iMul** |
| Function | Computes the multiplication of two integer inputs. |
| Inputs | In1: 'T1 and In2: 'T2 , with 'T1 = 'T2 and 'T1, 'T2 as int8, int16, int32, uint8, uint16, or uint32 type |
| Outputs | `Out1`: 'T2 |
| Caution | Can overflow |

# Integer Multiplication (Int16)

**Symbol 3.6:**


| | |
|---|---|
| SCADE name | **iMul16** |
| Function | Computes the multiplication of two int8 inputs and converts its output into int16. |
| Inputs | `In1`: 'T1 and `In2`: 'T2 , with 'T1 = 'T2 = int8 |
| Outputs | `Out1`: imported int16 |

# Integer Multiplication (Uint16)

**Symbol 3.7:**


| | |
|---|---|
| SCADE name | **iMul16U** |
| Function | Computes the multiplication of two uint8 inputs and converts its output into uint16. |
| Inputs | `In1`: 'T1 and `In2`: 'T2 , with 'T1 = 'T2 = uint8 |
| Outputs | `Out1`: imported uint16 |

# Integer Multiplication (Int32)

**Symbol 3.8:**


| | |
|---|---|
| SCADE name | **iMul32** |
| Function | Computes the multiplication of two int16 inputs and converts its output into int32. |
| Inputs | `In1`: 'T1 and `In2`: 'T2 , with 'T1 = 'T2 = int16 |
| Outputs | `Out1`: imported int32 |

# Integer Multiplication (Uint32)

**Symbol 3.9:**


| | |
|---|---|
| SCADE name | **iMul32U** |
| Function | Computes the multiplication of two uint16 inputs and converts its output into uint32. |
| Inputs | `In1`: 'T1 and `In2`: 'T2 , with 'T1 = 'T2 = uint16 |
| Outputs | `Out1`: imported uint32 |

# Integer Multiplication (Saturation)

**Symbol 3.10:**


| | |
|---|---|
| SCADE name | **iMulSat** |
| Function | Computes the saturated multiplication of two integer inputs to avoid overflows (if Out1 > max_int, then Out1 = max_int; if Out1 < min_int, then Out1 = min_int). |
| Inputs | In1: 'T1 and In2: 'T2 , with 'T1 = 'T2 and 'T1, 'T2 as int8, int16, int32, uint8, uint16, or uint32 type |
| Outputs | `Out1`: 'T1 |
| Caution | If the operation overflows relatively to the input type (size and signed/unsigned), the result is the max of the type. If it overflows below the lower limit, the result is the min of the type. |

# Integer Negation (Saturation)

**Symbol 3.11:**


| | |
|---|---|
| SCADE name | **iNegSat** |
| Function | Computes the saturated negative of an integer input (if Out1 > max_int, then Out1 = max_int). |
| Inputs | `In1`: 'T1 , with 'T1 = int8, int16, or int32 |
| Outputs | `Out1`: 'T1 |
| Caution | Implemented only for signed types int8, int16 and int32. iNegSat(min_type) is saturated to max_type, the largest possible value for the type. |

# Integer Subtraction (Saturation)

**Symbol 3.12:**


| | |
|---|---|
| SCADE name | **iSubSat** |
| Function | Subtracts an integer input from another with saturation (if Out1 > max_int, then Out1 = max_int; if Out1 < min_int, then Out1 = min_int). |
| Inputs | In1: 'T1 and In2: 'T2, with 'T1 = 'T2 and 'T1, 'T2 as int8, int16, int32, uint8, uint16, or uint32 type |
| Outputs | `Out1`: 'T1 |
| Caution | If the operation overflows relatively to the input type (size and signed/unsigned), the result is the max of the type. If it overflows below the lower limit, the result is the min of the type. |

# Integer Sum (Saturation)

**Symbol 3.13:**


| | |
|---|---|
| SCADE name | **iSumSat** |
| Function | Computes the addition of two integer inputs with saturation (if Out1 > max_int, then Out1 = max_int; if Out1 < min_int, then Out1 = min_int). |
| Inputs | In1: 'T1 and In2: 'T2, with 'T1 = 'T2 and 'T1, 'T2 as int8, int16, int32, uint8, uint16, or uint32 type |
| Outputs | `Out1`: 'T1 |
| Caution | If the operation overflows relatively to the input type (size and signed/unsigned), the result is the max of the type. If it overflows below the lower limit, the result is the min of the type. |

# Comparison Operators

All comparison operators defined in this library have as argument the following types: 'T1 and 'T2. This allows mixing types of different sizes, but does not support mixing signed and unsigned types. Types 'T1 and 'T2 can be any type from int8, int16, int32, or any type from uint8, uint16, uint32.

Mixing different size of implementation types is allowed for efficiency reason: avoiding a cast in SCADE models to make the types match may save an intermediate variable in comparison to the implicit cast performed in C code.

Read the description of the following operators:


"Strictly Less Than"
"Less Than or Equal"
"Equal"
"Different"
"Strictly Greater Than"
"Greater Than or Equal"


## Strictly Less Than

**Symbol 3.14:**

| | |
|---|---|
| SCADE name | **iLess** |
| Function | Compares two integer inputs to check whether the second one is strictly less than the first one and returns the corresponding Boolean data (if In1 < In2, then Out1 = `true`, else Out1 = `false`). |
| Inputs | `In1`: 'T1 and `In2`: 'T2, with 'T1 and 'T2 equal to any type from {int8, int16, int32) or 'T1 and 'T2 equal to any type from {uint8, uint16, uint32} |
| Outputs | `Out1`: bool |

# Less Than or Equal

**Symbol 3.15:**


| | |
|---|---|
| SCADE name | **iLessEq** |
| Function | Compares two integer inputs to check whether the second one is less than or equal to the first one and returns the corresponding Boolean data (if In1 <= In2, then Out1 = `true`, else Out1 = `false`). |
| Inputs | `In1`: 'T1 and `In2`: 'T2, with 'T1 and 'T2 equal to any type from {int8, int16, int32) or 'T1 and 'T2 equal to any type from {uint8, uint16, uint32} |
| Outputs | `Out1`: bool |

# Equal

**Symbol 3.16:**


| | |
|---|---|
| SCADE name | **iEqual** |
| Function | Compares equality between two integer inputs and returns the corresponding Boolean data (if In1 = In2, then Output = `true`, else output = `false`). |
| Inputs | `In1`: 'T1 and `In2`: 'T2, with 'T1 and 'T2 equal to any type from {int8, int16, int32) or 'T1 and 'T2 equal to any type from {uint8, uint16, uint32} |
| Outputs | `Out1`: bool |

# Different

**Symbol 3.17:**


| | |
|---|---|
| SCADE name | **iDifferent** |
| Function | Compares difference between two integer inputs and returns the corresponding Boolean data (if In1 != In2, then Output = `true`, else output = `false`). |
| Inputs | `In1`: 'T1 and `In2`: 'T2, with 'T1 and 'T2 equal to any type from {int8, int16, int32) or 'T1 and 'T2 equal to any type from {uint8, uint16, uint32} |
| Outputs | `Out1`: bool |

# Strictly Greater Than

**Symbol 3.18:**


| | |
|---|---|
| SCADE name | **iGreater** |
| Function | Compares two integer inputs to check whether the second one is strictly greater than the first one and returns the corresponding Boolean data (if In1 > In2, then Out1 = `true`, else Out1 = `false`). |
| Inputs | `In1`: 'T1 and `In2`: 'T2, with 'T1 and 'T2 equal to any type from {int8, int16, int32) or 'T1 and 'T2 equal to any type from {uint8, uint16, uint32} |
| Outputs | `Out1`: bool |

# Greater Than or Equal

**Symbol 3.19:**


| | |
|---|---|
| SCADE name | **iGreaterEq** |
| Function | Compares two integer inputs to check whether the second one is greater than or equal to the first one and returns the corresponding Boolean data (if In1 >= In2, then Out1 = `true`, else Out1 = `false`). |
| Inputs | `In1`: 'T1 and `In2`: 'T2 , with 'T1 and 'T2 equal to any type from {int8, int16, int32) or 'T1 and 'T2 equal to any type from {uint8, uint16, uint32} |
| Outputs | `Out1`: bool |

# Bitshift and Bitwise Operators

All bitshift and bitwise operators defined in this library can have their first operand of any type from int8, uint8, int16, uint16, int32, or uint32. The second operand is a hidden parameter of type 'T2. It can be either:

1. Type “int” that is not an “implemented type”. The objective is to use only literals and not variables of type int. A direct value can be set as a parameter from the graphical symbol.

   Example: `iShiftRight(_L1,2)`. This literal value is interpreted as the type of the first operand, that is, implementation casts it to the type of the first operand. No overflow check is performed on this cast.

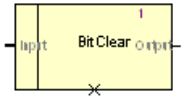
2. Same type as the first operand.

Read the description of the following operators:


“Bit Clear”
“Bit Set”
“Decrease LSB”
“Extract Bit Range”
“Increase LSB to Ceiling”
“Increase LSB to Fix”
“Increase LSB to Floor”
“Shift Left”
“Shift Right”


## Bit Clear

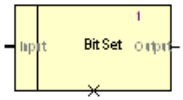
**Symbol 3.20:**




| | |
|---|---|
| SCADE name | **BitClear** |
| Function | Returns the input where the bit at position `Index` was set to 0 |
| Inputs | `Input`: 'T where 'T is unsigned |
| Hidden inputs | `Index`: 'T where 'T is unsigned |
| Outputs | `Output`: 'T where 'T is unsigned |

# Bit Set

**Symbol 3.21:**




| | |
|---|---|
| SCADE name | **BitSet** |
| Function | Returns the input where the bit at position `Index` was set to 1 |
| Inputs | `Input`: 'T where 'T is unsigned |
| Hidden inputs | `Index`: 'T where 'T is unsigned |
| Outputs | `Output`: 'T where 'T is unsigned |

# Decrease LSB

**Symbol 3.22:**

| | |
|---|---|
| SCADE name | **iDecLsb** |
| Function | Computes a multiplication by a power of 2 (implemented as a logical bit shift to the left). Same behavior as "Shift Left". |
| Inputs | `In1`: 'T1 , with 'T1 = int8, int16, int32, uint8, uint16 or uint32 |
| Hidden inputs | `Shift` : uint8 |
| Outputs | `Out1`: 'T1 |
| Caution | Overflow if one of the bits shifted outside the binary representation of the type has value 1. Used for decreasing the scale of fixed-point data flow (*e.g.,* from lsb -2 to lsb -4) corresponds to a multiplication by a power of 2, implemented by a shift left. |

# Extract Bit Range

**Symbol 3.23:**

| | |
|---|---|
| SCADE name | **ExtractBitRange** |
| Function | Takes the bits of `Input` from index `Start` to index `End` and returns a value that has exactly those bits (filling the beginning with zeros) |
| Inputs | `Input`: uint32 |
| Hidden inputs | `Start`: uint32<br>`End`: uint32 |
| Outputs | `Output`: uint32 |

# Increase LSB to Ceiling

**Symbol 3.24:**

| | |
|---|---|
| SCADE name | **iIncLsbCeil** |
| Function | Computes a division by a power of 2 (implemented as a logical bit shift to the right) with rounding toward ceiling. |
| Inputs | `In1`: 'T1 , with 'T1 = int8, int16, int32, uint8, uint16 or uint32 |
| Hidden inputs | `Shift` : uint8 |
| Outputs | `Out1`: 'T1 |
| Caution | Used for increasing the scale of fixed-point data flow (e.g., from lsb -4 to lsb -2) corresponds to a division by a power of 2, implemented by a shift right. |

# Increase LSB to Fix

**Symbol 3.25:**


| | |
|---|---|
| SCADE name | **iIncLsbFix** |
| Function | Computes a division by a power of 2 (implemented as a logical bit shift to the right) with rounding toward zero. |
| Inputs | `In1`: 'T1 , with 'T1 = int8, int16, int32, uint8, uint16, or uint32 |
| Hidden inputs | `Shift` : uint8 |
| Outputs | `Out1`: 'T1 |
| Caution | Used for increasing the scale of fixed-point data flow (e.g., from lsb -4 to lsb -2) corresponds to a division by a power of 2, implemented by a shift right. |

# Increase LSB to Floor

**Symbol 3.26:**


| | |
|---|---|
| SCADE name | **iIncLsbFloor** |
| Function | Computes a division by a power of 2 (implemented as a logical bit shift to the right) with rounding toward floor. Same operator behavior as "Shift Right". |
| Inputs | `In1`: 'T1 , with 'T1 = int8, int16, int32, uint8, uint16, or uint32 |
| Hidden inputs | `Shift` : uint8 |
| Outputs | `Out1`: 'T1 |
| Caution | Used for increasing the scale of fixed-point data flow (e.g., from lsb -4 to lsb -2) corresponds to a division by a power of 2, implemented by a shift right. |

# Shift Left

**Symbol 3.27:**

| | |
|---|---|
| SCADE name | **iShiftLeft** |
| Function | A logical bit shift to the left. |
| Inputs | `In1`: 'T1 , with 'T1 = int8, int16, int32, uint8, uint16, or uint32 |
| Hidden inputs | `Shift` : uint8 |
| Outputs | `Out1`: 'T1 |

# Shift Right

**Symbol 3.28:**

| | |
|---|---|
| SCADE name | **iShiftRight** |
| Function | A logical bit shift to the right. |
| Inputs | `In1`: 'T1 , with 'T1 = int8, int16, int32, uint8, uint16, or uint32 |
| Hidden inputs | `Shift` : uint8 |
| Outputs | `Out1`: 'T1 |
| Comment | Corresponds to an arithmetic division rounded to floor with no underflow check.<br>Result of division can be rounded in different ways using other operators like "Increase LSB to Ceiling", "Increase LSB to Fix", or "Increase LSB to Floor". |

# 4 / Library liblinear

This library contains operators that compute linear functions, discrete filters, or discrete transfer functions. You can read the technical description of the library components by package:

- "Filter and Transfer Function Operators" in **`filters`** package
- "Linear Function Operators" in **`linear`** package

Access the description of library operators in alphabetical order:


"Backlash"
"Derivative"
"Detect (Change)"
"Detect (Decrease)"
"Detect (Fall Negative)"
"Detect (Fall Non-Positive)"
"Detect (Increase)"
"Detect (Rise Non-Negative)"
"Detect (Rise Positive)"
"Discrete Filter (Numerator scalar, Denominator deg. 1)"
"Discrete Filter (Numerator scalar, Denominator deg. 1 normalized)"
"Discrete Filter (Numerator scalar, Denominator deg. 2)"
"Discrete Filter (Numerator scalar, Denominator deg. 2 normalized)"
"Discrete Filter (Numerator scalar, Denominator deg. Ds)"
"Discrete Filter (Numerator scalar, Denominator deg. Ds normalized)"
"Discrete Filter (Numerator deg. 1, Denominator deg. 1)"
"Discrete Filter (Numerator deg. 1, Denominator deg. 1 normalized)"
"Discrete Filter (Numerator deg. 1, Denominator deg. 2)"
"Discrete Filter (Numerator deg. 1, Denominator deg. 2 normalized)"
"Discrete Filter (Numerator deg. Ns, Denominator deg. Ds)"
"Discrete Filter (Numerator deg. Ns, Denominator deg. Ds normalized)"
"Discrete Filter (Numerator deg. Ns, Denominator deg. Ns)"
"Discrete Filter (Numerator deg. Ns, Denominator deg. Ns normalized)"
"Filter (First Order Loop)"
"Gain"
"Hit Crossing (Either Direction)"
"Hit Crossing (Falling Direction)"
"Hit Crossing (Rising Direction)"
"Integrator (Backward)"
"Integrator (Forward)"
"Integrator (Trapezoidal)"
"Mean Cycle (2 values)"
"Mean Cycle (3 values)"
"Memory"
"Memory (Basic)"
"Transfer Function (Numerator scalar, Denominator deg. 1)"
"Transfer Function (Numerator scalar, Denominator deg. 2)"
"Transfer Function (Numerator scalar, Denominator deg. Ds)"
"Transfer Function (Numerator deg. 1, Denominator deg. 2)"
"Transfer Function (Numerator deg. Ns, Denominator deg. Ds)"

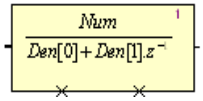
# Filter and Transfer Function Operators

Read the description of the following operators:


“Discrete Filter (Numerator scalar, Denominator deg. 1)”
“Discrete Filter (Numerator scalar, Denominator deg. 1 normalized)”
“Discrete Filter (Numerator scalar, Denominator deg. 2)”
“Discrete Filter (Numerator scalar, Denominator deg. 2 normalized)”
“Discrete Filter (Numerator scalar, Denominator deg. Ds)”
“Discrete Filter (Numerator scalar, Denominator deg. Ds normalized)”
“Discrete Filter (Numerator deg. 1, Denominator deg. 1)”
“Discrete Filter (Numerator deg. 1, Denominator deg. 1 normalized)”
“Discrete Filter (Numerator deg. 1, Denominator deg. 2)”
“Discrete Filter (Numerator deg. 1, Denominator deg. 2 normalized)”
“Discrete Filter (Numerator deg. Ns, Denominator deg. Ds)”
“Discrete Filter (Numerator deg. Ns, Denominator deg. Ds normalized)”
“Discrete Filter (Numerator deg. Ns, Denominator deg. Ns)”
“Discrete Filter (Numerator deg. Ns, Denominator deg. Ns normalized)”
“Transfer Function (Numerator scalar, Denominator deg. 1)”
“Transfer Function (Numerator scalar, Denominator deg. 2)”
“Transfer Function (Numerator scalar, Denominator deg. Ds)”
“Transfer Function (Numerator deg. 1, Denominator deg. 2)”
“Transfer Function (Numerator deg. Ns, Denominator deg. Ds)”

# Discrete Filter (Numerator scalar, Denominator deg. 1)

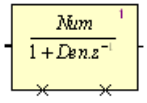
**Symbol 4.1:**




| | |
|---|---|
| SCADE name | **Filter01** |
| Package | filters |
| Function | Discrete filter expressed as a Z-transform function: `Num / (Den[0] + Den[1]*z^-1)`<br>Optimization of FilterND when numerator order is 0 and denominator order is 1. |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T<br>`Den`: 'T ^2 where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = (Num*InX)/Den[0]. |

# Discrete Filter (Numerator scalar, Denominator deg. 1 normalized)

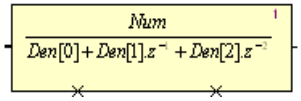
**Symbol 4.2:**




| | |
|---|---|
| SCADE name | **Filter01Norm** |
| Package | filters |
| Function | Normalized discrete filter expressed as a Z-transform function: `Num / (1 + Den*z^-1)`<br>Optimization of FilterND when numerator order is 0 and denominator order is 1. |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T<br>`Den`: 'T where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = Num*InX. |

# Discrete Filter (Numerator scalar, Denominator deg. 2)

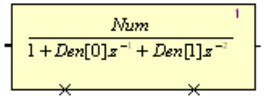
**Symbol 4.3:**




| | |
|---|---|
| SCADE name | **Filter02** |
| Package | filters |
| Function | Discrete filter expressed as a Z-transform function:<br>`Num / (Den[0] + Den[1]*z^-1  + Den[2]*z^-2)`<br>Optimization of FilterND when numerator order is 0 and denominator order is 2. |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T and `Den`: 'T ^3 where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = (Num*InX)/Den[0]. |

# Discrete Filter (Numerator scalar, Denominator deg. 2 normalized)

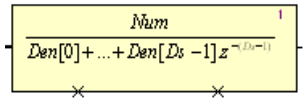
**Symbol 4.4:**




| | |
|---|---|
| SCADE name | **Filter02Norm** |
| Package | filters |
| Function | Normalized discrete filter expressed as a Z-transform function:<br>`Num / (1 + Den[0]*z^-1 + Den[1]*z^-2)`<br>Optimization of FilterND when numerator order is 0 and denominator order is 2. |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T and `Den`: 'T ^2 where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = Num*InX. |

# Discrete Filter (Numerator scalar, Denominator deg. Ds)

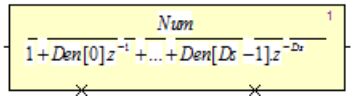
**Symbol 4.5:**




| | |
|---|---|
| SCADE name | **Filter0D** |
| Package | filters |
| Function | Discrete filter expressed as a Z-transform function:<br>`Num / (Den[0] + Den[1]*z^-1  + ... + Den[Ds - 1]*z^-(Ds - 1))`<br>Optimization of FilterND when numerator order is 0 and denominator order is Ds - 1. |
| Size parameters | `Ds` is denominator's array size (Ds > 2) |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T and `Den`: 'T ^Ds where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = (Num*InX)/Den[0]. |

# Discrete Filter (Numerator scalar, Denominator deg. Ds normalized)

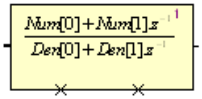
**Symbol 4.6:**




| | |
|---|---|
| SCADE name | **Filter0DNorm** |
| Package | filters |
| Function | Normalized discrete filter expressed as a Z-transform function:<br>`Num / (1 + Den[0]*z^-1  + ... + Den[Ds - 1]*z^-(Ds-2))`<br>Optimization of FilterND when numerator order is 0 and denominator order is Ds. |
| Size parameters | `Ds` is denominator's array size (Ds > 2) |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T and `Den`: 'T ^Ds where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = Num*InX. |

# Discrete Filter (Numerator deg. 1, Denominator deg. 1)

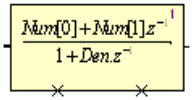
**Symbol 4.7:**



| | |
|---|---|
| SCADE name | **Filter11** |
| Package | filters |
| Function | Discrete filter expressed as a Z-transform function:<br>`(Num[0] + Num[1]*z^-1) / (Den[0] + Den[1]*z^-1)`<br>Optimization of FilterND when numerator order is 1 and denominator order is 1. |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T ^2 and `Den`: 'T ^2 where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = (Num*InX)/Den[0]. |

# Discrete Filter (Numerator deg. 1, Denominator deg. 1 normalized)

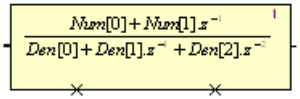
**Symbol 4.8:**



| | |
|---|---|
| SCADE name | **Filter11Norm** |
| Package | filters |
| Function | Normalized discrete filter expressed as a Z-transform function:<br>`(Num[0] + Num[1]*z^-1) / ( 1 + Den*z^-1)`<br>Optimization of FilterND when numerator order is 1 and denominator order is 1. |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T ^2 and `Den`: 'T where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = Num*InX. |

# Discrete Filter (Numerator deg. 1, Denominator deg. 2)

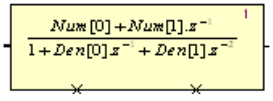
**Symbol 4.9:**




| | |
|---|---|
| SCADE name | **Filter12** |
| Package | filters |
| Function | Discrete filter expressed as a Z-transform function:<br>`(Num[0] + Num[1]*z^-1) / (Den[0] + Den[1]*z^-1  + Den[2]*z^-2)`<br>Optimization of FilterND when numerator order is 1 and denominator order is 2. |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T ^2 and `Den`: 'T ^3 where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = (Num*InX)/Den[0]. |

# Discrete Filter (Numerator deg. 1, Denominator deg. 2 normalized)

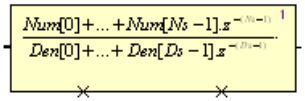
**Symbol 4.10:**




| | |
|---|---|
| SCADE name | **Filter12Norm** |
| Package | filters |
| Function | Normalized discrete filter expressed as a Z-transform function:<br>`(Num[0] + Num[1]*z^-1) / (1 + Den[0]*z^-1 + Den[1]*z^-2)`<br>Optimization of FilterND when numerator order is 1 and denominator order is 2. |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T ^2 and `Den`: 'T ^2 where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = Num*InX. |

# Discrete Filter (Numerator deg. Ns, Denominator deg. Ds)

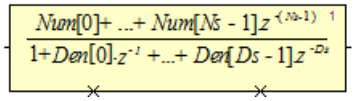
**Symbol 4.11:**



| | |
|---|---|
| SCADE name | **FilterND** |
| Package | filters |
| Function | Discrete filter expressed as a Z-transform function:<br>`(Num[0] + Num[1]*z^-1 + Num[Ns - 1]*z^(Ns - 1)) / (Den[0] + Den[1]*z^-1 +... + Den[Ds - 1]*z^-(Ds - 1))` |
| Size parameters | `Ns` is the numerator array size and `Ds` is the denominator array size (Ds > Ns > 1) |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T ^Ns and `Den`: 'T ^Ds where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = (Num*InX)/Den[0]. |

# Discrete Filter (Numerator deg. Ns, Denominator deg. Ds normalized)

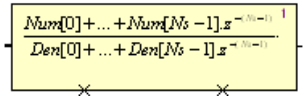
**Symbol 4.12:**




| | |
|---|---|
| SCADE name | **FilterNDNorm** |
| Package | filters |
| Function | Normalized discrete filter expressed as a Z-transform function:<br>`(Num[0] + Num[1]*z^-1 + Num[Ns - 1]*z^-(Ns - 1)) / (1 + Den[0]*z^-1 +... + Den[Ds - 1]*z^-Ds)` |
| Size parameters | `Ns` is the numerator array size and `Ds` is the denominator array size (Ds => Ns > 1) |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T ^Ns and `Den`: 'T ^Ds where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = Num*InX. |

# Discrete Filter (Numerator deg. Ns, Denominator deg. Ns)

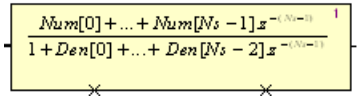
**Symbol 4.13:**



| | |
|---|---|
| SCADE name | **FilterNN** |
| Package | filters |
| Function | Discrete filter expressed as a Z-transform function:<br>`(Num[0] + Num[1]*z^-1 +...+ Num[Ns - 1]*z^-(Ns - 1)) / (Den[0] + Den[1]*z^-1 + ... + Den[Ns - 1]*z^-(Ns - 1))` |
| Size parameters | `Ns` is numerator's and denominator's array size (Ns > 1) |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T ^Ns and `Den`: 'T ^Ns where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = (Num*InX)/Den[0]. |

# Discrete Filter (Numerator deg. Ns, Denominator deg. Ns normalized)

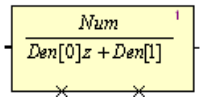
**Symbol 4.14:**



| | |
|---|---|
| SCADE name | **FilterNNNorm** |
| Package | filters |
| Function | Normalized discrete filter expressed as a Z-transform function:<br>`(Num[0] + Num[1]*z^-1 + Num[Ns - 1]*z^-(Ns - 1)) / (1 + Den[0]*z^-1 + ... + Den[Ns - 2]*z^-(Ns - 1))` |
| Size parameters | `Ns` is numerator's array size and `Ns - 1` is denominator's array size (Ns > 1) |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T ^Ns and `Den`: 'T ^(Ns - 1) where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = Num*InX. |

# Transfer Function (Numerator scalar, Denominator deg. 1)

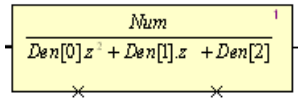
**Symbol 4.15:**



| | |
|---|---|
| SCADE name | **TransferFcn01** |
| Package | filters |
| Function | Discrete transfer function expressed as a Z-transform function:<br>`(Num) / (Den[0]*z + Den[1])`<br>Optimization of TransferFcnND when numerator order is 0 and the denominator order is 1. |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T and `Den`: 'T ^2 where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = 0. |

# Transfer Function (Numerator scalar, Denominator deg. 2)

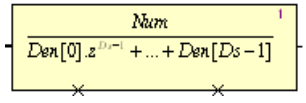
**Symbol 4.16:**




| | |
|---|---|
| SCADE name | **TransferFcn02** |
| Package | filters |
| Function | Discrete transfer function expressed as a Z-transform function:<br>`(Num) / (Den[0]*z^2 + Den[1]*z + Den[2])`<br>Optimization of TransferFcnND when numerator order is 0 and the denominator order is 2. |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T and `Den`: 'T ^3 where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = 0. |

# Transfer Function (Numerator scalar, Denominator deg. Ds)

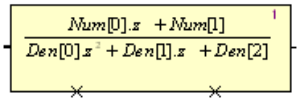
**Symbol 4.17:**




| | |
|---|---|
| SCADE name | **TransferFcn0D** |
| Package | filters |
| Function | Discrete transfer function expressed as a Z-transform function:<br>`(Num) / (Den[0]*z^(Ds - 1) + ... + Den[Ds - 2]*z + Den[Ds-1])`<br>Optimization of TransferFcnND when numerator order is 0 and the denominator order is Ds - 1. |
| Size parameters | `Ds` is denominator's array size (Ds > 2) |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T and `Den`: 'T ^Ds where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = 0. |

# Transfer Function (Numerator deg. 1, Denominator deg. 2)

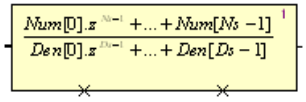
**Symbol 4.18:**




| | |
|---|---|
| SCADE name | **TransferFcn12** |
| Package | filters |
| Function | Discrete transfer function expressed as a Z-transform function:<br>`(Num[0]*z + Num[1]) / (Den[0]*z^2 + Den[1]*z + Den[2])`<br>Optimization of TransferFcnND when numerator order is 1 and the denominator order is 2. |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T ^2 and `Den`: 'T ^3 where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = 0. |

# Transfer Function (Numerator deg. Ns, Denominator deg. Ds)

**Symbol 4.19:**



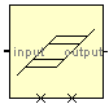

| | |
|---|---|
| SCADE name | **TransferFcnND** |
| Package | filters |
| Function | Discrete transfer function expressed as a Z-transform function:<br>`(Num[0]*z^(Ns - 1) + ... + Num[Ns - 2]*z + Num[Ns - 1]) / (Den[0]*z^(Ds - 1) + ... + Den[Ds - 2]*z + Den[Ds - 1])` |
| Size parameters | `Ns` is the numerator array size and `Ds` is the denominator array size (Ds > Ns > 1) |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T ^Ns and `Den`: 'T ^Ds where 'T is float |
| Outputs | `Out`: 'T where 'T is float |
| Comment | At first tick, Out = 0. |

# Linear Function Operators

Read the description of the following operators:


"Backlash"
"Derivative"
"Detect (Change)"
"Detect (Decrease)"
"Detect (Fall Negative)"
"Detect (Fall Non-Positive)"
"Detect (Increase)"
"Detect (Rise Non-Negative)"
"Detect (Rise Positive)"
"Filter (First Order Loop)"
"Gain"
"Hit Crossing (Either Direction)"
"Hit Crossing (Falling Direction)"
"Hit Crossing (Rising Direction)"
"Integrator (Backward)"
"Integrator (Forward)"
"Integrator (Trapezoidal)"
"Mean Cycle (2 values)"
"Mean Cycle (3 values)"
"Memory"
"Memory (Basic)"

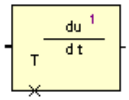

## Backlash

**Symbol 4.20:** 


| | |
|---|---|
| SCADE name | **Backlash** |
| Package | linear |
| Function | The behavior is as follows: if the input was decreasing before and is now increasing, the output remains constant until it has crossed `deadband_width`, and then it increases at the same rhythm as the input. And vice versa. |
| Inputs | `input`: 'T where 'T is float |
| Hidden inputs | `deadband_width`: 'T and `initial_output`: 'T where 'T is float |
| Outputs | `output`: 'T where 'T is float |

# Derivative

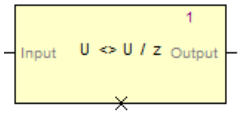
**Symbol 4.21:**




| | |
|---|---|
| SCADE name | **Derivative** |
| Package | linear |
| Function | Computes a simple approximation of the input derivative:<br>Output(0)=0.0<br>Output(k)=(u(k)-u(k-1))/TimeCycle |
| Inputs | `U`: 'T where 'T is float |
| Hidden inputs | `TimeCycle`: 'T where 'T is float |
| Outputs | `Derivative`: 'T where 'T is float |
| Caution | Very sensitive to noise |

# Detect (Change)

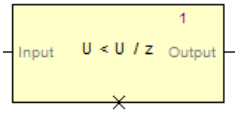
**Symbol 4.22:**




| | |
|---|---|
| SCADE name | **DetectChange** |
| Package | linear |
| Function | The output is `true` if the value of the input is different from that of previous cycle |
| Inputs | `Input`: 'T where 'T is numeric |
| Hidden inputs | `Init`: 'T where 'T is numeric |
| Outputs | `Output`: bool |

# Detect (Decrease)

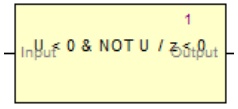
**Symbol 4.23:**




| | |
|---|---|
| SCADE name | **DetectDecrease** |
| Package | linear |
| Function | The output is `true` if the value of the input is strictly lower than that of previous cycle |
| Inputs | `Input`: 'T where 'T is numeric |
| Hidden inputs | `Init`: 'T where 'T is numeric |
| Outputs | `Output`: bool |

# Detect (Fall Negative)

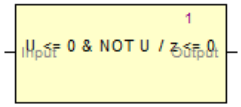
**Symbol 4.24:**




| | |
|---|---|
| SCADE name | **DetectFallNegative** |
| Package | linear |
| Function | The output is `true` if the input is negative and was non-negative on previous cycle |
| Inputs | `Input`: 'T where 'T is numeric |
| Outputs | `Output`: bool |

# Detect (Fall Non-Positive)

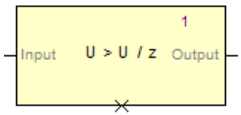
**Symbol 4.25:**




| | |
|---|---|
| SCADE name | **DetectFallNonPositive** |
| Package | linear |
| Function | The output is `true` if the input is non-positive and was positive on previous cycle |
| Inputs | `Input`: 'T where 'T is numeric |
| Outputs | `Output`: bool |

# Detect (Increase)

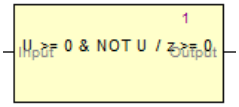
**Symbol 4.26:**




| | |
|---|---|
| SCADE name | **DetectIncrease** |
| Package | linear |
| Function | The output is `true` if the value of the input is strictly greater than that of previous cycle |
| Inputs | `Input`: 'T where 'T is numeric |
| Hidden inputs | `Init`: 'T where 'T is numeric |
| Outputs | `Output`: bool |

# Detect (Rise Non-Negative)

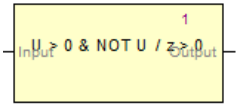
**Symbol 4.27:**




| | |
|---|---|
| SCADE name | **DetectRiseNonNegative** |
| Package | linear |
| Function | The output is `true` if the input is non-negative and was negative on previous cycle |
| Inputs | `Input`: 'T where 'T is numeric |
| Outputs | `Output`: bool |

# Detect (Rise Positive)

**Symbol 4.28:**




| | |
|---|---|
| SCADE name | **DetectRisePositive** |
| Package | linear |
| Function | The output is `true` if the input is positive and was non-positive on previous cycle |
| Inputs | `Input`: 'T where 'T is numeric |
| Outputs | `Output`: bool |

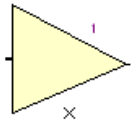
# Filter (First Order Loop)

**Symbol 4.29:**

| | |
|---|---|
| SCADE name | **Filter1stOrderLoop** |
| Package | linear |
| Function | Creates a customizable first order filter working on two cycles.<br>At initialization, output equals Init. After initialization, output equals K1*Input(k)-k2*Output_R(k-1) where k stands for the current cycle. |
| Inputs | `F1OL_Input`: 'T where 'T is numeric |
| Hidden inputs | `K1`: 'T - `K2`: 'T - `Init`: 'T where 'T is numeric |
| Outputs | `F1OL_Output`: 'T where 'T is numeric |
| KCG pragma | Operator expansion |
| Comment | Filter created using transfer function: K1/(1+k2*$z^{-1}$). |

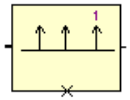
# Gain

**Symbol 4.30:**



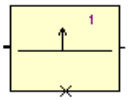
| | |
|---|---|
| SCADE name | **Gain** |
| Package | linear |
| Function | Computes the value of an input multiplied by a set gain value. G_Output equals G_Input multiplied by Gain. |
| Inputs | `G_Input`: 'T |
| Hidden inputs | `Gain`: 'T |
| KCG pragma | Operator expansion |
| Outputs | `G_Output`: 'T where 'T is numeric |

# Hit Crossing (Either Direction)

**Symbol 4.31:** 

| | |
|---|---|
| SCADE name | **HitCrossingEither** |
| Package | linear |
| Function | Detects crossing of a specified value by a signal from either direction. Output is `true` when the input reaches the offset by growing or decreasing. |
| Inputs | `HCE_Input`: 'T |
| Hidden inputs | `Offset`: 'T where 'T is numeric |
| Outputs | `HCE _Output`: bool |

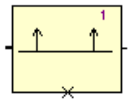
# Hit Crossing (Falling Direction)

**Symbol 4.32:** 

| | |
|---|---|
| SCADE name | **HitCrossingFalling** |
| Package | linear |
| Function | Detects crossing of a specified value by a falling signal. Output is `true` when the input reaches the offset by decreasing. |
| Inputs | `HCF_Input`: 'T |
| Hidden inputs | `Offset`: 'T where 'T is numeric |
| Outputs | `HCF_Output`: bool |

# Hit Crossing (Rising Direction)

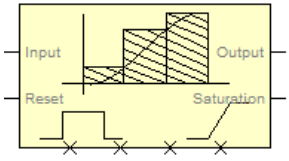
**Symbol 4.33:**




| | |
|---|---|
| SCADE name | **HitCrossingRising** |
| Package | linear |
| Function | Detects crossing of a specified value by a rising signal. Output is `true` when the input reaches the offset by growing. |
| Inputs | `HCR_Input`: 'T where 'T is numeric |
| Hidden inputs | `Offset`: 'T where 'T is numeric |
| Outputs | `HCR _Output`: bool |

# Integrator (Backward)

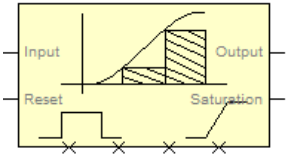
**Symbol 4.34:**




| | |
|---|---|
| SCADE name | **IntegralBackward** |
| Package | linear |
| Function | Backward Euler discrete-time integrator.<br>Computes the integral of `Input` using the backward rule. If the integral is outside of the `]LowLimit, Highlimit[` range, it is saturated to this range. If saturation is active, the `Saturation` output is `true`. |
| Inputs | `Input`: 'T where 'T is numeric |
| Hidden inputs | `Reset`: bool - `HighLimit`: 'T (upper bound of integral) - `LowLimit`: 'T (lower bound of integral)<br>`init`: 'T - `deltaT`: 'T where 'T is numeric |
| Outputs | `Output`: 'T and `Saturation`: bool where 'T is numeric |

# Integrator (Forward)

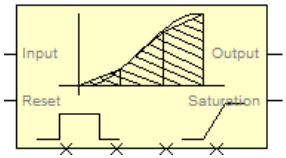
**Symbol 4.35:**




| | |
|---|---|
| SCADE name | **IntegralForward** |
| Package | linear |
| Function | Forward Euler discrete-time integrator.<br>The **IntegralForward** operator computes the integral of `Input` using the forward rule. If the integral is outside of the `]LowLimit, HighLimit[` range, it is saturated to this range. If saturation is active, the `Saturation` output is `true`. |
| Inputs | `Input`: 'T where 'T is numeric and `Reset`: bool |
| Hidden inputs | `HighLimit`: 'T (upper bound of integral)- `LowLimit`: 'T (lower bound of integral)<br>`init`: 'T - `deltaT`: 'T where 'T is numeric where 'T is numeric |
| Outputs | `Output`: 'T and `Saturation`: bool where 'T is numeric |

# Integrator (Trapezoidal)

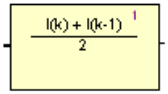
**Symbol 4.36:**




| | |
|---|---|
| SCADE name | **IntegralTrapezoid** |
| Package | linear |
| Function | Trapezoidal discrete-time integrator.<br>The **IntegralTrapezoid** operator computes the integral of `Input` using the trapezoid rule. If the integral is outside of the `]LowLimit, HighLimit[` range, it is saturated to this range. If saturation is active, the `Saturation` output is `true`. |
| Inputs | `Input`: 'T where 'T is numeric and `Reset`: bool |
| Hidden inputs | `HighLimit`: 'T (upper bound of integral) - `LowLimit`: 'T (lower bound of integral)<br>`init`: 'T - `deltaT`: 'T where 'T is numeric |
| Outputs | `Output`: 'T and `Saturation`: bool where 'T is numeric |
| Comment | Forward integrator y(k) = y(k-1) + T/2 * (u(k) + u(k-1)). Output is limited to the range [LowLimit,HighLimit]. Output is reset to zero at first cycle and when reset is `true`. |

# Mean Cycle (2 values)

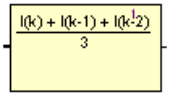
**Symbol 4.37:**




| | |
|---|---|
| SCADE name | **MeanCycle2** |
| Package | linear |
| Function | Computes the mean of two iterative values of the signal input.<br>MeanOn2Step (0) = $\frac{(U(0) + U(0))}{2.0}$<br>MeanOn2Step (k) = $\frac{(U(k) + U(k-1))}{2.0}$ |
| Inputs | U: 'T where 'T is float |
| Outputs | `MeanOn2Step`: 'T where 'T is float |

# Mean Cycle (3 values)

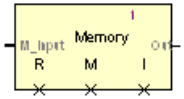
**Symbol 4.38:**




| | |
|---|---|
| SCADE name | **MeanCycle3** |
| Package | linear |
| Function | Computes the mean of three iterative values of the signal input.<br>MeanOn3Steps (0) = $\frac{(U(0) + U(0) + U(0))}{3.0}$<br>MeanOn3Steps (1) = $\frac{(U(1) + U(0) + U(0))}{3.0}$<br>MeanOn3Steps (k) = $\frac{(U(k) + U(k-1) + U(k-2))}{3.0}$ |
| Inputs | U: 'T where 'T is float |
| Outputs | `MeanOn2Step`: 'T where 'T is float |

# Memory

**Symbol 4.39:**




| | |
|---|---|
| SCADE name | **Memory** |
| Package | linear |
| Function | Stores a value with resettable initial value. If Reset is `true`, then Out = InitVal , else if MemCond is `true`, Out = Input, else Out equals last stored value. |
| Inputs | `M_Input`: 'T where 'T numeric |
| Hidden inputs | R (`Reset`): bool - `M` (`MemCond`): bool - `I` (`InitVal`): 'T where 'T numeric |
| Outputs | `Out`: 'T where 'T numeric |
| KCG pragma | Operator expansion |
| Comment | Initialization condition overcomes memorization condition. For any value of MemCond, if Reset is `true`, Output equals InitVal. |

# Memory (Basic)

**Symbol 4.40:**

| | |
|---|---|
| SCADE name | **MemoryBasic** |
| Package | linear |
| Function | Stores a value without resetting memory to its initial value. If Write condition is `true`, then Out = Input, else Out equals last output. Initial value is set to Init. |
| Inputs | `BM_Input`: 'T where 'T numeric |
| Hidden inputs | `Init`: 'T where 'T numeric and `Write`: bool |
| Outputs | `Memorized`: 'T where 'T numeric |
| KCG pragma | Operator expansion |
| Comment | At initialization, if Write equals `false`, the output value and the memory equal Init. Init and BM_Input are often linked to the same source. |

# 5 / Library libmath

This library contains basic mathematical operators and mathematical operations on vectors and matrices. You can read the technical description of the library components by package:

- "Arithmetic Operators" in **`math`** package
- "Conversion Operators" in **`math`** package
- "Interval Operators" in **`math`** package
- "Vector and Matrix Operators" in **`vect`** package

Access the description of library operators in alphabetical order:


“Absolute Value”
“Boolean to Float32”
“Boolean to Float64”
“Boolean to Int8”
“Boolean to Int16”
“Boolean to Int32”
“Boolean to Int64”
“Boolean to Uint8”
“Boolean to Uint16”
“Boolean to Uint32”
“Boolean to Uint16”
“BuildArray”
“Determinant of Square Matrix (2x2)”
“Determinant of Square Matrix (3x3)”
“Determinant of Square Matrix (4x4)”
“In Range (In-In)”
“In Range (In-Out)”
“In Range (Out-In)”
“In Range (Out-Out)”
“Integer to Boolean”
“Inverse of Square Matrix (2x2)”
“Inverse of Square Matrix (3x3)”
“Inverse of Square Matrix (4x4)”
“Matrix Addition”
“Matrix Difference”
“Matrix Product”
“Matrix Product by Vector”
“Maximum (2 inputs)”
“Maximum (3 inputs)”
“Mean (2 inputs)”
“Mean (3 inputs)”
“Minimum (2 inputs)”
“Minimum (3 inputs)”
“Mod”
“Octet to Uint8”
“Polynomial”
“Real to Boolean”
“Rounding Function”
“Rounding to Ceiling Function”
“Rounding to Floor Function”
“Scalar Product”
“Selector”
“Sign”
“Vector Addition”
“Vector Difference”
“Vector Product by Matrix”

# Arithmetic Operators

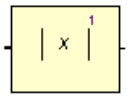
The **`libmath`** library contains several basic arithmetic operators that allow to handle integer or real values.

Read the description of the following operators:


"Absolute Value"
"Maximum (2 inputs)"
"Maximum (3 inputs)"
"Mean (2 inputs)"
"Mean (3 inputs)"
"Minimum (2 inputs)"
"Minimum (3 inputs)"
"Mod"
"Polynomial"
"Sign"


## Absolute Value

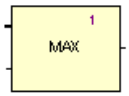
**Symbol 5.1:**




| | |
|---|---|
| SCADE name | **Abs** |
| Package | math |
| Function | Computes the absolute value of real or integer values. |
| Inputs | `A_Input`: 'T where 'T float |
| Outputs | `A_Output`: 'T where 'T float |
| KCG pragma | Operator expansion |
| Comment | If the input is positive or null then the output is equal to the input, else the output is equal to the opposite of input. |

# Maximum (2 inputs)

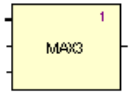
**Symbol 5.2:**




| | |
|---|---|
| SCADE name | **Max** |
| Package | math |
| Function | Returns the maximum output from two real or integer values (if In1 >= In2, return In1, else return In2). |
| Inputs | `I1` and `I2`: 'T where 'T numeric |
| Outputs | `Ma_Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |

# Maximum (3 inputs)

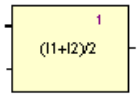
**Symbol 5.3:**




| | |
|---|---|
| SCADE name | **Max3** |
| Package | math |
| Function | Returns the maximum output from three real or integer values (if In1 >= In2 AND In1 >= In3, return In1, else if In2 >= In3, return In2, else return In3). |
| Inputs | `I1`, `I2`, and `I3`: 'T where 'T numeric |
| Outputs | `Ma3_Output`: 'T where 'T numeric |

# Mean (2 inputs)

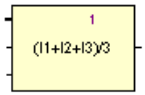
**Symbol 5.4:**




| | |
|---|---|
| SCADE name | **Mean** |
| Package | math |
| Function | Computes the mean of two real inputs. ( I1 + I2 ) / 2.0 |
| Inputs | `I1` and `I2`: 'T where 'T numeric |
| Outputs | `Me_output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |

# Mean (3 inputs)

**Symbol 5.5:**




| | |
|---|---|
| SCADE name | **Mean3** |
| Package | math |
| Function | Computes the mean of three real inputs. (I1 + I2 + I3) / 3.0 |
| Inputs | `I1` and `I2`: 'T where 'T numeric |
| Outputs | `Me3_output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |

# Minimum (2 inputs)

**Symbol 5.6:**

| | |
|---|---|
| SCADE name | **Min** |
| Package | math |
| Function | Returns the minimum output from two real or integer values (if In1 <= In2, return In1, else return In2). |
| Inputs | `I1` and `I2`: 'T where 'T numeric |
| Outputs | `Mi_Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |

# Minimum (3 inputs)

**Symbol 5.7:**

| | |
|---|---|
| SCADE name | **Min3** |
| Package | math |
| Function | Returns the minimum output from three real or integer values (if In1 <= In2 AND In1 <= In3, return In1, else if In2 <= In3, return In2, else return In3). |
| Inputs | `I1`, `I2`, and `I3`: 'T where 'T numeric |
| Outputs | `Mi3_Output`: 'T where 'T numeric |

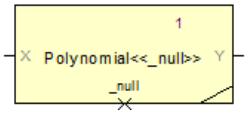
# Mod

**Symbol 5.8:**

| | |
|---|---|
| SCADE name | **Mod** |
| Package | math |
| Function | Computes the modulo function.<br>If the divisor (`modulus`) is strictly equal to 0, a default value is returned instead. |
| Inputs | `value`: 'T where 'T numeric<br>`modulus`: 'T where 'T numeric |
| Outputs | `remain`: 'T where 'T numeric |

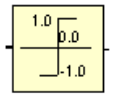
# Polynomial

**Symbol 5.9:**




| | |
|---|---|
| SCADE name | **Polynomial** |
| Package | math |
| Function | If C is the list [$C_{(N-1)}$, $C_{(N-2)}$, ..., $C_0$,], the operator computes the following:<br>$C_{(N-1)} \cdot X^{(N-1)} + C_{(N-2)} \cdot X^{(N-2)} + \ldots + C_0$ |
| Inputs | X: 'T |
| Hidden inputs | C: 'T ^N where 'T numeric |
| Outputs | Y: 'T where 'T numeric |

# Sign

**Symbol 5.10:** 


| | |
|---|---|
| SCADE name | **Sign** |
| Package | math |
| Function | Extracts the sign (+ or -) of a real input and returns a real value from set {-1.0, 0.0, 1.0} based on the sign of the input. If input is positive, then output is equal to 1. If input is negative, then output is equal to -1. If input is 0, then output is equal to 0. |
| Inputs | `S_Input`: 'T where 'T numeric |
| Outputs | `S_Output`: 'T where 'T numeric |
| Caution | Relies on a strict equality between real numbers for the 0.0 value. |

# Conversion Operators

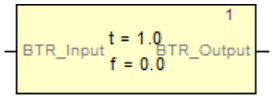
The **`libmath`** library contains operators that allow to convert some input values or round real numbers.

Read the description of the following operators:


"Boolean to Float32"
"Boolean to Float64"
"Boolean to Int8"
"Boolean to Int16"
"Boolean to Int32"
"Boolean to Int64"
"Boolean to Uint8"
"Boolean to Uint16"
"Boolean to Uint32"
"Boolean to Uint64"
"Integer to Boolean"
"Octet to Uint8"
"Real to Boolean"
"Rounding Function"
"Rounding to Ceiling Function"
"Rounding to Floor Function"


## Boolean to Float32

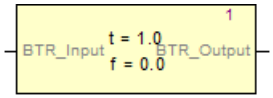
**Symbol 5.11:**




| | |
|---|---|
| SCADE name | **BoolToFloat32** |
| Package | math |
| Function | Converts a Boolean value into a real value.<br>When the input is `true`, the output is equal to 1.0, otherwise it is equal to 0.0. |
| Inputs | `BTI_Input`: bool |
| Outputs | `BTI_Output`: float32 |
| KCG pragma | Operator expansion |

# Boolean to Float64

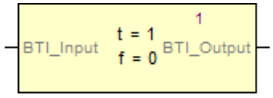
**Symbol 5.12:**




| | |
|---|---|
| SCADE name | **BoolToFloat64** |
| Package | math |
| Function | Converts a Boolean value into a real value.<br>When the input is `true`, the output is equal to 1.0, otherwise it is equal to 0.0. |
| Inputs | `BTR_Input`: bool |
| Outputs | `BTR_Output`: float64 |
| KCG pragma | Operator expansion |

# Boolean to Int8

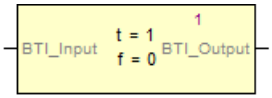
**Symbol 5.13:**




| | |
|---|---|
| SCADE name | **BoolToInt8** |
| Package | math |
| Function | Converts a Boolean input into an integer value.<br>When the input is `true`, the output is equal to 1, otherwise the output is equal to 0. |
| Inputs | `BTI_Input`: bool |
| Outputs | `BTI_Output`: int8 |
| KCG pragma | Operator expansion |

# Boolean to Int16

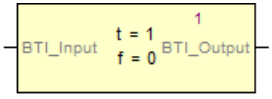
**Symbol 5.14:**




| | |
|---|---|
| SCADE name | **BoolToInt16** |
| Package | math |
| Function | Converts a Boolean value into an integer value.<br>When the input is `true`, the output is equal to 1, otherwise the output is equal to 0. |
| Inputs | `BTI_Input`: bool |
| Outputs | `BTI_Output`: int16 |
| KCG pragma | Operator expansion |

# Boolean to Int32

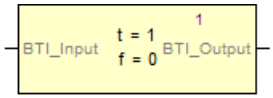
**Symbol 5.15:**




| | |
|---|---|
| SCADE name | **BoolToInt32** |
| Package | math |
| Function | Converts a Boolean value into an integer value.<br>When the input is `true`, the output is equal to 1, otherwise the output is equal to 0. |
| Inputs | `BTI_Input`: bool |
| Outputs | `BTI_Output`: int32 |
| KCG pragma | Operator expansion |

# Boolean to Int64

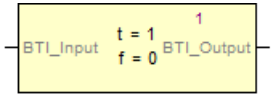
**Symbol 5.16:**




| | |
|---|---|
| SCADE name | **BoolToInt64** |
| Package | math |
| Function | Converts a Boolean value into an integer value.<br>When the input is `true`, the output is equal to 1, otherwise the output is equal to 0. |
| Inputs | `BTI_Input`: bool |
| Outputs | `BTI_Output`: int64 |
| KCG pragma | Operator expansion |

# Boolean to Uint8

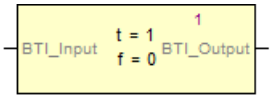
**Symbol 5.17:**




| | |
|---|---|
| SCADE name | **BoolToUint8** |
| Package | math |
| Function | Converts a Boolean value into an integer value.<br>When the input is `true`, the output is equal to 1, otherwise the output is equal to 0. |
| Inputs | `BTI_Input`: bool |
| Outputs | `BTI_Output`: uint8 |
| KCG pragma | Operator expansion |

# Boolean to Uint16

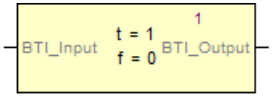
**Symbol 5.18:**




| | |
|---|---|
| SCADE name | **BoolToUint16** |
| Package | math |
| Function | Converts a Boolean value into an integer value.<br>When the input is `true`, the output is equal to 1, otherwise the output is equal to 0. |
| Inputs | `BTI_Input`: bool |
| Outputs | `BTI_Output`: uint16 |
| KCG pragma | Operator expansion |

# Boolean to Uint32

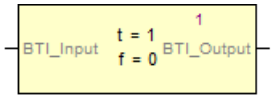
**Symbol 5.19:**




| | |
|---|---|
| SCADE name | **BoolToUint32** |
| Package | math |
| Function | Converts a Boolean value into an integer value.<br>When the input is `true`, the output is equal to 1, otherwise the output is equal to 0. |
| Inputs | `BTI_Input`: bool |
| Outputs | `BTI_Output`: uint32 |
| KCG pragma | Operator expansion |

# Boolean to Uint64

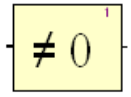
**Symbol 5.20:**




| | |
|---|---|
| SCADE name | **BoolToUint64** |
| Package | math |
| Function | Converts a Boolean value into an integer value.<br>When the input is `true`, the output is equal to 1, otherwise the output is equal to 0. |
| Inputs | `BTI_Input`: bool |
| Outputs | `BTI_Output`: uint64 |
| KCG pragma | Operator expansion |

# Integer to Boolean

**Symbol 5.21:**




| | |
|---|---|
| SCADE name | **IntToBool** |
| Package | math |
| Function | Converts an integer value into a Boolean value.<br>When the input is 0, the output is equal to `false`, otherwise the output is equal to `true`. |
| Inputs | `ITB_Input`: 'T where 'T is integer |
| Outputs | `ITB_Output`: bool |
| KCG pragma | Operator expansion |

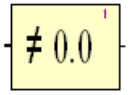
# Octet to Uint8

**Symbol 5.22:**

| | |
|---|---|
| SCADE | **OctetToUint8** |
| Package | math |
| Function | Interprets 8 bits (Boolean inputs) as an unsigned integer value between 0 and 255. |
| Inputs | `b1`, `b2`, `b3`, `b4`, `b5`, `b6`, `b7`, and `b8`: bool |
| Outputs | `OTI_Output`: uint8 |

# Real to Boolean

**Symbol 5.23:**




| | |
|---|---|
| SCADE name | **RealToBool** |
| Package | math |
| Function | Converts a real value into a Boolean value.<br>When the input is 0.0, the output is equal to `false`, otherwise the output is equal to `true`. |
| Inputs | `RTB_Input`: 'T where 'T is float |
| Outputs | `RTB_Output`: bool |
| KCG pragma | Operator expansion |
| Caution | Relies on a strict equality between real numbers. |

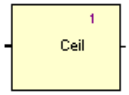
# Rounding Function

**Symbol 5.24:**

| | |
|---|---|
| SCADE name | **Round** |
| Package | math |
| Function | Rounds a real input. R_Output = x when R_Input [ x-0.5, x+0.5 [ |
| Inputs | `R_Input`: 'T where 'T is float |
| Outputs | `R_Output`: int32 |

# Rounding to Ceiling Function

**Symbol 5.25:**




| | |
|---|---|
| SCADE name | **RoundCeil** |
| Package | math |
| Function | Rounds a real input up to the next integer value.<br>RC_Output = x when R_Input ]x–1, x] |
| Inputs | `RC_Input`: 'T where 'T is float |
| Outputs | `RC_Output`: int32 |

# Rounding to Floor Function

**Symbol 5.26:**

| | |
|---|---|
| SCADE name | **RoundFloor** |
| Package | math |
| Function | Rounds a real input down to the next integer value.<br>RF_Output = x when R_Input [x, x+1[ |
| Inputs | `RF_Input`: 'T where 'T is float |
| Outputs | `RF_Output`: int32 |

# Interval Operators

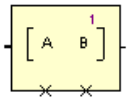
The **`libmath`** library contains operators that allow to compute various types of intervals between input values.

Read the description of the following operators:

"In Range (In-In)"
"In Range (In-Out)"
"In Range (Out-In)"
"In Range (Out-Out)"

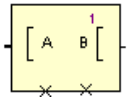
## In Range (In-In)

**Symbol 5.27:**



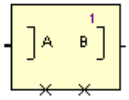

| | |
|---|---|
| SCADE name | **InRangeInIn** |
| Package | math |
| Function | Returns a `true` output if the input belongs to the [A , B ] range; otherwise it is `false`. |
| Inputs | `IRII_Input`: 'T where 'T numeric |
| Hidden inputs | A: 'T (lower border) and B: 'T (upper border) where 'T numeric |
| Outputs | `IRII_Output`: bool |
| KCG pragma | Operator expansion |

# In Range (In-Out)

**Symbol 5.28:** 

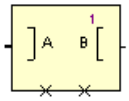

| | |
|---|---|
| SCADE name | **InRangeInOut** |
| Package | math |
| Function | Returns a `true` output if the input belongs to the [A , B [ range; otherwise it is `false`. |
| Inputs | `IRIO_Input`: 'T where 'T numeric<br>Value in the [-MaximumValue ; MaximumValue] range. |
| Hidden inputs | A: 'T (lower border) and B: 'T (upper border) where 'T numeric |
| Outputs | `IRIO_Output`: bool |
| KCG pragma | Operator expansion |

# In Range (Out-In)

**Symbol 5.29:** 


| | |
|---|---|
| SCADE name | **InRangeOutIn** |
| Package | math |
| Function | Returns a `true` output if the input belongs to the ] A , B ] range; otherwise it is `false`. |
| Inputs | `IROI_Input`: 'T where 'T numeric |
| Hidden inputs | A: 'T (lower border) and B: 'T (upper border) where 'T numeric |
| Outputs | `IROI_Output`: bool |
| KCG pragma | Operator expansion |

# In Range (Out-Out)

**Symbol 5.30:** 


| | |
|---|---|
| SCADE name | **InRangeOutOut** |
| Package | math |
| Function | Returns a `true` output if the input belongs to the ] A , B [ range; otherwise it is `false`. |
| Inputs | `IROO_Input`: 'T where 'T numeric |
| Hidden inputs | A: 'T (lower border) and B: 'T (upper border) where 'T numeric |
| Outputs | `IROO_Output`: bool |
| KCG pragma | Operator expansion |

# Vector and Matrix Operators

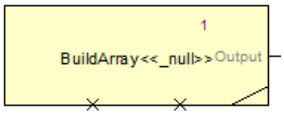
The **`libmath`** library contains several basic operators that allow to handle vectors or matrices.

Read the description of the following operators:


"BuildArray"
"Determinant of Square Matrix (2x2)"
"Determinant of Square Matrix (3x3)"
"Determinant of Square Matrix (4x4)"
"Inverse of Square Matrix (2x2)"
"Inverse of Square Matrix (3x3)"
"Inverse of Square Matrix (4x4)"
"Matrix Addition"
"Matrix Difference"
"Matrix Product"
"Matrix Product by Vector"
"Scalar Product"
"Selector"
"Vector Addition"
"Vector Difference"
"Vector Product by Matrix"


## BuildArray

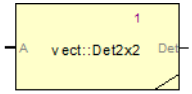
**Symbol 5.31:**




| | |
|---|---|
| SCADE name | **BuildArray** |
| Package | vect |
| Function | Builds an array of values where the first value is `StartValue` and every other value is "Increment"-higher than the previous one.<br>For instance: if `StartValue` is 4.5 and `Increment` is 0.5, the array is [4.5, 5.0, 5.5, 6.0...]. |
| Hidden inputs | `StartValue`: 'T where 'T numeric<br>`Increment`: 'T where 'T numeric |
| Outputs | `Output`: 'T ^N where 'T numeric |

# Determinant of Square Matrix (2x2)

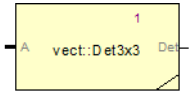
**Symbol 5.32:**




| | |
|---|---|
| SCADE name | **Det2x2** |
| Package | vect |
| Function | Computes the determinant of a square matrix of type 'T^2^2 |
| Inputs | A: 'T ^2^2 |
| Outputs | `Det`: 'T where 'T numeric |

# Determinant of Square Matrix (3x3)

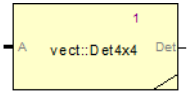
**Symbol 5.33:**




| | |
|---|---|
| SCADE name | **Det3x3** |
| Package | vect |
| Function | Computes the determinant of a square matrix of type 'T^3^3 |
| Inputs | A: 'T ^3^3 |
| Outputs | `Det`: 'T where 'T numeric |

# Determinant of Square Matrix (4x4)

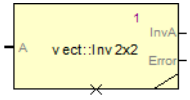
**Symbol 5.34:**




| | |
|---|---|
| SCADE name | **Det4x4** |
| Package | vect |
| Function | Computes the determinant of a square matrix of type 'T^4^4 |
| Inputs | A: 'T ^4^4 |
| Outputs | `Det`: 'T where 'T numeric |

# Inverse of Square Matrix (2x2)

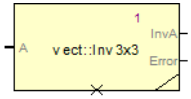
**Symbol 5.35:**




| | |
|---|---|
| SCADE name | **Inv2x2** |
| Package | vect |
| Function | Inverses a square matrix of type 'T^2^2 |
| Inputs | A: 'T^2^2 where 'T is float |
| Hidden Inputs | `Epsilon`: 'T where 'T is float |
| Outputs | `InvA`: 'T ^2^2 where 'T is float<br>`Error`: bool |
| Comment | Error output is set to true when the determinant of the matrix is lower or equal to Epsilon input. It is assumed that Epsilon is a positive value. |

# Inverse of Square Matrix (3x3)

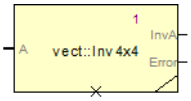
**Symbol 5.36:**




| | |
|---|---|
| SCADE name | **Inv3x3** |
| Package | vect |
| Function | Inverses a square matrix of type 'T^3^3 |
| Inputs | A: 'T^3^3 where 'T is float |
| Hidden Inputs | `Epsilon`: 'T where 'T is float |
| Outputs | `InvA`: 'T ^3^3 where 'T is float<br>`Error`: bool |
| Comment | Error output is set to true when the determinant of the matrix is lower or equal to Epsilon input. It is assumed that Epsilon is a positive value. |

# Inverse of Square Matrix (4x4)

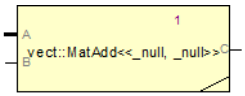
**Symbol 5.37:**




| | |
|---|---|
| SCADE name | **Inv4x4** |
| Package | vect |
| Function | Inverses a square matrix of type 'T^4^4 |
| Inputs | A: 'T^4^4 where 'T is float |
| Hidden Inputs | `Epsilon`: 'T where 'T is float |
| Outputs | `InvA`: 'T ^3^3 where 'T is float<br>`Error`: bool |
| Comment | Error output is set to true when the determinant of the matrix is lower or equal to Epsilon input. It is assumed that Epsilon is a positive value. |

# Matrix Addition

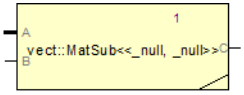
**Symbol 5.38:**




| | |
|---|---|
| SCADE name | **MatAdd** |
| Package | vect |
| Function | Computes the sum of two matrices of type 'T^n^m |
| Size parameters | m and n |
| Inputs | A: 'T ^n^m<br>B: 'T ^n^m where 'T numeric |
| Outputs | C: 'T ^n^m |

# Matrix Difference

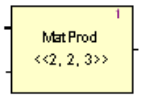
**Symbol 5.39:**




| | |
|---|---|
| SCADE name | **MatSub** |
| Package | vect |
| Function | Computes the difference of two matrices of type 'T^n^m |
| Size parameters | m and n |
| Inputs | A: 'T ^n^m<br>B: 'T ^n^m where 'T numeric |
| Outputs | C: 'T ^n^m |

# Matrix Product

**Symbol 5.40:**




| | |
|---|---|
| SCADE name | **MatProd** |
| Package | vect |
| Function | Computes the matrix product A*B, matrixes (m,n) implemented as 'T^n^m |
| Size parameters | m, n, and p |
| Inputs | A: 'T ^n^m<br>B: 'T ^p^n where 'T numeric |
| Outputs | C: 'T ^p^m |

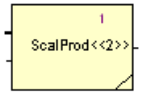
# Matrix Product by Vector

**Symbol 5.41:**

| | |
|---|---|
| SCADE name | **MatVectProd** |
| Package | vect |
| Function | Computes the product of a matrix by a vector, matrix (m,n) implemented as 'T^n^m |
| Size parameters | m and n |
| Inputs | A: 'T ^n^m<br>V: 'T ^n where 'T numeric |
| Outputs | R: 'T ^m |

# Scalar Product

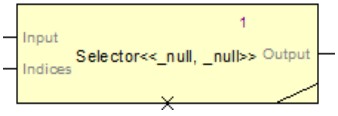
**Symbol 5.42:**




| | |
|---|---|
| SCADE name | **ScalProd** |
| Package | vect |
| Function | Computes the scalar product of vectors V and W:<br>`SP = V[0]*W[0]+...+V[N-1]*W[N-1]` |
| Size parameters | n |
| Inputs | V: 'T ^n<br>W: 'T ^n where 'T numeric |
| Outputs | SP: 'T |
| KCG pragma | Operator expansion |

# Selector

**Symbol 5.43:**




| | |
|---|---|
| SCADE name | **Selector** |
| Package | vect |
| Function | Takes an array `Input` of size `SrcSize` and outputs an array of size `DstSize` where the contents are the elements of the input array at the indices given by `Indices`. If any of the `Indices` is outside of the range, the value `Default` is set in the output array instead. |
| Inputs | `Input`: 'T ^SrcSize where 'T numeric<br>`Indices`: 'I where 'I integer |
| Hidden inputs | `Default`: 'T where 'T numeric |
| Outputs | `Output`: 'T ^DstSize where 'T numeric |

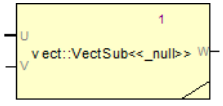
# Vector Addition

**Symbol 5.44:**

| | |
|---|---|
| SCADE name | **VectAdd** |
| Package | vect |
| Function | Computes the addition of two vectors of type 'T^n |
| Size parameters | n |
| Inputs | U: 'T ^n<br>V: 'T ^n where 'T numeric |
| Outputs | W: 'T ^n |

# Vector Difference

**Symbol 5.45:**




| | |
|---|---|
| SCADE name | **VectSub** |
| Package | vect |
| Function | Computes the difference of two vectors of type 'T^n |
| Size parameters | n |
| Inputs | U: 'T ^n<br>V: 'T ^n where 'T numeric |
| Outputs | W: 'T ^n |

# Vector Product by Matrix

**Symbol 5.46:**

| | |
|---|---|
| SCADE name | **VectMatProd** |
| Package | vect |
| Function | Computes the product of a vector by a matrix, matrix (m,n) implemented as 'T^n^m |
| Size parameters | m and n |
| Inputs | V: 'T ^m<br>A: 'T ^n^m where 'T numeric |
| Outputs | R: 'T ^n |

# 6 / Library libmathext

This library contains trigonometric functions, power-based functions, as well as some advanced functions. All C macro functions necessary for this library are defined in `macro_libmathext32.h` and `macro_libmathext64.h` files located under `SCADE/libraries/SC65/libmathext`.

---

**Note**

Depending on the target compiler, you must modify this file.

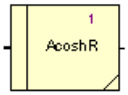
---


- "Trigonometric and Power-Based Functions"
- "Advanced Mathematical Operators"
- "Conversion Operators"

# Trigonometric and Power-Based Functions

All operators corresponding to these functions are provided for a single output.

**Symbol 6.1:**




| | | | | |
|---|---|---|---|---|
| SCADE name | See below | | | |
| Package | mathext | | | |
| Function | <MathFunction> | SCADE name | <MathFunction> | SCADE name |
| | cos (x) | **CosR** | atan2 (x) | **Atan2R** |
| | sin (x) | **SinR** | acosh (x) | **AcoshR** |
| | tan (x) | **TanR** | asinh (x) | **AsinhR** |
| | cosh (x) | **CoshR** | atanh (x) | **AtanhR** |
| | sinh (x) | **SinhR** | ln (x) | **LnR** |
| | tanh (x) | **TanhR** | $e^x$ | **ExpR** |
| | acos (x) | **AcosR** | log (x) | **LogR** |
| | asin (x) | **AsinR** | $\sqrt{x}$ | **SqrtR** |
| | atan (x) | **AtanR** | $10^x$ | **TenPowR** |
| | | **HypotR** | (sin(x), cos(x)) | **SinCosR** |
| Inputs | `Input1`: 'T<br>`Input1` and `Input2`: 'T for **Atan2R**<br>`HypotR_I1` and `HypotR_I2`: 'T for **HypotR** where 'T is float | | | |
| Outputs | `Output1`: 'T<br>`Output1` and `Output2`: 'T for **SinCosR**<br>`HypotR_O`: 'T for **HypotR** where 'T is float | | | |
| Comment | Behavior may vary depending on compiler math libraries. | | | |

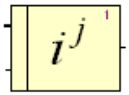
# Advanced Mathematical Operators

The **`libmath`** library contains advanced mathematical functions that allow to handle integer and/or real values.

Read the description of the following operators:

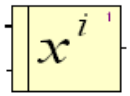
"Integer Power for Integers"
"Integer Power for Reals"
"Inverse"
"Real Power for Reals"
"Square"

## Integer Power for Integers

**Symbol 6.2:** 

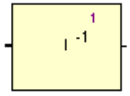
| | |
|---|---|
| SCADE name | **PowerI** |
| Package | mathext |
| Function | `P_Ouput` equals `P_Input` powered to `Power`. |
| Dependencies | C macro defined in macro_libmathext32.h and macro_libmathext64.h files under SCADE/libraries/SC65/libmathext |
| Inputs | `P_Input`: int32 and `Power`: int32 |
| Outputs | `P_Output`: int32 |

# Integer Power for Reals

**Symbol 6.3:** 

| | |
|---|---|
| SCADE name | **PowerR** |
| Package | mathext |
| Function | `P_Ouput` equals `P_Input` powered to `Power`. |
| Dependencies | C macro defined in macro_libmathext32.h and macro_libmathext64.h files under SCADE/libraries/SC65/libmathext |
| Inputs | `P_Input`: 'T and `Power`: int32 where 'T is float |
| Outputs | `P_Output`: 'T where 'T is float |

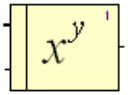
# Inverse

**Symbol 6.4:** 

| | |
|---|---|
| SCADE name | **Inv** |
| Package | mathext |
| Function | Computes the reciprocal of a real or int number to return a real output |
| Inputs | `Inv_In`: 'T where 'T is float |
| Outputs | `Inv_Out`: 'T where 'T is float |
| Caution | If Inv_In equals 0, division by zero. |

# Real Power for Reals

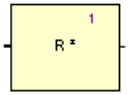
**Symbol 6.5:**



| | |
|---|---|
| SCADE name | **PowerRR** |
| Package | mathext |
| Function | `P_Ouput` equals `P_Input` powered to `Power`, `Power` being a real.<br>It is assumed that: "P_Input > 0" or "(P_Input == 0 && Power > 0)" |
| Dependencies | C macro defined in macro_libmathext32.h and macro_libmathext64.h files under SCADE/libraries/SC65/libmathext |
| Inputs | `P_Input`: 'T and `Power`: 'T where 'T is float |
| Outputs | `P_Output`: 'T where 'T is float |
| Comment | According to compiler and function. SCADE simulator checks that `P_input` is strictly positive, or if null that `Power` is strictly positive. If it is not the case (*i.e.*, P_input is negative), SCADE Simulator raises an exception. |

# Square

**Symbol 6.6:**




| | |
|---|---|
| SCADE name | **Square** |
| Package | mathext |
| Function | Computes the square value of real or integer numbers. |
| Inputs | `Square_In`: 'T where 'T numeric |
| Outputs | `Square_Out`: 'T |

# Conversion Operators

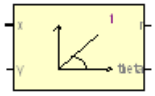
The **`libmathext`** library contains conversion functions.

Read the description of the following operators:


"Cartesian to Polar"
"Cartesian to Spherical"
"Celsius to Fahrenheit"
"Degrees to Radians"
"Fahrenheit to Celsius"
"Polar to Cartesian"
"Radians to Degrees"
"Spherical to Cartesian"


## Cartesian to Polar

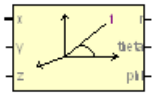
**Symbol 6.7:**




| | |
|---|---|
| SCADE name | **CartesianToPolar** |
| Package | libmathext |
| Function | Conversion from Cartesian coordinates (x,y) into polar coordinates |
| Inputs | `x`: 'T and `y`: 'T where 'T is float |
| Outputs | `r`: 'T and `theta`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# Cartesian to Spherical

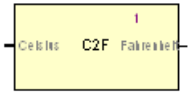
**Symbol 6.8:**




| | |
|---|---|
| SCADE name | **CartesianToSpherical** |
| Package | libmathext |
| Function | Conversion from Cartesian coordinates (x,y,z) into spherical coordinates |
| Inputs | `x`: 'T , `y`: 'T, and `z`: 'T where 'T is float |
| Outputs | `r`: 'T, `theta`: 'T, and `phi`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# Celsius to Fahrenheit

**Symbol 6.9:**




| | |
|---|---|
| SCADE name | **CelsiusToFahrenheit** |
| Package | libmathext |
| Function | Conversion from Celsius degrees to Fahrenheit degrees |
| Inputs | `Celsius`: 'T where 'T is float |
| Outputs | `Fahrenheit`: 'T where 'T is float |
| KCG pragma | Operator expansion |

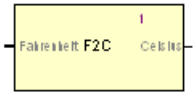
# Degrees to Radians

**Symbol 6.10:**

| | |
|---|---|
| SCADE name | **DegreesToRadians** |
| Package | libmathext |
| Function | Conversion from degrees to radians |
| Inputs | `Degrees`: 'T where 'T is float |
| Outputs | `Radians`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# Fahrenheit to Celsius

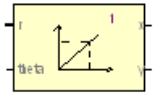
**Symbol 6.11:**




| | |
|---|---|
| SCADE name | **FahrenheitToCelsius** |
| Package | libmathext |
| Function | Conversion from Fahrenheit degrees to Celsius degrees |
| Inputs | `Fahrenheit`: 'T where 'T is float |
| Outputs | `Celsius`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# Polar to Cartesian

**Symbol 6.12:**



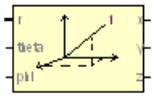

| | |
|---|---|
| SCADE name | **PolarToCartesian** |
| Package | libmathext |
| Function | Conversion from polar coordinates (r,θ) into Cartesian coordinates |
| Inputs | `r`: 'T and `theta`: 'T where 'T is float |
| Outputs | `x`: 'T and `y`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# Radians to Degrees

**Symbol 6.13:**

| | |
|---|---|
| SCADE name | **RadiansToDegrees** |
| Package | libmathext |
| Function | Conversion from radians to degrees |
| Inputs | `Radians`: 'T where 'T is float |
| Outputs | `Degrees`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# Spherical to Cartesian

**Symbol 6.14:** 

| | |
|---|---|
| SCADE name | **SphericalToCartesian** |
| Package | libmathext |
| Function | Conversion from spherical coordinates (r,θ, φ) into Cartesian coordinates |
| Inputs | `r`: 'T, `theta`: 'T, and `phi`: 'T where 'T is float |
| Outputs | `x`: 'T , `y`: 'T, and `z`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# 7 / Library libpwlinear

This library contains operators that compute piecewise linear functions and LookUp Table operations. You can read the technical description of the library components by package:

- "Piecewise Linear Functions" in **`pwlinear`** package
- "Look-Up Table Operators" in **`lut`** package

Access the description of library operators in alphabetical order:


"CheckSlope"
"Clock Counter"
"Counter"
"Dead Band (Asymmetrical)"
"Dead Band (Symmetrical)"
"Hysteresis (Falling)"
"Hysteresis (Rising)"
"Interpolation 1D"
"Interpolation 1D (Floor)"
"Interpolation 2D"
"Interpolation 2D (Floor)"
"Limiter (Asymmetrical)"
"Limiter (Symmetrical)"
"LUT 1D"
"LUT 1D (Value above)"
"LUT 1D (Value below)"
"LUT 1D (Nearest value)"
"LUT 2D"
"LUT 2D (Value above)"
"LUT 2D (Value below)"
"LUT 2D (Nearest value)"
"LUT 3D"
"LUT 3D (Nearest value)"
"MaxReset"
"MinReset"
"Pre-load (Asymmetrical)"
"Pre-load (Symmetrical)"
"Pre-LUT"
"Pre-LUT (Direct)"
"Quantizer"
"Rate Limiter"

# Piecewise Linear Functions

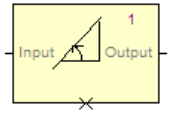
The **`libpwlinear`** library contains several basic operators that compute piecewise linear functions.

Read the description of the following operators:


"CheckSlope"
"Clock Counter"
"Counter"
"Dead Band (Asymmetrical)"
"Dead Band (Symmetrical)"
"Hysteresis (Falling)"
"Hysteresis (Rising)"
"Limiter (Asymmetrical)"
"Limiter (Symmetrical)"
"MaxReset"
"MinReset"
"Pre-load (Asymmetrical)"
"Pre-load (Symmetrical)"
"Quantizer"
"Rate Limiter"


## CheckSlope

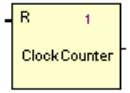
**Symbol 7.1:**




| | |
|---|---|
| SCADE name | **CheckSlope** |
| Package | pwlinear |
| Function | Returns true at first cycle if the difference between the value of `Input` and that of previous cycle is greater than `Max` (in absolute value) |
| Inputs | `Input`: 'T where 'T numeric |
| Hidden inputs | `Max`: T where 'T numeric |
| Outputs | `Output`: bool |

# Clock Counter

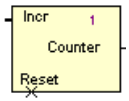
**Symbol 7.2:**




| | |
|---|---|
| SCADE name | **ClockCounter** |
| Package | pwlinear |
| Function | Basic counter. Increments the output by one at each cycle. At initialization, or if Reset equals `true`, output is set to 0. |
| Inputs | `Reset`: bool |
| Outputs | `Count`: int32 |

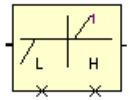
# Counter

**Symbol 7.3:**



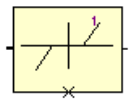

| | |
|---|---|
| SCADE name | **Counter** |
| Package | pwlinear |
| Function | The output increments at each cycle by Incr.<br>At initialization or if Reset equals `true`, the output is set to 0. |
| Inputs | `Incr`: 'T where 'T numeric |
| Hidden inputs | `Reset`: bool |
| Outputs | `Count`: 'T where 'T numeric |

# Dead Band (Asymmetrical)

**Symbol 7.4:** 

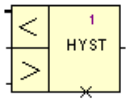

| | |
|---|---|
| SCADE name | **DeadBandUnSymmetrical** |
| Package | pwlinear |
| Function | Asymmetrical deadband. Keeps output equal to 0.0 as long as input is in [lower tolerance, upper tolerance] range. When input reaches either limit, output is equal to input minus corresponding limit. |
| Inputs | `DBUS_Input`: 'T where 'T numeric |
| Hidden inputs | `LowTol`: 'T and `HiTol`: 'T where 'T numeric |
| Outputs | `DBUS_Output`: 'T where 'T numeric |
| Comment | For a correct behavior, `HiTol` must be larger than `LowTol`. |

# Dead Band (Symmetrical)

**Symbol 7.5:** 

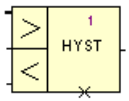
| | |
|---|---|
| SCADE name | **DeadBandSymmetrical** |
| Package | pwlinear |
| Function | Basic symmetrical deadband. Keeps output equal to 0.0 as long as input is within [-Tolerance, +Tolerance] range. Before the lower bound, output is equal to the input plus the tolerance; after the higher bound, it is equal to input minus tolerance. |
| Inputs | `DBS_Input`: 'T where 'T numeric |
| Hidden inputs | `Tolerance`: 'T where 'T numeric |
| Outputs | `DBS_Output`: 'T where 'T numeric |
| Comment | For a correct behavior, `Tolerance` must be positive. |

# Hysteresis (Falling)

**Symbol 7.6:** 

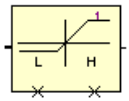

| | |
|---|---|
| SCADE name | **FallingHysteresis** |
| Package | pwlinear |
| Function | Detects with hystererisis if the input is above a threshold. Returns a Boolean value (if FH_Input < LL, then output is `true`, else if FH_Input > UL, then output is `false`, else output equals previous output). |
| Inputs | `LL` (Lower limit), `UL` (Upper limit), and `FH_Input`: 'T where 'T numeric |
| Hidden inputs | `Init`: bool |
| Outputs | `FH_Output`: bool |
| Comment | On first step, output equals `Init` if (LL <= FH_Input <= UL). For a correct behavior, `UL` must be larger than `LL`. |

# Hysteresis (Rising)

**Symbol 7.7:** 

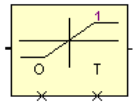

| | |
|---|---|
| SCADE name | **RisingHysteresis** |
| Package | pwlinear |
| Function | Detects with hysteresis if the input is below a threshold. Returns a Boolean value (if A > UL, then output is `true`, else if A < LL, then output is `false`, else output equals previous output). |
| Inputs | A, `UL` (Upper limit), and `LL` (Lower limit): 'T where 'T numeric |
| Hidden inputs | `Init`: bool |
| Outputs | S: bool |
| Comment | On first step, output equals `Init` if (LL <= A <= UL). For a correct behavior, `UL` must be larger than `LL`. |

# Limiter (Asymmetrical)

**Symbol 7.8:** 


| | |
|---|---|
| SCADE name | **LimiterUnSymmetrical** |
| Package | pwlinear |
| Function | Asymmetrical limiting band with specified upper and lower limits, not necessarily including 0. If the input is lower than L, the output is L; if the input is greater than H, the output is H; otherwise the output is equal to the input. |
| Inputs | `LUS_Input`: 'T where 'T numeric |
| Hidden inputs | `Lowlimit`: 'T and `HighLimit`: 'T where 'T numeric |
| Outputs | `LUS_Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |
| Comment | For a correct behavior, `HighLimit` must be larger than `LowLimit`. |

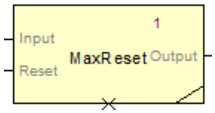
# Limiter (Symmetrical)

**Symbol 7.9:** 


| | |
|---|---|
| SCADE name | **LimiterSymmetrical** |
| Package | pwlinear |
| Function | Basic symmetrical limiting band for some arbitrary values, not necessarily 0. Let's call L the value (origin - tolerance) and U the value (origin + tolerance). If the input is lower than L, the output is L; if the input is greater than U, the output is U; if not output is equal to the input. |
| Inputs | `LS_Input`: 'T where 'T numeric |
| Hidden inputs | `BandOrigin`: 'T and `Tolerance`: 'T where 'T numeric |
| Outputs | `LS_Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |
| Comment | For a correct behavior, `Tolerance` should be positive. |

# MaxReset

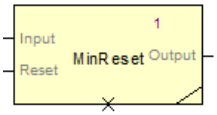
**Symbol 7.10:**




| | |
|---|---|
| SCADE name | **MaxReset** |
| Package | pwlinear |
| Function | When `Reset` is true and before `Init`, `Output` is set to `Init`. Whenever `Input` is greater than the previous `Output`, `Output` is set to `Input`. |
| Inputs | `Input`: 'T where 'T numeric<br>`Reset`: bool |
| Hidden inputs | `Init`: 'T where 'T numeric |
| Outputs | `Output`: 'T where 'T numeric |

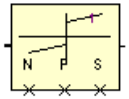
# MinReset

**Symbol 7.11:**



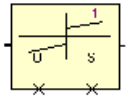

| | |
|---|---|
| SCADE name | **MinReset** |
| Package | pwlinear |
| Function | When `Reset` is true and before `Init`, `Output` is set to `Init`. Whenever `Input` is lower than the previous `Output`, `Output` is set to `Input`. |
| Inputs | `Input`: 'T where 'T numeric<br>`Reset`: bool |
| Hidden inputs | `Init`: 'T where 'T numeric |
| Outputs | `Output`: 'T where 'T numeric |

# Pre-load (Asymmetrical)

**Symbol 7.12:** 

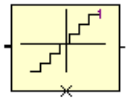

| | |
|---|---|
| SCADE name | **PreLoadUnSymmetrical** |
| Package | pwlinear |
| Function | Asymmetrical preload with specified offset when negative or positive, and slope. The output is equal to (input*Slope)+PosOffset if the input is positive or zero, otherwise equal to (input*Slope)+NegOffset if the input is negative. |
| Inputs | `PLUS_Input`: 'T where 'T numeric |
| Hidden inputs | `NegOffset`: 'T, `PosOffset`: 'T, and `Slope`: 'T |
| Outputs | `PLUS_Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |

# Pre-load (Symmetrical)

**Symbol 7.13:** 

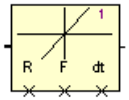

| | |
|---|---|
| SCADE name | **PreLoadSymmetrical** |
| Package | pwlinear |
| Function | Basic symmetrical preload with specified offset and slope. The output is equal to (input*slope) + offset if the input is strictly positive, otherwise the output is equal to (input * slope) - offset. |
| Inputs | `PLS_Input`: 'T where 'T numeric |
| Hidden inputs | `Offset`: 'T and `Slope`: 'T where 'T numeric |
| Outputs | `PLS_Output`: 'T where 'T numeric |

# Quantizer

**Symbol 7.14:** 

| | |
|---|---|
| SCADE name | **Quantizer** |
| Package | pwlinear |
| Function | Basic quantizer based on specified interval.<br>Discretizes the input at a specified interval and produces a stair-step output. Q_Output = Interval* Round (Q_Input/ Interval) |
| Inputs | `Q_Input`: 'T where 'T is float |
| Hidden inputs | `Interval`: 'T where 'T is float |
| Outputs | `Q_Output`: 'T where 'T is float |
| Comment | For a correct behavior, `Interval` must never be strictly positive. If `Interval` equals 0, there is a division by 0. |

# Rate Limiter

**Symbol 7.15:** 


| | |
|---|---|
| SCADE name | **RateLimiter** |
| Package | pwlinear |
| Function | Limits the first derivative of the input signal. The output does not change faster than the specified limit. Let's call Rate the derivative ((RL_Input-pre(RL_Output))/DeltaT. If Rate > Rising, then output equals (Rising* DeltaT + pre(RL_Output)), else if Rate < Falling, then output equals (Falling* DeltaT + pre(RL_Output)), otherwise, output equals input. |
| Inputs | `RL_Input`: 'T where 'T is float |
| Hidden inputs | `Rising`: 'T (Rising slew rate), `Falling`: 'T (Falling slew rate), and `deltaT`: 'T where 'T is float |
| Outputs | `RL_Output`: 'T where 'T is float |
| Comment | For a correct behavior, `deltaT` must be strictly positive. If it equals 0, there is a division by 0. |

# Look-Up Table Operators

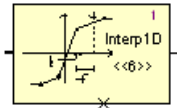
The **`libpwlinear`** library contains several basic operators that compute Look-Up Table (LUT) operations.

Read the description of the following operators:


"Interpolation 1D"
"Interpolation 1D (Floor)"
"Interpolation 2D"
"Interpolation 2D (Floor)"
"LUT 1D"
"LUT 1D (Value above)"
"LUT 1D (Value below)"
"LUT 1D (Nearest value)"
"LUT 2D"
"LUT 2D (Value above)"
"LUT 2D (Value below)"
"LUT 2D (Nearest value)"
"LUT 3D"
"LUT 3D (Nearest value)"
"Pre-LUT"
"Pre-LUT (Direct)"


## Interpolation 1D

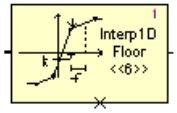
**Symbol 7.16:**




| | |
|---|---|
| SCADE name | **Interp1D** |
| Package | lut |
| Function | Uses the pre-calculated index and interval fraction from the PreLut function to interpolate output value from <Xs> points set as vector Y. |
| Size parameters | Xs |
| Inputs | `IdxX`: LutIndex = {k:int32, f:'T} where 'T is float |
| Hidden inputs | Y: 'T ^Xs where 'T is float |
| Outputs | `OutY`: 'T where 'T is float |

# Interpolation 1D (Floor)

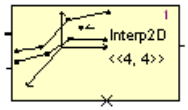
**Symbol 7.17:**




| | |
|---|---|
| SCADE name | **Interp1DFloor** |
| Package | lut |
| Function | Uses the calculated index and interval fraction from the PreLut function to return corresponding output value to nearest and below y element from vector Y of size <Xs>. If no value is found then the nearest is returned. |
| Size parameters | Xs |
| Inputs | `IdxX`: LutIndex = {k:int32, f:'T} where 'T is float |
| Hidden inputs | Y: 'T ^Xs where 'T is float |
| Outputs | `OutY`: 'T where 'T is float |

# Interpolation 2D

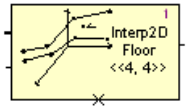
**Symbol 7.18:**




| | |
|---|---|
| SCADE name | **Interp2D** |
| Package | lut |
| Function | Uses two precalculated indices and interval fractions from the PreLut function to interpolate output value from <Xs>*<Ys> points set as array Z. |
| Size parameters | Xs and Ys |
| Inputs | `IdxX`: LutIndex = {k:int32, f:'T} and `IdxY`: LutIndex = {k:int32, f:'T} |
| Hidden inputs | Z: 'T ^Ys^Xs where 'T is float |
| Outputs | `OutZ`: 'T where 'T is float |

# Interpolation 2D (Floor)

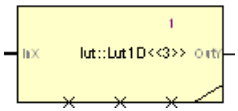
**Symbol 7.19:**




| | |
|---|---|
| SCADE name | **Interp2DFloor** |
| Package | lut |
| Function | Uses two precalculated indices and interval fractions from the PreLut function to interpolate output value from <Xs>*<Ys> points set as array Z. |
| Size parameters | Xs and Ys |
| Inputs | `IdxX`: LutIndex = {k:int32, f:'T} and `IdxY`: LutIndex = {k:int32, f:'T} |
| Hidden inputs | Z: 'T ^Ys^Xs where 'T is float |
| Outputs | `OutZ`: 'T where 'T is float |

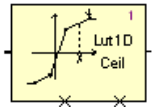
# LUT 1D

**Symbol 7.20:**




| | |
|---|---|
| SCADE name | **Lut1D** |
| Package | lut |
| Function | Performs piecewise linear interpolation to approximate y = f(x) with <Xs> points set as vectors X and Y. X values must increase strictly monotonically. For out-of-bounds input, `Extrapol` parameter allows linear extrapolation, otherwise end-point values are output. |
| Size parameters | Xs |
| Inputs | `InX`: 'T where 'T is float |
| Hidden inputs | X: 'T ^Xs, Y: 'T ^Xs, and `Extrapol`: bool where 'T is float |
| Outputs | `OutY`: 'T where 'T is float |
| Comment | An "assume" assertion in all LutXXX operators checks that the X vector in 1D case, and X and Y vectors in 2D case increase strictly monotonically. |
| KCG pragma | Operator expansion |

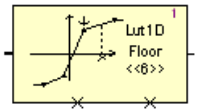
# LUT 1D (Value above)

**Symbol 7.21:** 


| | |
|---|---|
| SCADE name | **Lut1DCeil** |
| Package | lut |
| Function | Returns the output value corresponding to nearest and above x element from <Xs> points set as vectors X and Y. If no value is found then the nearest is returned. X values must increase strictly monotonically. |
| Size parameters | Xs |
| Inputs | `InX`: 'T where 'T is float |
| Hidden inputs | X: 'T ^Xs and Y: 'T ^Xs where 'T is float |
| Outputs | `OutY`: 'T where 'T is float |
| Comment | An "assume" assertion in all LutXXX operators checks that the X vector in 1D case, and X and Y vectors in 2D case increase strictly monotonically. |

# LUT 1D (Value below)

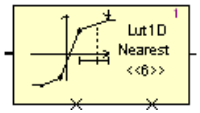
**Symbol 7.22:**




| | |
|---|---|
| SCADE name | **Lut1DFloor** |
| Package | lut |
| Function | Returns the output value corresponding to nearest and below x element from <Xs> points set as vectors X and Y. If no value is found then the nearest is returned. X values must increase strictly monotonically. |
| Size parameters | Xs |
| Inputs | `InX`: 'T where 'T is float |
| Hidden inputs | X: 'T ^Xs and Y: 'T ^Xs where 'T is float |
| Outputs | `OutY`: 'T where 'T is float |
| Comment | An "assume" assertion in all LutXXX operators checks that the X vector in 1D case, and X and Y vectors in 2D case increase strictly monotonically. |

# LUT 1D (Nearest value)

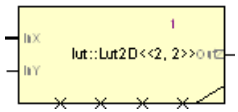
**Symbol 7.23:**




| | |
|---|---|
| SCADE name | **Lut1DNearest** |
| Package | lut |
| Function | Returns the output value corresponding to nearest x element from <Xs> points set as vectors X and Y. X values must increase strictly monotonically. |
| Size parameters | Xs |
| Inputs | `InX`: 'T where 'T is float |
| Hidden inputs | X: 'T ^Xs and Y: 'T ^Xs where 'T is float |
| Outputs | `OutY`: 'T where 'T is float |
| Comment | An "assume" assertion in all LutXXX operators checks that the X vector in 1D case, and X and Y vectors in 2D case increase strictly monotonically. |

# LUT 2D

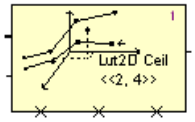
**Symbol 7.24:**




| | |
|---|---|
| SCADE name | **Lut2D** |
| Package | lut |
| Function | Performs piecewise linear interpolation to approximate z = f(x,y) with <Xs>, <Ys> points set as vectors X, Y, and <Xs>*<Ys> points set as array Z. X and Y values must increase strictly monotonically. For out-of-bounds inputs, Extrapol parameter allows linear extrapolation, otherwise the end-point values are output. |
| Size parameters | Xs and Ys |
| Inputs | `InX`: 'T and `InY`: 'T where 'T is float |
| Hidden inputs | X: 'T ^Xs, Y: 'T ^Ys, Z: 'T ^Ys^Xs, `Extrapol`: bool where 'T is float |
| Outputs | `OutZ`: 'T where 'T is float |
| Comment | An "assume" assertion in all LutXXX operators checks that the X vector in 1D case, and X and Y vectors in 2D case increase strictly monotonically. |
| KCG pragma | Operator expansion |

# LUT 2D (Value above)

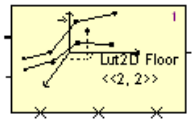
**Symbol 7.25:**




| | |
|---|---|
| SCADE name | **Lut2DCeil** |
| Package | lut |
| Function | Returns output value corresponding to nearest and above x and y elements from <Xs>, <Ys> points set as vectors X, Y, and <Xs>*<Ys> points set as array Z. If no value is found then the nearest is returned. X and Y values must increase strictly monotonically. |
| Size parameters | Xs and Ys |
| Inputs | `InX`: 'T and `InY`: 'T where 'T is float |
| Hidden inputs | X: 'T ^Xs, Y: 'T ^Ys, and Z: 'T ^Ys^Xs where 'T is float |
| Outputs | `OutZ`: 'T where 'T is float |
| Comment | An "assume" assertion in all LutXXX operators checks that the X vector in 1D case, and X and Y vectors in 2D case increase strictly monotonically. |

# LUT 2D (Value below)

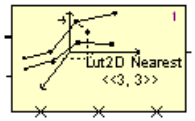
**Symbol 7.26:**




| | |
|---|---|
| SCADE name | **Lut2DFloor** |
| Package | lut |
| Function | Returns output value corresponding to nearest and below x and y elements from <Xs>, <Ys> points set as vectors X, Y, and <Xs>*<Ys> points set as array Z. If no value is found then the nearest is returned. X and Y values must increase strictly monotonically. |
| Size parameters | Xs and Ys |
| Inputs | `InX`: 'T and `InY`: 'T where 'T is float |
| Hidden inputs | X: 'T ^Xs, Y: 'T ^Ys, and Z: 'T ^Ys^Xs where 'T is float |
| Outputs | `OutZ`: 'T where 'T is float |
| Comment | An "assume" assertion in all LutXXX operators checks that the X vector in 1D case, and X and Y vectors in 2D case increase strictly monotonically. |

# LUT 2D (Nearest value)

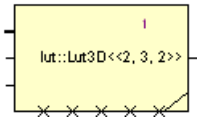
**Symbol 7.27:**




| | |
|---|---|
| SCADE name | **Lut2DNearest** |
| Package | lut |
| Function | Returns output value corresponding to nearest x and y elements from <Xs>, <Ys> points set as vectors X, Y, and <Xs>*<Ys> points set as array Z. X and Y values must increase strictly monotonically. |
| Size parameters | Xs and Ys |
| Inputs | `InX`: 'T and `InY`: 'T where 'T is float |
| Hidden inputs | X: 'T ^Xs, Y: 'T ^Ys, and Z: 'T ^Ys^Xs where 'T is float |
| Outputs | `OutZ`: 'T where 'T is float |
| Comment | An "assume" assertion in all LutXXX operators checks that the X vector in 1D case, and X and Y vectors in 2D case increase strictly monotonically. |

# LUT 3D

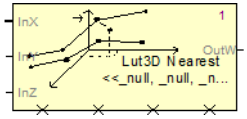
**Symbol 7.28:**




| | |
|---|---|
| SCADE name | **Lut3D** |
| Package | lut |
| Function | Performs piecewise linear interpolation to approximate w = f(x,y,z) with <Xs>, <Ys>, <Zs> points set as vectors X, Y, Z, and <Xs>*<Ys>*<Zs> points set as array W. X, Y, and Z values must increase strictly monotonically. For out-of-bounds inputs, Extrapol parameter allows linear extrapolation, otherwise the end-point values are output. |
| Size parameters | Xs, Ys, and Zs |
| Inputs | `InX`: 'T, `InY`: 'T, and `InZ`: 'T where 'T is float |
| Hidden inputs | `X`: 'T ^Xs, `Y`: 'T ^Ys, `Z`: 'T ^Zs, `W`: 'T ^Zs^Ys^Xs, and `Extrapol`: bool where 'T is float |
| Outputs | `OutW`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# LUT 3D (Nearest value)

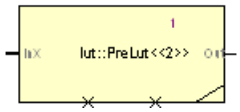
**Symbol 7.29:**




| | |
|---|---|
| SCADE name | **Lut3DNearest** |
| Package | lut |
| Function | Returns output value corresponding to nearest x, y, z elements from <Xs>, <Ys>, and <Zs> points set as vectors X, Y, Z, and <Xs>*<Ys>*<Zs> points set as array W. X, Y, and Z values must increase strictly monotonically. |
| Size parameters | Xs, Ys, and Zs |
| Inputs | `InX`: 'T, `InY`: 'T, and `InZ`: 'T where 'T is float |
| Hidden inputs | `X`: 'T ^Xs, `Y`: 'T ^Ys, `Z`: 'T ^Zs, and `W`: 'T ^Zs^Ys^Xs where 'T is float |
| Outputs | `OutW`: 'T where 'T is float |
| Comment | An "assume" assertion in all LutXXX operators checks that the X vector in 1D case, and X and Y vectors in 2D case, and X, Y, and Z vectors in 3D case increase strictly monotonically. |

# Pre-LUT

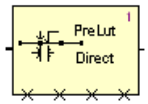
**Symbol 7.30:**




| | |
|---|---|
| SCADE name | **PreLut** |
| Package | lut |
| Function | Performs index search and interval fraction calculation from vector X for input InX.<br>X values must increase strictly monotonically. For out-of-bounds input, the computation of interval fraction corresponds to linear extrapolation or to end-point values depending on parameter Extrapol. |
| Size parameters | Xs |
| Inputs | `InX`: 'T where 'T is float |
| Hidden inputs | `X`: 'T ^Xs and `Extrapol`: bool where 'T is float |
| Outputs | `Out`: `LutIndex` = {k:'T, f:int32} where 'T is float |

# Pre-LUT (Direct)

**Symbol 7.31:**




| | |
|---|---|
| SCADE name | **PreLutDirect** |
| Package | lut |
| Function | Performs index search and interval fraction calculation for input InX when X points are evenly spaced. X points are defined by the first point (Orig), interval between 2 points (step), number of elements (NbElem). For out-of-bounds input, computation of interval fraction corresponds to linear extrapolation or to end-point values depending on parameter Extrapol. |
| Inputs | `InX`: 'T where 'T is float |
| Hidden inputs | `Orig`: 'T , `Step`: 'T , `NbElem`: int32, and `Extrapol`: bool where 'T is float |
| Outputs | `OutIndex`: `LutIndex` = {k:int32, f:'T} where 'T is float |

# 8 / Library libsmlk

This library contains operators designed to reproduce the behavior of Simulink blocks with respect to SCADE formalism when translating Simulink models.


- “Discrete Filter Operators”
- “Discrete-Time Integrator Operators”
- “Arithmetic Operators”
- “Time Operators”
- “Bitwise Logical Operators”
- “Conversion and Transformation Operators”
- “Flip-Flop Operators”
- “Logical Operators”
- “Model Verification Operators”
- “Signal Routing Operators”
- “Subsystem Operators”
- “Signal Generator Operators”
- “Discontinuity Operators”
- “LUT, Pre-LUT and Interpolation Operators”
- “Miscellaneous Operators”

Access the description of library operators in alphabetical order:


"AND"
"Assertion"
"Assertion (Inverse)"
"Backlash"
"Bias"
"Bit Clear"
"Bit Set"
"Bitwise AND"
"Bitwise Exclusive OR"
"Bitwise Nand"
"Bitwise Nor"
"Bitwise NOT"
"Bitwise OR"
"Bitwise Shift"
"Bitwise Shift Left"
"Bitwise Shift Right"
"Boolean to Boolean"
"Border Crossing Detection"
"Cartesian to Polar"
"Cartesian to Spherical"
"Celsius to Fahrenheit"
"Check Bounds"
"Check Bounds (Extended)"
"Check Gap"
"Check Gap (Extended)"
"Check Gradient"
"Check Gradient (Extended)"
"Check Lower Bound"
"Check Lower Bound (Extended)"
"Check Upper Bound"
"Check Upper Bound (Extended)"
"Clock"
"Clock Generator"
"Compare (all operations)"
"Compare Enumeration"
"Counter (limited)"
"Dead Zone Dynamic"
"Decrement Time to Zero (int/int32)"
"Decrement Time to Zero (real/float)"
"Decrement to Zero"
"Degrees to Radians"
"Detect Change"
"Detect Decrease"
"Detect Falling Edge (negative)"
"Detect Falling Edge (non-positive)"
"Detect Increase"
"Detect Rising Edge (non-negative)"
"Detect Rising Edge (positive)"
"Discrete Filter"
"Discrete Filter (Normalized)"
"Discrete-Time Integrator (Backward)"
"Discrete-Time Integrator (Forward)"
"Discrete-Time Integrator (Trapezoidal)"
"DLatch"
"Entry Detection"
"Extract Bit Range"
"Enumeration to Integer"
"Fahrenheit to Celsius"
"Flip-Flop D"
"Flip-Flop JK"
"Flip-Flop SR"
"Ground (Boolean)"
"Ground (int/int32)"
"Ground (real/float)"
"Identity"
"If Then Else"
"Ignore First Input"
"Inactive Cycles"
"Inactive Time"
"Integer to Boolean"
"Integer to Enumeration"
"In Range (In-In)"
"In Range (In-Out)"
"In Range (Out-In)"
"In Range (Out-Out)"
"Interpolation1D2"
"Interpolation1D (Floor)"
"Interpolation2D2"
"Interpolation2D (Floor)"
"Inverse (int/int32)"
"LUT 1D"
"Maximum (Resettable)"
"Merge (2 inputs)"
"Merge (3 inputs)"
"Minimum (Resettable)"
"Modulo (int/int32)"
"Modulo (real/float)"
"NOT"
"OR"
"Polar to Cartesian"
"Polynomial"
"Pre-LUT2"
"Pre-LUT2 (Direct)"
"Quantizer"
"Radians to Degrees"
"Rate Limiter"
"Rate Limiter Dynamic"
"Real to Boolean"
"Relay"
"Remainder (real/float)"
"Sample Time (all math operations)"
"Saturate Dynamic"
"Scalar to Boolean"
"Sign (int/int32)"
"Spherical to Cartesian"
"Switch"
"Switch (GTT)"
"Switch (NEZ)"
"Switch Enumeration"
"Switch Enumeration (GTT)"
"Transfer Function (First Order)"
"Transfer Function (Lead or Lag)"
"Transfer Function (Real Zero)"
"Trigger Either Edge"
"Trigger Either Edge (Extended)"
"Trigger Falling Edge"
"Trigger Falling Edge (Extended)"
"Trigger Rising Edge"
"Trigger Rising Edge (Extended)"
"Unit Delay (Helper)"
"Unit Delay Enabled"
"Unit Delay Enabled (Init)"
"Unit Delay Enabled Resettable"
"Unit Delay Enabled Resettable (Init)"
"Unit Delay (Init)"
"Unit Delay Resettable"
"Unit Delay Resettable (Init)"
"Width (Matrix)"
"Wrap To Zero"

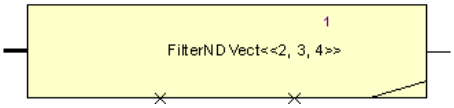
# Discrete Filter Operators

Read the description of the following operators:

"Discrete Filter" "Discrete Filter (Normalized)"

## Discrete Filter

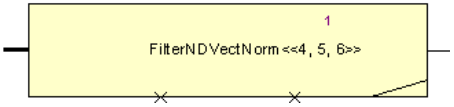
**Symbol 8.1:**




| | |
|---|---|
| SCADE names | **FilterNDVect** |
| Package | smlk |
| Function | Vectorized version of filters::FilterND (see "Filter and Transfer Function Operators"). |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T ^Ns^Vect and `Den:` 'T ^Ds where 'T is float |
| Outputs | `Out:` 'T ^Vect where 'T is float |

## Discrete Filter (Normalized)

**Symbol 8.2:**




| | |
|---|---|
| SCADE names | **FilterNDVectNorm** |
| Package | smlk |
| Function | Vectorized version of filters::FilterNDNorm (see "Filter and Transfer Function Operators"). |
| Inputs | `In`: 'T where 'T is float |
| Hidden inputs | `Num`: 'T^Ns^Vect and `Den:` 'T^Ds where 'T is float |
| Outputs | `Out:` 'T^Vect where 'T is float |

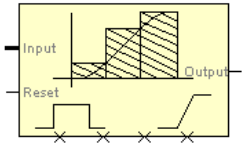
# Discrete-Time Integrator Operators

Read the description of the following operators:


"Discrete-Time Integrator (Backward)"
"Discrete-Time Integrator (Forward)"
"Discrete-Time Integrator (Trapezoidal)"
"Transfer Function (First Order)"
"Transfer Function (Lead or Lag)"
"Transfer Function (Real Zero)"


## Discrete-Time Integrator (Backward)

**Symbol 8.3:**

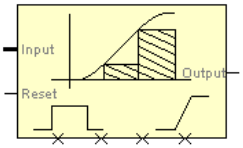



(DtiBwdLimRst)

| | |
|---|---|
| SCADE names | Several backward types are available in the library: **DtiBwdLimRst**; **DtiBwdLimRstE**; **DtiBwdLimRstF**; **DtiBwdLimRstL**; **DtiBwdLimRstN**; **DtiBwdLimRstR**; **DtiBwdLimSRst**; **DtiBwdLimSRstE**; **DtiBwdLimSRstF**; **DtiBwdLimSRstL**; **DtiBwdLimSRstN**; **DtiBwdLimSRstR**; **DtiBwdRst**; **DtiBwdRstE**; **DtiBwdRstF**; **DtiBwdRstL**; **DtiBwdRstN**; **DtiBwdRstR**<br>Naming convention:<br>**Bwd** indicates Backward Euler Integrator method<br>**Lim** indicates operators with limits on outputs<br>**S** indicates saturared operators **RstF** indicates falling external reset<br>**RstR** for rising external reset, **RstE** for either external reset, **RstN** for none, **RstL** for external reset level with type inferred for second reset input as int32 or 'T where 'T is float, and **Rst** for external reset level with type inferred for second reset input as bool. |
| Package | dti |
| Function | Signal discrete-time integration or accumulation based on Backward Euler method. |
| Inputs | `Input`: 'T where 'T is float<br>`Reset`: 'T or bool (except **DtiBwdRstN**, **DtiBwdLimRstN**, **DtiBwdLimSRstN**) |
| Hidden inputs | `init`: 'T and `deltaT`: 'T<br>`HighLimit`: 'T and `LowLimit`: 'T where 'T is float |
| Outputs | `Output`: 'T where 'T is float<br>`Saturation`: int32 (for saturated operators) |
| KCG pragma | Operator expansion of **DtiBwdRstN** only |

# Discrete-Time Integrator (Forward)

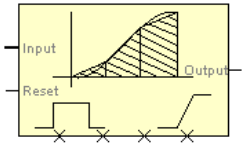
**Symbol 8.4:**




(DtiFwdLimRst)

| | |
|---|---|
| SCADE name | Several forward types are available in the library:<br>**DtiFwdLimRst**; **DtiFwdLimRstE**; **DtiFwdLimRstF**; **DtiFwdLimRstL**; **DtiFwdLimRstN**; **DtiFwdLimRstR**; **DtiFwdLimSRst**; **DtiFwdLimSRstE**; **DtiFwdLimSRstF**; **DtiFwdLimSRstL**; **DtiFwdLimSRstN**; **DtiFwdLimSRstR**; **DtiFwdRst**; **DtiFwdRstE**; **DtiFwdRstF**; **DtiFwdRstL**; **DtiFwdRstN**; **DtiFwdRstR**<br>Naming convention:<br>**Fwd** indicates Forward Euler Integrator method<br>**Lim** indicates operators with limits on outputs<br>**S** indicates saturared operators<br>**RstF** indicates falling external reset, **RstR** for rising external reset, **RstE** for either external reset, **RstN** for none, **RstL** for external reset level with type inferred for second reset input as int32 or 'T, and **Rst** for external reset level with type inferred for second reset input as bool. |
| Package | dti |
| Function | Signal discrete-time integration or accumulation based on Forward Euler method. |
| Dependencies | n/a |
| Inputs | `Input`: 'T where 'T is float<br>`Reset`: 'T or bool (except **DtiFwdRstN**, **DtiFwdLimRstN**, **DtiFwdLimSRstN**) |
| Hidden inputs | `init`: 'T and `deltaT`: 'T<br>`HighLimit`: 'T and `LowLimit`: 'T where 'T is float |
| Outputs | `Output`: 'T where 'T is float<br>`Saturation`: int32 (for saturated operators) |

# Discrete-Time Integrator (Trapezoidal)

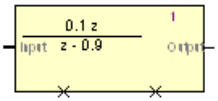
**Symbol 8.5:** 

(DtiTpzLimRst)

| | |
|---|---|
| SCADE name | Several trapezoidal types are available in the library:<br>**DtiTpzLimRst**; **DtiTpzLimRstE**; **DtiTpzLimRstF**; **DtiTpzLimRstL**; **DtiTpzLimRstN**; **DtiTpzLimRstR**; **DtiTpzLimSRst**; **DtiTpzLimSRstE**; **DtiTpzLimSRstF; DtiTpzLimSRstL; DtiTpzLimSRstN; DtiTpzLimSRstR; DtiTpzRst**; **DtiTpzRstE**; **DtiTpzRstF**; **DtiTpzRstL**; **DtiTpzRstN**; **DtiTpzRstR**<br>Naming convention:<br>**Tpz** indicates Trapezoidal Integrator method<br>**Lim** indicates operators with limits on outputs<br>**S** indicates saturared operators<br>**RstF** indicates falling external reset, **RstR** for rising external reset, **RstE** for either external reset, **RstN** for none, **RstL** for external reset level with type inferred for second reset input as int32 or 'T, and **Rst** for external reset level with type inferred for second reset input as bool. |
| Package | dti |
| Function | Signal discrete-time integration or accumulation based on Trapezoidal method. |
| Inputs | `Input`: 'T where 'T is float<br>`Reset`: 'T or bool (except **DtiTpzRstN**, **DtiTpzLimRstN**, **DtiTpzLimSRstN**) |
| Hidden inputs | `init`: 'T and `deltaT`: 'T<br>`HighLimit`: 'T and `LowLimit`: 'T where 'T is float |
| Outputs | `Output`: 'T where 'T is float<br>`Saturation`: int32 (for saturated operators) |

# Transfer Function (First Order)

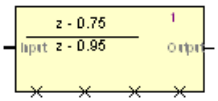
**Symbol 8.6:**




| | |
|---|---|
| SCADE name | **TransferFcnFirstOrder** |
| Package | smlk |
| Function | Discrete-time first order transfer function. |
| Inputs | `Input`: 'T<br>`Pole`: 'T where 'T is float |
| Hidden inputs | `IC`: 'T where 'T is float |
| Outputs | `Output`: 'T where 'T is float |

# Transfer Function (Lead or Lag)

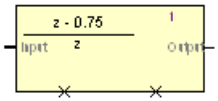
**Symbol 8.7:**




| | |
|---|---|
| SCADE name | **TransferFcnLeadOrLag** |
| Package | smlk |
| Function | Discrete-time lead or lag compensator. |
| Inputs | `Input`: 'T where 'T is float |
| Hidden inputs | `Pole`: 'T<br>`Zero`: 'T<br>`IC_Input`: 'T<br>`IC_Output`: 'T where 'T is float |
| Outputs | `Output`: 'T where 'T is float |

# Transfer Function (Real Zero)

**Symbol 8.8:**




| | |
|---|---|
| SCADE name | **TransferFcnRealZero** |
| Package | smlk |
| Function | Discrete-time transfer function that has real zero and no pole. |
| Inputs | `Input`: 'T where 'T is float |
| Hidden inputs | `Zero`: 'T<br>`IC`: 'T where 'T is float |
| Outputs | `Output`: 'T where 'T is float |

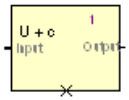
# Arithmetic Operators

Read the description of the following operators:


“Bias”
“Inverse (int/int32)”
“Maximum (Resettable)”
“Minimum (Resettable)”
“Modulo (int/int32)”
“Modulo (real/float)”
“Polynomial”
“Remainder (real/float)”
“Sign (int/int32)”


## Bias

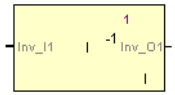
**Symbol 8.9:**




| | |
|---|---|
| SCADE name | **Bias** |
| Package | smlk |
| Function | Adds bias to the input. |
| Inputs | `Input`: 'T |
| Hidden inputs | `Bias`: 'T |
| Outputs | `Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |

# Inverse (int/int32)

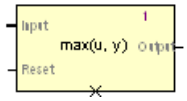
**Symbol 8.10:**




| | |
|---|---|
| SCADE name | **InvI** |
| Package | smlk |
| Function | Returns the inverse of an integer input. |
| Inputs | `Inv_I1`: int32 |
| Outputs | `Inv_O1`: int32 |
| KCG pragma | Operator expansion |

# Maximum (Resettable)

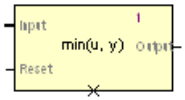
**Symbol 8.11:**




| | |
|---|---|
| SCADE name | **MaxReset** |
| Package | smlk |
| Function | Outputs the maximum of all past inputs. State can be reset. |
| Inputs | `Input`: 'T<br>`Reset`: 'T |
| Hidden inputs | `IC`: 'T2 |
| Outputs | `Output`: 'T where 'T numeric |

# Minimum (Resettable)

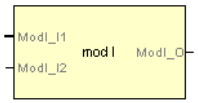
**Symbol 8.12:**




| | |
|---|---|
| SCADE name | **MinReset** |
| Package | smlk |
| Function | Outputs the minimum of all past inputs. State can be reset. |
| Inputs | `Input`: 'T<br>`Reset`: 'T |
| Hidden inputs | `IC`: 'T2 |
| Outputs | `Output`: 'T where 'T numeric |

# Modulo (int/int32)

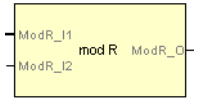
**Symbol 8.13:**




| | |
|---|---|
| SCADE name | **ModI** |
| Package | smlk |
| Function | Computes the modulus of two integer inputs. |
| Inputs | `ModI_I1`: int32 and `ModI_I2`: int32 |
| Outputs | `ModI_O`: int32 |

# Modulo (real/float)

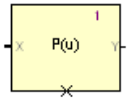
**Symbol 8.14:**




| | |
|---|---|
| SCADE name | **ModR** |
| Package | smlk |
| Function | Computes the modulus of two real/float inputs. |
| Inputs | `ModR_I1`: 'T and `ModR_I2`: 'T where 'T is float |
| Outputs | `ModR_O`: 'T where 'T is float |

# Polynomial

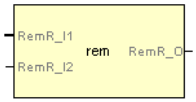
**Symbol 8.15:**




| | |
|---|---|
| SCADE name | **Polynomial <<N>>** |
| Package | smlk |
| Function | Performs the evaluation of polynomial coefficients on input values. |
| Inputs | X: 'T where 'T is numeric |
| Hidden inputs | C: 'T ^N where 'T is numeric |
| Outputs | Y: 'T where 'T is numeric |
| KCG pragma | Operator expansion |

# Remainder (real/float)

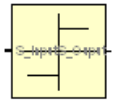
**Symbol 8.16:**




| | |
|---|---|
| SCADE name | **RemR** |
| Package | smlk |
| Function | Computes the remainder of a division. |
| Inputs | `RemR_O`: 'T and `RemR_I2`: 'T where 'T is float |
| Outputs | `RemR_O`: 'T where 'T is float |

# Sign (int/int32)

**Symbol 8.17:**



| | |
|---|---|
| SCADE name | **SignInt** |
| Package | smlk |
| Function | Returns an integer sign value based on the value of a numeric input. |
| Inputs | `S_Input`: 'T where 'T numeric |
| Outputs | `S_Output`: int32 |

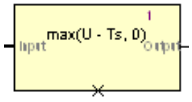
# Time Operators

Read the description of the following operators:


"Decrement Time to Zero (int/int32)"
"Decrement Time to Zero (real/float)"
"Decrement to Zero"
"Sample Time (all math operations)"
"Unit Delay Enabled"
"Unit Delay Enabled (Init)"
"Unit Delay Enabled Resettable"
"Unit Delay Enabled Resettable (Init)"
"Unit Delay (Init)"
"Unit Delay Resettable"
"Unit Delay Resettable (Init)"


## Decrement Time to Zero (int/int32)

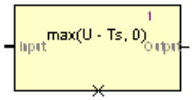
**Symbol 8.18:**




| | |
|---|---|
| SCADE name | **DecrementTimeToZeroI** |
| Package | smlk |
| Function | Decreases real-world value of signal by sample time, but only to zero. |
| Inputs | `Input`: 'T where 'T is integer |
| Hidden inputs | `Ts`: float64 |
| Outputs | `Output`: int32 |
| KCG pragma | Operator expansion |

# Decrement Time to Zero (real/float)

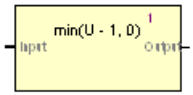
**Symbol 8.19:**




| | |
|---|---|
| SCADE name | **DecrementTimeToZeroR** |
| Package | smlk |
| Function | Decreases real-world value of signal by sample time, but only to zero. |
| Inputs | `Input`: 'T where 'T is float |
| Hidden inputs | `Ts`: 'T where 'T is float |
| Outputs | `Output`: 'T where 'T is float |
| KCG pragma | Operator expansion |

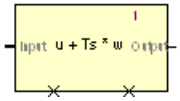
# Decrement to Zero

**Symbol 8.20:**




| | |
|---|---|
| SCADE name | **DecrementToZero** |
| Package | smlk |
| Function | Decreases real-world value of signal by sample time, but only to zero. |
| Inputs | `Input`: 'T |
| Outputs | `Output`: 'T where 'T numeric |

# Sample Time (all math operations)

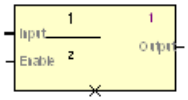
**Symbol 8.21:** 

SampleTimeAdd

| | |
|---|---|
| SCADE name | Several operations on sample time are available in the library:<br>• Addition: **SampleTimeAdd**<br>• Division: **SampleTimeDiv**<br>• Inverse: **SampleTimeInv**<br>• Multiplication: **SampleTimeMult**<br>• Subtraction: **SampleTimeSub** |
| Package | smlk |
| Function | Computes operations involving sample time. |
| Inputs | For **SampleTimeAdd**, **SampleTimeMult**, **SampleTimeSub**:<br>`Input`: 'T where 'T is numeric<br>For **SampleTimeInv**<br>`Input`: 'T where 'T is float<br>For **SampleTimeDiv**:<br>`Input`: 'T where 'T is float |
| Hidden inputs | For **SampleTimeAdd**, **SampleTimeMult**, **SampleTimeSub**:<br>`W`: 'T<br>`dT`: 'T<br>For **SampleTimeDiv**:<br>`W`: 'T<br>`dT`: 'T<br>For **SampleTimeInv**:<br>`W`: 'U<br>`dT`: 'U where 'U is float |
| Outputs | `Output`: 'T where 'T numeric<br>`Output`: 'T (for **SampleTimeDiv**) or 'U (for **SampleTimeInv**) where 'T and 'U are float |
| KCG pragma | Operator expansion |

# Unit Delay Enabled

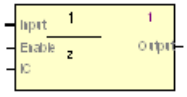
**Symbol 8.22:**




| | |
|---|---|
| SCADE name | **UDEnabled** |
| Package | smlk |
| Function | Delays input signal by one tick when Enable input is on, otherwise holds the current state and outputs that value. |
| Inputs | `Input`: 'T<br>`Enable`: 'T1 |
| Hidden inputs | `IC`: 'T |
| Outputs | `Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |
| Comment | Needed for graphical layout purposes of initial condition (IC) input. |

# Unit Delay Enabled (Init)

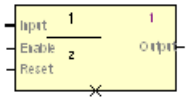
**Symbol 8.23:**




| | |
|---|---|
| SCADE name | **UDEnabledInit** |
| Package | smlk |
| Function | Delays input signal by one tick when Enable input is on, otherwise holds the current state and outputs that value. |
| Inputs | `Input`: 'T, `Enable`: 'T1, and `IC`: 'T |
| Outputs | `Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |
| Comment | Initial condition (IC) input is not an hidden input. |

# Unit Delay Enabled Resettable

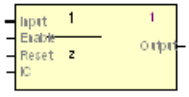
**Symbol 8.24:**




| | |
|---|---|
| SCADE name | **UDEnabledReset** |
| Package | smlk |
| Function | Delays input signal by one tick when Enable input is on, otherwise holds the current state and outputs that value. State can be reset based on reset input. |
| Inputs | `Input`: 'T, `Enable`: 'T1, and `Reset`: 'T2 |
| Hidden inputs | `IC`: 'T |
| Outputs | `Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |
| Comment | Needed for graphical layout purposes of initial condition (IC) input. |

# Unit Delay Enabled Resettable (Init)

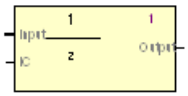
**Symbol 8.25:**




| | |
|---|---|
| SCADE name | **UDEnabledResetInit** |
| Package | smlk |
| Function | Delays input signal by one tick when Enable input is on, otherwise holds the current state and outputs that value. State can be reset based on reset input. |
| Inputs | `Input`: 'T, `Enable`: 'T1, `Reset`: 'T2, and `IC`: 'T |
| Outputs | `Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |
| Comment | Initial condition (IC) input is not an hidden input. |

# Unit Delay (Init)

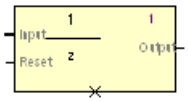
**Symbol 8.26:**




| | |
|---|---|
| SCADE name | **UDInit** |
| Package | smlk |
| Function | Delays input signal by one tick. |
| Inputs | `Input`: 'T<br>`IC`: 'T |
| Outputs | `Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |
| Comment | Initial condition (IC) input is not an hidden input. |

# Unit Delay Resettable

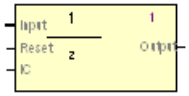
**Symbol 8.27:**




| | |
|---|---|
| SCADE name | **UDReset** |
| Package | smlk |
| Function | Delays input signal by one tick. State can be reset based on reset input. |
| Inputs | `Input`: 'T<br>`Reset`: 'T1 |
| Hidden inputs | `IC`: 'T |
| Outputs | `Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |
| Comment | Needed for graphical layout purposes of initial condition (IC) input. |

# Unit Delay Resettable (Init)

**Symbol 8.28:**




| | |
|---|---|
| SCADE name | **UDResetInit** |
| Package | smlk |
| Function | Delays input signal by one tick. State can be reset based on reset input. |
| Inputs | `Input`: 'T<br>`Reset`: 'T1<br>`IC`: 'T |
| Outputs | `Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |
| Comment | Initial condition (IC) input is not an hidden input. |

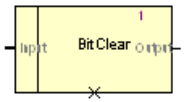
# Bitwise Logical Operators

Read the description of the following operators:


"Bit Clear"
"Bit Set"
"Bitwise AND"
"Bitwise Nand"
"Bitwise Nor"
"Bitwise NOT"
"Bitwise OR"
"Bitwise Exclusive OR"
"Bitwise Shift"
"Bitwise Shift Left"
"Bitwise Shift Right"
"Extract Bit Range"


## Bit Clear

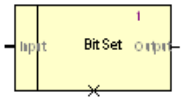
**Symbol 8.29:**




| | |
|---|---|
| SCADE name | **BitClear** (imported function) |
| Package | bits |
| Function | Sets specified bit of input value to `false`. |
| Inputs | `Input`: int32 |
| Hidden inputs | `Index`: int32 |
| Outputs | `Output`: int32 |

# Bit Set

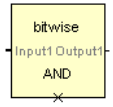
**Symbol 8.30:**




| | |
|---|---|
| SCADE name | **BitSet** (imported function) |
| Package | bits |
| Function | Sets specified bit of input value to `true`. |
| Inputs | `Input`: int32 |
| Hidden inputs | `Index`: int32 |
| Outputs | `Output`: int32 |

# Bitwise AND

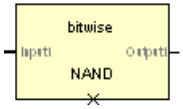
**Symbol 8.31:**




| | |
|---|---|
| SCADE name | **BitwiseAnd** |
| Package | bits |
| Function | Performs a bitwise logical operation AND. |
| Inputs | `Input1`: int32 |
| Hidden inputs | `Operand2`: int32 |
| Outputs | `Output1`: int32 |

# Bitwise Nand

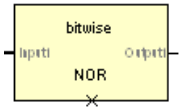
**Symbol 8.32:**




| | |
|---|---|
| SCADE name | **BitwiseNand** |
| Package | bits |
| Function | Performs a bitwise logical operation NOT AND. |
| Inputs | `Input1`: int32 |
| Hidden inputs | `Operand2`: int32 |
| Outputs | `Output1`: int32 |

# Bitwise Nor

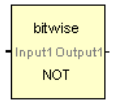
**Symbol 8.33:**




| | |
|---|---|
| SCADE name | **BitwiseNor** |
| Package | bits |
| Function | Performs a bitwise logical operation NOT OR. |
| Inputs | `Input1`: int32 |
| Hidden inputs | `Operand2`: int32 |
| Outputs | `Output1`: int32 |

# Bitwise NOT

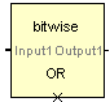
**Symbol 8.34:**




| | |
|---|---|
| SCADE name | **BitwiseNot** |
| Package | bits |
| Function | Performs a bitwise logical operation NOT. |
| Inputs | `Input1`: int32 |
| Outputs | `Output1`: int32 |

# Bitwise OR

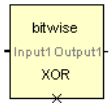
**Symbol 8.35:**




| | |
|---|---|
| SCADE name | **BitwiseOr** |
| Package | bits |
| Function | Performs a bitwise logical operation OR. |
| Inputs | `Input1`: int32 |
| Hidden inputs | `Operand2`: int32 |
| Outputs | `Output1`: int32 |

# Bitwise Exclusive OR

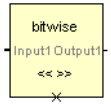
**Symbol 8.36:**




| | |
|---|---|
| SCADE name | **BitwiseXor** |
| Package | bits |
| Function | Performs a bitwise exclusive logical operation OR. |
| Inputs | `Input1`: int32 |
| Hidden inputs | `Operand2`: int32 |
| Outputs | `Output1`: int32 |

# Bitwise Shift

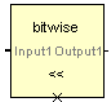
**Symbol 8.37:**




| | |
|---|---|
| SCADE name | **BitwiseShift** |
| Package | bits |
| Function | Shifts a specified number of bits. |
| Inputs | `Input1`: int32 |
| Hidden inputs | `BitShiftRight`: int32 |
| Outputs | `Output1`: int32 |

# Bitwise Shift Left

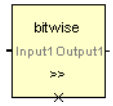
**Symbol 8.38:**




| | |
|---|---|
| SCADE name | **BitwiseShiftLeft** |
| Package | bits |
| Function | Shifts left a specified number of bits. |
| Inputs | `Input1`: int32 |
| Hidden inputs | `Operand2`: int32 |
| Outputs | `Output1`: int32 |

# Bitwise Shift Right

**Symbol 8.39:**




| | |
|---|---|
| SCADE name | **BitwiseShiftRight** |
| Package | bits |
| Function | Shifts right a specified number of bits. |
| Inputs | `Input1`: int32 |
| Hidden inputs | `Operand2`: int32 |
| Outputs | `Output1`: int32 |

# Extract Bit Range

**Symbol 8.40:**

| | |
|---|---|
| SCADE name | **ExtractBitRange** (imported function) |
| Package | bits |
| Function | Outputs a selection of contiguous bits from the input signal. |
| Inputs | `Input`: int32 |
| Hidden inputs | `Start`: int32<br>`End`: int32 |
| Outputs | `Output`: int32 |

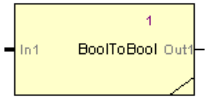
# Conversion and Transformation Operators

Read the description of the following operators:


"Boolean to Boolean"
"Cartesian to Polar"
"Cartesian to Spherical"
"Celsius to Fahrenheit"
"Degrees to Radians"
"Enumeration to Integer"
"Fahrenheit to Celsius"
"Integer to Boolean"
"Integer to Enumeration"
"Polar to Cartesian"
"Radians to Degrees"
"Real to Boolean"
"Scalar to Boolean"
"Spherical to Cartesian"


## Boolean to Boolean

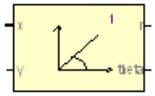
**Symbol 8.41:**




| | |
|---|---|
| SCADE name | **BoolToBool** (specialization of `ScalarToBool`) |
| Package | smlkutils |
| Function | Conversion of scalar values into Boolean values. |
| Inputs | In1: `bool` |
| Outputs | Out1: `bool` |
| KCG pragma | Operator expansion |

# Cartesian to Polar

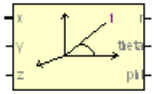
**Symbol 8.42:**



| | |
|---|---|
| SCADE name | **Cartesian2Polar** |
| Package | conv |
| Function | Transformation from cartesian to polar coordinates. |
| Inputs | `x`: 'T<br>`y`: 'T where 'T is float |
| Outputs | `r`: 'T<br>`theta`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# Cartesian to Spherical

**Symbol 8.43:**



| | |
|---|---|
| SCADE name | **Cartesian2Spherical** |
| Package | conv |
| Function | Transformation from cartesian to spherical coordinates. |
| Inputs | `x`: 'T<br>`y`: 'T<br>`z`: 'T where 'T is float |
| Outputs | `r`: 'T<br>`theta`: 'T<br>`phi`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# Celsius to Fahrenheit

**Symbol 8.44:**

| | |
|---|---|
| SCADE name | **Celsius2Fahrenheit** |
| Package | conv |
| Function | Conversion from Celsius degrees to Fahrenheit degrees. |
| Inputs | `Celsius`: 'T where 'T is float |
| Outputs | `Fahrenheit`: 'T where 'T is float |
| KCG pragma | Operator expansion |

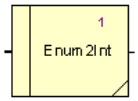
# Degrees to Radians

**Symbol 8.45:**

| | |
|---|---|
| SCADE name | **Degrees2Radians** |
| Package | conv |
| Function | Conversion from degrees to radians. |
| Inputs | `Degrees`: 'T where 'T is float |
| Outputs | `Radians`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# Enumeration to Integer

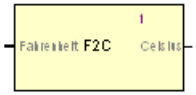
**Symbol 8.46:**




| | |
|---|---|
| SCADE name | **Enum2Int** |
| Package | conv |
| Function | Conversion of an enumerated type to its underlying integer value. |
| Inputs | `Input1:` 'T |
| Outputs | `Output1:` int32 |
| Comment | Specialization operators shall be provided by users. |

# Fahrenheit to Celsius

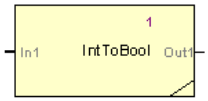
**Symbol 8.47:**




| | |
|---|---|
| SCADE name | **Fahrenheit2Celsius** |
| Package | conv |
| Function | Conversion from Fahrenheit degrees to Celsius degrees. |
| Inputs | `Fahrenheit`: 'T where 'T is float |
| Outputs | `Celsius`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# Integer to Boolean

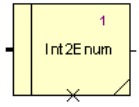
**Symbol 8.48:**




| | |
|---|---|
| SCADE name | **IntToBool** (specialization of `ScalarToBool`) |
| Package | smlkutils |
| Function | Conversion of scalar values into Boolean values. |
| Inputs | In1: 'T where 'T is integer |
| Outputs | `Out1: bool` |
| KCG pragma | Operator expansion |

# Integer to Enumeration

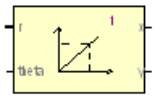
**Symbol 8.49:**




| | |
|---|---|
| SCADE name | **Int2Enum** |
| Package | conv |
| Function | Conversion of an integer value to the corresponding enumerated value if any, or to Default otherwise. |
| Inputs | `Input1`: int32 |
| Hidden inputs | `Default: 'T` |
| Outputs | `Output1: 'T` |
| Comment | Specialization operators shall be provided by users. |

# Polar to Cartesian

**Symbol 8.50:**




| | |
|---|---|
| SCADE name | **Polar2Cartesian** |
| Package | conv |
| Function | Transformation from polar to cartesian coordinates. |
| Inputs | `r`: 'T<br>`theta`: 'T where 'T is float |
| Outputs | `x`: 'T<br>`y`: 'T where 'T is float |
| KCG pragma | Operator expansion |

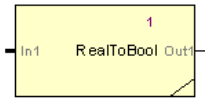
# Radians to Degrees

**Symbol 8.51:**

| | |
|---|---|
| SCADE name | **Radians2Degrees** |
| Package | conv |
| Function | Conversion from radians to degrees. |
| Inputs | `Radians`: 'T where 'T is float |
| Outputs | `Degrees`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# Real to Boolean

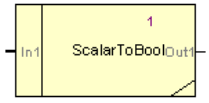
**Symbol 8.52:**




| | |
|---|---|
| SCADE name | **RealToBool** (specialization of `ScalarToBool`) |
| Package | smlkutils |
| Function | Conversion of scalar values into Boolean values. |
| Inputs | `In1`: 'T where 'T is float |
| Outputs | `Out1: bool` |
| KCG pragma | Operator expansion |

# Scalar to Boolean

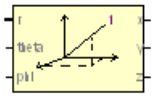
**Symbol 8.53:**




| | |
|---|---|
| SCADE name | **ScalarToBool** (specialized operator) |
| Package | smlkutils |
| Function | Conversion of scalar values into Boolean values. |
| Inputs | `In1`: 'T where 'T is float |
| Outputs | `Out1: bool` |

# Spherical to Cartesian

**Symbol 8.54:**




| | |
|---|---|
| SCADE name | **Spherical2Cartesian** |
| Package | conv |
| Function | Transformation from spherical to cartesian coordinates. |
| Inputs | `r`: 'T , `theta`: 'T, and `phi`: 'T where 'T is float |
| Outputs | `x`: 'T , `y`: 'T, and `z`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# Flip-Flop Operators

Read the description of the following operators:


"Clock"
"DLatch"
"Flip-Flop D"
"Flip-Flop JK"
"Flip-Flop SR"

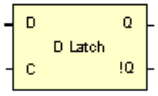

## Clock

**Symbol 8.55:**

| | |
|---|---|
| SCADE name | **Clock** |
| Package | smlk |
| Function | Boolean output toggles at each tick. |
| Outputs | `Output1`: bool |

## DLatch

**Symbol 8.56:**




| | |
|---|---|
| SCADE name | **DLatch** |
| Package | flipflop |
| Function | Latches the data input when the clock input is true. |
| Inputs | `D`: bool and `C`: bool |
| Outputs | `Q`: bool and `notQ`: bool |
| Comment | Output Q equals input D if the input C changes its value.<br>Output notQ is equal to (not Q). |

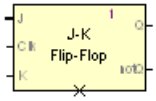
# Flip-Flop D

**Symbol 8.57:**

| | |
|---|---|
| SCADE name | **FlipFlopD** |
| Package | flipflop |
| Function | Basic D flip-flop of Boolean values. |
| Inputs | `D`: bool, `Clk`: bool, and `notClr`: bool |
| Outputs | `Q`: bool and `notQ`: bool |
| Comment | Output Q is equal to Input D if there is a rising edge on input Clk and if the notClr input is true, otherwise Output Q is false. Output notQ is equal to (not Q). |

# Flip-Flop JK

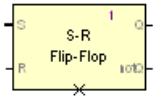
**Symbol 8.58:**




| | |
|---|---|
| SCADE name | **FlipFlopJK** |
| Package | flipflop |
| Function | Basic JK flip-flop of Boolean values. |
| Inputs | `J`: bool, `Clk`: bool, and `K`: bool |
| Hidden inputs | `IC`: bool |
| Outputs | `Q`: bool and `notQ`: bool |
| Comment | On falling edge of Clk input, the following computation is done:<br>• if not J and not K, Qn= Qn-1<br>• if not J and K, Qn = false<br>• if J and not K, Qn = true<br>• if J and K, Qn = not Qn-1<br>Otherwise, Qn = Qn-1<br>Output notQ is equal to (not Q). |

# Flip-Flop SR

**Symbol 8.59:**




| | |
|---|---|
| SCADE name | **FlipFlopSR** |
| Package | flipflop |
| Function | Basic Set-Reset flip-flop of Boolean values. |
| Inputs | `S`: bool and `R`: bool |
| Hidden inputs | `IC`: bool |
| Outputs | `Q`: bool and `notQ`: bool |
| Comment | On falling edge of Clk input, the following computation is done:<br>• if not S and not R, Qn= Qn-1<br>• if not S and R, Qn = false<br>• if S and not R, Qn = true<br>• if S and R, Qn = false<br>Output notQ = if (R and notR) false else (not Qn). |

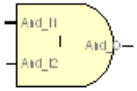
# Logical Operators

Read the description of the following operators:


"AND"
"Compare (all operations)"
"Compare Enumeration"
"Detect Change"
"Detect Decrease"
"Detect Falling Edge (negative)"
"Detect Falling Edge (non-positive)"
"Detect Increase"
"Detect Rising Edge (non-negative)"
"Detect Rising Edge (positive)"
"In Range (In-In)"
"In Range (In-Out)"
"In Range (Out-In)"
"In Range (Out-Out)"
"NOT"
"OR"

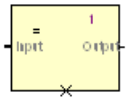

## AND

**Symbol 8.60:**




| | |
|---|---|
| SCADE name | **AndNum** |
| Package | smlk |
| Function | Performs logical AND operation with numeric inputs. |
| Inputs | `And_I1` and `And_I2`: 'T where 'T numeric |
| Outputs | `And_O`: bool |
| KCG pragma | Operator expansion |

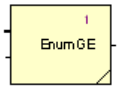
# Compare (all operations)

**Symbol 8.61:** 

Equal

| | |
|---|---|
| SCADE name | Several comparison operations are available in the library:<br>• Equal: **Equal**<br>• Greater or equal: **GreaterEqual**<br>• Greater than: **GreaterThan**<br>• Less or equal: **LessEqual**<br>• Less than: **LessThan**<br>• Not equal: **NotEqual** |
| Package | cmp |
| Function | Compares a signal to a specified constant. |
| Inputs | `Input`: 'T |
| Hidden inputs | `Value`: 'T where 'T numeric |
| Outputs | `Output`: bool |
| KCG pragma | Operator expansion |

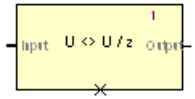
# Compare Enumeration

**Symbol 8.62:** 

(EnumGE)

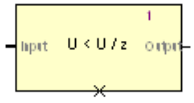
| | |
|---|---|
| SCADE name | Several comparison operations are available in the library:<br>• Greater or equal: **EnumGE**<br>• Greater than: **EnumGT**<br>• Less or equal: **EnumLE**<br>• Less than: **EnumLT** |
| Package | cmp |
| Function | Comparison of enumerated values to their conversion into integers (using `smlk::EnumToInt` specialization) |
| Inputs | `Input1: 'T`<br>`Input2: 'T` |
| Outputs | `Output1: bool` |
| Comment | cmp::EnumLT: Output1 = smlk::EnumToInt(Input1) < smlk::EnumToInt(Input2)<br>cmp::EnumLE: Output1 = smlk::EnumToInt(Input1) <= smlk::EnumToInt(Input2)<br>cmp::EnumGT: Output1 = smlk::EnumToInt(Input1) > smlk::EnumToInt(Input2)<br>cmp::EnumGE: Output1 = smlk::EnumToInt(Input1) >= smlk::EnumToInt(Input2) |

# Detect Change

**Symbol 8.63:** 


| | |
|---|---|
| SCADE name | **DetectChange** |
| Package | smlk |
| Function | Detects if the input value has changed with respect to previous step. |
| Inputs | `Input`: 'T |
| Hidden inputs | `IC`: 'T where 'T numeric |
| Outputs | `Output`: bool |
| KCG pragma | Operator expansion |

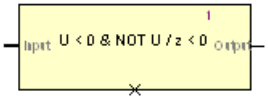
# Detect Decrease

**Symbol 8.64:** 


| | |
|---|---|
| SCADE name | **DetectDecrease** |
| Package | smlk |
| Function | Detects decrease in signal value. |
| Inputs | `Input`: 'T |
| Hidden inputs | `IC`: 'T where 'T numeric |
| Outputs | `Output`: bool |
| KCG pragma | Operator expansion |

# Detect Falling Edge (negative)

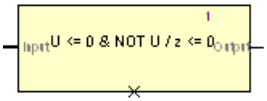
**Symbol 8.65:**




| | |
|---|---|
| SCADE name | **DetectFallNeg** |
| Package | smlk |
| Function | Detects falling edge when signal value decreases to strictly negative value and its previous value was non-negative. |
| Inputs | `Input`: 'T |
| Hidden inputs | `IC`: 'T where 'T numeric |
| Outputs | `Output`: bool |
| KCG pragma | Operator expansion |

# Detect Falling Edge (non-positive)

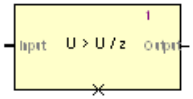
**Symbol 8.66:**




| | |
|---|---|
| SCADE name | **DetectFallNonPos** |
| Package | smlk |
| Function | Detects falling edge when signal value decreases to non-positive value and its previous value was strictly positive. |
| Inputs | `Input`: 'T |
| Hidden inputs | `IC`: 'T where 'T numeric |
| Outputs | `Output`: bool |
| KCG pragma | Operator expansion |

# Detect Increase

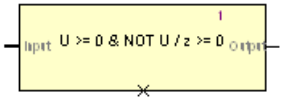
**Symbol 8.67:**




| | |
|---|---|
| SCADE name | **DetectIncrease** |
| Package | smlk |
| Function | Detects increase in signal value. |
| Inputs | `Input`: 'T |
| Hidden inputs | `IC`: 'T where 'T numeric |
| Outputs | `Output`: bool |
| KCG pragma | Operator expansion |

# Detect Rising Edge (non-negative)

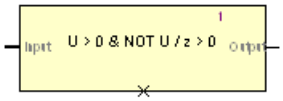
**Symbol 8.68:**




| | |
|---|---|
| SCADE name | **DetectRiseNonNeg** |
| Package | smlk |
| Function | Detects rising edge when signal value increases to non-negative value and its previous value was strictly negative. |
| Inputs | `Input`: 'T |
| Hidden inputs | `IC`: 'T where 'T numeric |
| Outputs | `Output`: bool |
| KCG pragma | Operator expansion |

# Detect Rising Edge (positive)

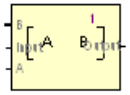
**Symbol 8.69:**




| | |
|---|---|
| SCADE name | **DetectRisePositive** |
| Package | smlk |
| Function | Detects rising edge when signal value increases to strictly positive value and its previous value was non-positive. |
| Inputs | `Input`: 'T |
| Hidden inputs | `IC`: 'T where 'T numeric |
| Outputs | `Output`: bool |
| KCG pragma | Operator expansion |

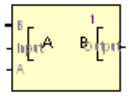
# In Range (In-In)

**Symbol 8.70:**



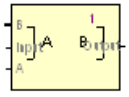

| | |
|---|---|
| SCADE name | **InRangeInIn** |
| Package | smlk |
| Function | Returns a `true` output if the input belongs to the [A , B] range; otherwise it is `false`. |
| Inputs | B: 'T<br>`Input`: 'T<br>A: 'T where 'T numeric |
| Outputs | `Output`: bool |
| KCG pragma | Operator expansion |

# In Range (In-Out)

**Symbol 8.71:** 

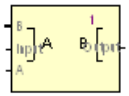
| | |
|---|---|
| SCADE name | **InRangeInOut** |
| Package | smlk |
| Function | Returns a `true` output if the input belongs to the [A , B[ range; otherwise it is `false`. |
| Inputs | B: 'T<br>`Input`: 'T<br>A: 'T where 'T numeric |
| Outputs | `Output`: bool |
| KCG pragma | Operator expansion |

# In Range (Out-In)

**Symbol 8.72:** 

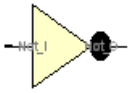
| | |
|---|---|
| SCADE name | **InRangeOutIn** |
| Package | smlk |
| Function | Returns a `true` output if the input belongs to the ]A , B] range; otherwise it is `false`. |
| Inputs | B: 'T<br>`Input`: 'T<br>A: 'T where 'T numeric |
| Outputs | `Output`: bool |
| KCG pragma | Operator expansion |

# In Range (Out-Out)

**Symbol 8.73:** 

| | |
|---|---|
| SCADE name | **InRangeOutOut** |
| Package | smlk |
| Function | Returns a `true` output if the input belongs to the ]A , B[ range; otherwise it is `false`. |
| Inputs | B: 'T<br>`Input`: 'T<br>A: 'T where 'T numeric |
| Outputs | `Output`: bool |
| KCG pragma | Operator expansion |

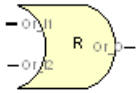
# NOT

**Symbol 8.74:** 

| | |
|---|---|
| SCADE name | **NotNum** |
| Package | smlk |
| Function | Performs logical NOT operation with numeric inputs. |
| Inputs | `Not_I1` and Not_I2: 'T where 'T numeric |
| Outputs | `Not_O`: bool |
| KCG pragma | Operator expansion |

# OR

**Symbol 8.75:**



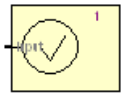

| | |
|---|---|
| SCADE name | **OrNum** |
| Package | smlk |
| Function | Performs logical OR operation with numeric inputs. |
| Inputs | `Or_I1` and `Or_I2`: 'T where 'T numeric |
| Outputs | `Or_O`: bool |
| KCG pragma | Operator expansion |

# Model Verification Operators

Read the description of the following operators:

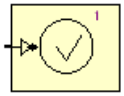

"Assertion"
"Assertion (Inverse)"
"Check Bounds"
"Check Bounds (Extended)"
"Check Gap"
"Check Gap (Extended)"
"Check Gradient"
"Check Gradient (Extended)"
"Check Lower Bound"
"Check Lower Bound (Extended)"
"Check Upper Bound"
"Check Upper Bound (Extended)"


## Assertion

**Symbol 8.76:** 

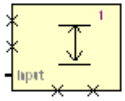
| | |
|---|---|
| SCADE name | **Assertion** |
| Package | chk |
| Function | Checks whether signal is nonzero. |
| Inputs | Input: 'T |
| Outputs | none |
| KCG pragma | Operator expansion |

# Assertion (Inverse)

**Symbol 8.77:** 

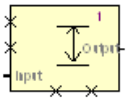
| | |
|---|---|
| SCADE name | **AssertionInv** |
| Package | chk |
| Function | Checks whether signal is zero. |
| Inputs | Input: 'T |
| Outputs | none |
| KCG pragma | Operator expansion |

# Check Bounds

**Symbol 8.78:** 

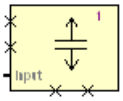

| | |
|---|---|
| SCADE name | **CheckBounds** |
| Package | chk |
| Function | Checks that an input is strictly lower than Min and strictly higher than Max. |
| Inputs | `Input`: 'T |
| Hidden inputs | `Max`: 'T and `Min`: 'T where 'T numeric<br>`MaxOpen`: bool<br>`MinOpen`: bool |
| Outputs | none |
| KCG pragma | Operator expansion |

# Check Bounds (Extended)

**Symbol 8.79:** 

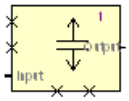

| | |
|---|---|
| SCADE name | **CheckBoundsEx** |
| Package | chk |
| Function | Checks that an input is strictly lower than Min and strictly higher than Max and returns test results. |
| Inputs | `Input`: 'T |
| Hidden inputs | `Max`: 'T and `Min`: 'T where 'T numeric<br>`MaxOpen`: bool<br>`MinOpen`: bool |
| Outputs | `Output`: bool |

# Check Gap

**Symbol 8.80:** 

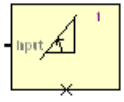

| | |
|---|---|
| SCADE name | **CheckGap** |
| Package | chk |
| Function | Checks that an input is strictly higher than Min and strictly lower than Max. |
| Inputs | `Input`: 'T |
| Hidden inputs | `Max`: 'T and `Min`: 'T where 'T numeric<br>`MaxOpen`: bool<br>`MinOpen`: bool |
| Outputs | none |
| KCG pragma | Operator expansion |

# Check Gap (Extended)

**Symbol 8.81:** 

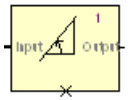

| | |
|---|---|
| SCADE name | **CheckGapEx** |
| Package | chk |
| Function | Checks that an input is strictly higher than Min and strictly lower than Max and returns test results. |
| Inputs | `Input`: 'T |
| Hidden inputs | `Max`: 'T and `Min`: 'T where 'T numeric<br>`MaxOpen`: bool<br>`MinOpen`: bool |
| Outputs | `Output`: bool |

# Check Gradient

**Symbol 8.82:** 

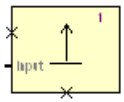

| | |
|---|---|
| SCADE name | **CheckGradient** |
| Package | chk |
| Function | Checks that the expression Input - fby(Input;1;0) is strictly lower than Max. |
| Inputs | `Input`: 'T |
| Hidden inputs | `Max`: 'T where 'T numeric |
| Outputs | none |
| KCG pragma | Operator expansion |

# Check Gradient (Extended)

**Symbol 8.83:** 

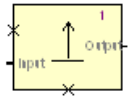

| | |
|---|---|
| SCADE name | **CheckGradientEx** |
| Package | chk |
| Function | Checks that the expression Input - fby(Input;1;0) is strictly lower than Max and returns test results. |
| Inputs | `Input`: 'T |
| Hidden inputs | `Max`: 'T where 'T numeric |
| Outputs | `Output`: bool |

# Check Lower Bound

**Symbol 8.84:** 

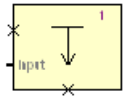

| | |
|---|---|
| SCADE name | **CheckLowerBound** |
| Package | chk |
| Function | Checks that an input is strictly higher than Min. |
| Inputs | `Input`: 'T |
| Hidden inputs | `Min`: 'T where 'T numeric<br>`MinOpen`: bool |
| Outputs | none |
| KCG pragma | Operator expansion |

# Check Lower Bound (Extended)

**Symbol 8.85:** 

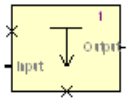

| | |
|---|---|
| SCADE name | **CheckLowerBoundEx** |
| Package | chk |
| Function | Checks that an input is strictly higher than Min and returns test results. |
| Inputs | `Input`: 'T |
| Hidden inputs | `Min`: 'T where 'T numeric<br>`MinOpen`: bool |
| Outputs | `Output`: bool |

# Check Upper Bound

**Symbol 8.86:** 


| | |
|---|---|
| SCADE name | **CheckUpperBound** |
| Package | chk |
| Function | Checks that an input is strictly lower than Max. |
| Inputs | `Input`: 'T |
| Hidden inputs | `Max`: 'T where 'T numeric<br>`MaxOpen`: boo |
| Outputs | none |
| KCG pragma | Operator expansion |

# Check Upper Bound (Extended)

**Symbol 8.87:** 


| | |
|---|---|
| SCADE name | **CheckUpperBoundEx** |
| Package | chk |
| Function | Checks that an input is strictly lower than Max and eturns test results. |
| Inputs | `Input`: 'T |
| Hidden inputs | `Max`: 'T where 'T numeric<br>`MaxOpen`: boo |
| Outputs | `Output`: bool |

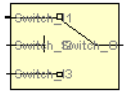
# Signal Routing Operators

Read the description of the following operators:


"If Then Else"
"Ignore First Input"
"Merge (2 inputs)"
"Merge (3 inputs)"
"Switch"
"Switch (GTT)"
"Switch (NEZ)"
"Switch Enumeration"
"Switch Enumeration (GTT)"


## If Then Else

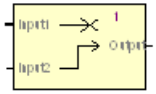
**Symbol 8.88:**




| | |
|---|---|
| SCADE name | **IfThenElse** |
| Package | smlk |
| Function | Switches output between the first and third input based on the criteria applied to the value of the second input when Boolean (if I2 >= threshold, then O=I1, else O=I3). |
| Inputs | `Switch_I1`: 'T<br>`Switch_I2`: bool<br>`Switch_I3`: 'T where 'T numeric |
| Outputs | `Switch_O`: 'T |
| KCG pragma | Operator expansion |

# Ignore First Input

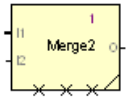
**Symbol 8.89:**




| | |
|---|---|
| SCADE name | **IgnoreFirst** |
| Package | smlk |
| Function | Connects the first input to a terminator and returns the second input to the output. |
| Inputs | `Input1`: 'T<br>`Input2`: 'T1 |
| Outputs | `Output`: 'T1 |
| KCG pragma | Operator expansion |

# Merge (2 inputs)

**Symbol 8.90:**




| | |
|---|---|
| SCADE name | **Merge2** |
| Package | smlk |
| Function | Merges two inputs into one output based on the Boolean conditions specified as hidden inputs. |
| Inputs | `I1` and `I2`: 'T where 'T numeric |
| Hidden inputs | `InitVal`: 'T<br>`C1` and `C2`: bool |
| Outputs | `O`: 'T |

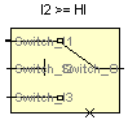
# Merge (3 inputs)

**Symbol 8.91:**

| | |
|---|---|
| SCADE name | **Merge3** |
| Package | smlk |
| Function | Merges three inputs into one output based on the Boolean conditions specified as hidden inputs. |
| Inputs | `I1`, `I2` and `I3`: 'T where 'T numeric |
| Hidden inputs | `InitVal`: 'T<br>`C1`, `C2` and `C3`: bool |
| Outputs | `O`: 'T |

# Switch

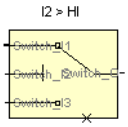
**Symbol 8.92:**




| | |
|---|---|
| SCADE name | **Switch** |
| Package | smlk |
| Function | Switches output between the first and the third input based on criteria applied to the value of the second input (if I2 >= threshold, then O=I1, else O=I3). |
| Inputs | `Switch_I1`: 'T<br>`Switch_I2`: 'T2 where 'T2 is numeric<br>`Switch_I3`: 'T |
| Hidden inputs | `Switch_HI`: 'T2 (threshold value) where 'T2 is numeric |
| Outputs | `Switch_O`: 'T |
| KCG pragma | Operator expansion |

# Switch (GTT)

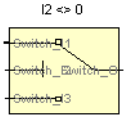
**Symbol 8.93:**




| | |
|---|---|
| SCADE name | **SwitchGTT** |
| Package | smlk |
| Function | Switches output between the first and the third input based on criteria applied to the value of the second input (if I2 > threshold, then O=I1, else O=I3). |
| Inputs | `Switch_I1`: 'T<br>`Switch_I2`: 'T2 where 'T2 is numeric<br>`Switch_I3`: 'T |
| Hidden inputs | `Switch_HI`: 'T2 (threshold value) where 'T2 is numeric |
| Outputs | `Switch_O`: 'T |
| KCG pragma | Operator expansion |

# Switch (NEZ)

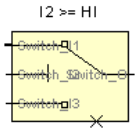
**Symbol 8.94:**




| | |
|---|---|
| SCADE name | **SwitchNEZ** |
| Package | smlk |
| Function | Switches output between the first and the third input based on criteria applied to the value of the second input (if I2 not equal to 0, then O=I1, else O=I3). |
| Inputs | `Switch_I1`: 'T<br>`Switch_I2`: 'T2 where 'T2 is numeric<br>`Switch_I3`: 'T |
| Outputs | `Switch_O`: 'T |
| KCG pragma | Operator expansion |

# Switch Enumeration

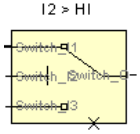
**Symbol 8.95:**




| | |
|---|---|
| SCADE name | **SwitchEnum** |
| Package | smlk |
| Function | Switches output between the first and the third input based on criteria applied to the value of the second input (if I2 >= threshold, then O=I1, else O=I3). |
| Inputs | Switch_I1: 'T, Switch_I2: 'T2, and Switch_I3: 'T |
| Hidden inputs | Switch_HI: 'T2 (threshold value) |
| Outputs | Switch_O: 'T |
| KCG pragma | Operator expansion |

# Switch Enumeration (GTT)

**Symbol 8.96:**




| | |
|---|---|
| SCADE name | **SwitchGTTEnum** |
| Package | smlk |
| Function | Switches output between the first and the third input based on criteria applied to the value of the second input (if I2 > threshold, then O=I1, else O=I3). |
| Inputs | Switch_I1: 'T, Switch_I2: 'T2, and Switch_I3: 'T |
| Hidden inputs | Switch_HI: 'T2 (threshold value) |
| Outputs | Switch_O: 'T |
| KCG pragma | Operator expansion |

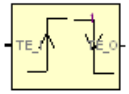
# Subsystem Operators

Read the description of the following operators:


"Trigger Either Edge"
"Trigger Either Edge (Extended)"
"Trigger Falling Edge"
"Trigger Falling Edge (Extended)"
"Trigger Rising Edge"
"Trigger Rising Edge (Extended)"

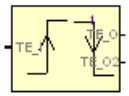

## Trigger Either Edge

**Symbol 8.97:**



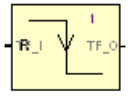

| | |
|---|---|
| SCADE name | **TriggerEither** |
| Package | smlk |
| Function | Detects rising or falling edge (compared to zero) of an input. |
| Inputs | `TE_I`: 'T where 'T numeric |
| Outputs | `TE_O`: bool |
| KCG pragma | Operator expansion |

# Trigger Either Edge (Extended)

**Symbol 8.98:** 


| | |
|---|---|
| SCADE name | **TriggerEitherEx** |
| Package | smlk |
| Function | Detects rising or falling edge (compared to zero) of an input. |
| Inputs | `TE_I`: 'T where 'T numeric |
| Outputs | `TE_O`: bool and `TE_O2`: int8 |
| Comment | An additional output indicates the kind of edge that is activated (-1 for falling, 1 for rising, and 0 for no edge). |

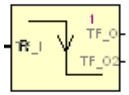
# Trigger Falling Edge

**Symbol 8.99:** 


| | |
|---|---|
| SCADE name | **TriggerFall** |
| Package | smlk |
| Function | Detects falling edge (compared to zero) of an integer input. |
| Inputs | `TF_I`: 'T where 'T numeric |
| Outputs | `TF_O`: bool |
| KCG pragma | Operator expansion |

# Trigger Falling Edge (Extended)

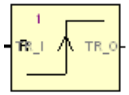
**Symbol 8.100:**




| | |
|---|---|
| SCADE name | **TriggerFallEx** |
| Package | smlk |
| Function | Detects falling edge (compared to zero) of a real/float input. |
| Inputs | `TF_I`: 'T where 'T numeric |
| Outputs | `TF_O`: bool<br>`TF_O2`: int8 |
| Comment | An additional output indicates if the edge is activated (-1 for falling or 0 for no edge). |

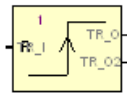
# Trigger Rising Edge

**Symbol 8.101:**




| | |
|---|---|
| SCADE name | **TriggerRise** |
| Package | smlk |
| Function | Detects rising edge (compared to zero) of an integer input. |
| Inputs | `TR_I`: 'T where 'T numeric |
| Outputs | `TR_O`: bool |
| KCG pragma | Operator expansion |

# Trigger Rising Edge (Extended)

**Symbol 8.102:** 


| | |
|---|---|
| SCADE name | **TriggerRiseEx** |
| Package | smlk |
| Function | Detects rising edge (compared to zero) of a real/float input. |
| Inputs | `TR_I`: 'T where 'T numeric |
| Outputs | `TR_O`: bool<br>`TR_O2`: int8 |
| Comment | An additional output indicates if the edge is activated (1 for rising or 0 for no edge). |

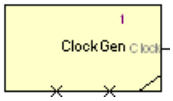
# Signal Generator Operators

Read the description of the following operators:


"Clock Generator"
"Ground (Boolean)"
"Ground (int/int32)"
"Ground (real/float)"


## Clock Generator

**Symbol 8.103:**




| SCADE name | **ClockGen** |
|---|---|
| Package | smlk |
| Function | Generates a Boolean clock signal based on two hidden integer inputs. |
| Inputs | None |
| Hidden inputs | `Period`: int32 and `Offset`: int32 |
| Outputs | `Clock`: bool |

## Ground (Boolean)

**Symbol 8.104:**

| SCADE name | **GroundB** |
|---|---|
| Package | smlk |
| Function | Generates a Boolean output set as `false`. |
| Inputs | None |
| Outputs | `GB_Out`: bool |
| KCG pragma | Operator expansion |

# Ground (int/int32)

**Symbol 8.105:**


| | |
|---|---|
| SCADE name | **GroundI** |
| Package | smlk |
| Function | Generates an integer output set to **0**. |
| Inputs | None |
| Outputs | `GI_Out`: int32 |
| KCG pragma | Operator expansion |

# Ground (real/float)

**Symbol 8.106:**


| | |
|---|---|
| SCADE name | **GroundR** |
| Package | smlk |
| Function | Generates a real/float64 output set to **0.0**. |
| Inputs | None |
| Outputs | `GR_Out`: float64 |
| KCG pragma | Operator expansion |

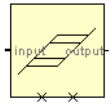
# Discontinuity Operators

Read the description of the following operators:


"Backlash"
"Dead Zone Dynamic"
"Rate Limiter"
"Rate Limiter Dynamic"
"Relay"
"Saturate Dynamic"
"Wrap To Zero"


## Backlash

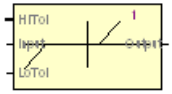
**Symbol 8.107:**




| | |
|---|---|
| SCADE name | **Backlash** |
| Package | smlk |
| Function | Returns a change in output equal to a change in input with a deadband in direction changes. |
| Inputs | `input`: 'T where 'T is float |
| Hidden inputs | `deadband_width`: 'T and `initial_output`: 'T where 'T is float |
| Outputs | `output`: 'T where 'T is float |

# Dead Zone Dynamic

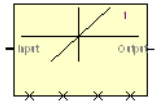
**Symbol 8.108:**




| | |
|---|---|
| SCADE name | **DeadZoneDynamic** |
| Package | smlk |
| Function | Keeps the output equal to 0.0 as long as the input is within the [lower tolerance, upper tolerance] range. When the input reaches beyond one limit, the output is equal to the input minus the limit. |
| Inputs | `HiTol`: 'T, `Input`: 'T, and `LoTol`: 'T |
| Outputs | `Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |

# Rate Limiter

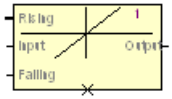
**Symbol 8.109:**




| | |
|---|---|
| SCADE name | **RateLimiter** |
| Package | smlk |
| Function | Limits the first derivative of the input signal. The output does not change faster than the specified limit. |
| Inputs | `input`: 'T where 'T is float |
| Hidden inputs | `RisingSlewLimit`: 'T, `FallingSlewLimit`: 'T, `TimeStep`: 'T, and `InitialOutput`: 'T where 'T is float |
| Outputs | `output`: 'T where 'T is float |

# Rate Limiter Dynamic

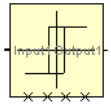
**Symbol 8.110:**




| | |
|---|---|
| SCADE name | **RateLimiterDynamic** |
| Package | smlk |
| Function | Limits the first derivative of the signal passing through. The output does not change faster than the specified limit. |
| Inputs | `Rising`: 'T, `Input`: 'T, and `Falling`: 'T where 'T is float |
| Hidden inputs | `deltaT`: 'T where 'T is float |
| Outputs | `Output`: 'T where 'T is float |
| KCG pragma | Operator expansion |
| Comment | Let us call Rate the derivative ((RL_Input-pre(RL_Output))/DeltaT.<br>If Rate > Rising, then RL_Output equals (Rising* DeltaT + pre(RL_Output))<br>If Rate < Falling, then RL_Output equals (Falling* DeltaT + pre(RL_Output))<br>Otherwise, RL_Output=RL_Input. |

# Relay

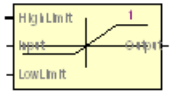
**Symbol 8.111:**




| | |
|---|---|
| SCADE name | **Relay** |
| Package | smlk |
| Function | Basic relay that generates discontinuous output values based on input. |
| Inputs | `Input1`: 'T where 'T numeric |
| Hidden inputs | `SwitchOn`: 'T and `SwitchOff`: 'T<br>`OutputOn`: 'T and `OutputOff`: 'T |
| Outputs | `Output1`: 'T |

# Saturate Dynamic

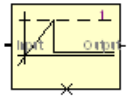
**Symbol 8.112:**




| | |
|---|---|
| SCADE name | **SaturateDynamic** |
| Package | smlk |
| Function | If the input is lower than a lower limit, the output is the lower limit; if the input is greater than a higher limit, the output is the higher limit; otherwise the output is equal to the input. |
| Inputs | `HighLimit`: 'T, `Input`: 'T, and `LowLimit`: 'T |
| Outputs | `Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |
| Comment | If Input is lower than LowLimit, Output is equal to LowLimit; if Input is greater than HighLimit, Output is equal to HighLimit; otherwise Output is equal to Input. |

# Wrap To Zero

**Symbol 8.113:**




| | |
|---|---|
| SCADE name | **WrapToZero** |
| Package | smlk |
| Function | Sets the output to zero if the input is above threshold. |
| Inputs | `Input`: 'T |
| Hidden inputs | `Threshold`: 'T |
| Outputs | `Output`: 'T where 'T numeric |
| KCG pragma | Operator expansion |

# LUT, Pre-LUT and Interpolation Operators

Read the description of the following operators:

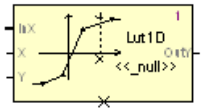

"LUT 1D"
"Pre-LUT2"
"Pre-LUT2 (Direct)"
"Interpolation1D2"
"Interpolation1D (Floor)"
"Interpolation2D2"
"Interpolation2D (Floor)"


**Note**

The Pre-Look-up and Interpolation operators from the `lut` package use a structured type, `LutIndex`. These operators map directly to Simulink blocks until release R2006a. From release R2006b, these Simulink operators output the index and factor in two outputs instead of a structure. The following operators are used in SCADE Suite Gateway for Simulink to map directly Simulink blocks.

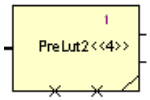
## LUT 1D

**Symbol 8.114:**



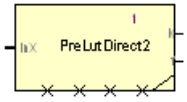

| | |
|---|---|
| SCADE name | **Lut1D** |
| Package | smlk |
| Size parameters | Xs |
| Inputs | `InX`: 'T where 'T is float<br>`X`: 'T^Xs and `Y`: 'T^Xs |
| Hidden inputs | `Extrapol`: bool |
| Outputs | `OutY`: 'T where 'T is float |
| KCG pragma | Operator expansion |

# Pre-LUT2

**Symbol 8.115:** 


| | |
|---|---|
| SCADE name | **PreLut2** |
| Package | smlk |
| Function | See introductory note above |
| Size parameters | Xs |
| Inputs | `InX`: 'T where 'T float |
| Hidden inputs | `X`: 'T ^Xs where 'T float<br>`Extrapol`: bool |
| Outputs | `k`: int32<br>`f`: 'T where 'T is float |

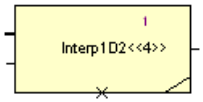
# Pre-LUT2 (Direct)

**Symbol 8.116:** 


| | |
|---|---|
| SCADE name | **PreLutDirect2** |
| Package | smlk |
| Function | See introductory note above |
| Inputs | `InX`: 'T where 'T is float |
| Hidden inputs | `Orig`: 'T<br>`Step`: 'T<br>`NbElem`: int32<br>`Extrapol`: bool where 'T is float |
| Outputs | `k`: int32<br>`f`: 'T where 'T is float |

# Interpolation1D2

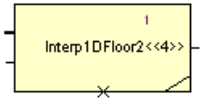
**Symbol 8.117:**




| | |
|---|---|
| SCADE name | **Interp1D2** |
| Package | smlk |
| Function | See introductory note above |
| Size parameters | Xs |
| Inputs | `k`: int32<br>`f`: 'T where 'T is float |
| Hidden inputs | `Y`: 'T^Xs where 'T is float |
| Outputs | `OutY`: 'T where 'T is float |

# Interpolation1D (Floor)

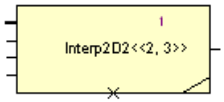
**Symbol 8.118:**




| | |
|---|---|
| SCADE name | **Interp1DFloor2** |
| Package | smlk |
| Function | See introductory note above |
| Size parameters | Xs |
| Inputs | `k`: int32<br>`f`: 'T where 'T is float |
| Hidden inputs | `Y`: 'T ^Xs where 'T is float |
| Outputs | `OutY`: 'T where 'T is float |

# Interpolation2D2

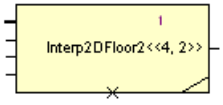
**Symbol 8.119:**




| | |
|---|---|
| SCADE name | **Interp2D2** |
| Package | smlk |
| Function | See introductory note above |
| Size parameters | Xs and Ys |
| Inputs | `k1`: int32, `f1`: 'F, `k2`: int32, and `f2`: 'F where 'F is float |
| Hidden inputs | `Z`: 'F^Ys^Xs where 'F is float |
| Outputs | `OutZ`: 'F where 'F is float |

# Interpolation2D (Floor)

**Symbol 8.120:**




| | |
|---|---|
| SCADE name | **Interp2DFloor2** |
| Package | smlk |
| Function | See introductory note above |
| Size parameters | Xs and Ys |
| Inputs | `k1`: int32, `f1`: 'T, `k2`: int32, and `f2`: 'T where 'T is float |
| Hidden inputs | `Z`: 'T ^Ys^Xs where 'T is float |
| Outputs | `OutZ`: 'T where 'T is float |

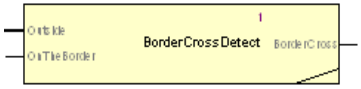
# Miscellaneous Operators

Read the description of the following operators:


"Border Crossing Detection"
"Counter (limited)"
"Entry Detection"
"Identity"
"Inactive Cycles"
"Inactive Time"
"Quantizer"
"Unit Delay (Helper)"
"Width (Matrix)"
"Width (Scalar)"
"Width (Vector)"


## Border Crossing Detection

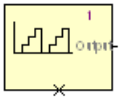
**Symbol 8.121:**




| | |
|---|---|
| SCADE name | **BorderCrossDetect** |
| Package | smlk |
| Function | Detects border crossing based on two Boolean inputs: one input indicates whether the signal is outside the border and the other one whether the signal is at the border. |
| Inputs | `Outside`: bool and `OnTheBorder`: bool |
| Outputs | `BorderCross`: bool |

## Counter (limited)

**Symbol 8.122:**




| | |
|---|---|
| SCADE name | **CounterLimited** |
| Package | smlk |
| Function | Counts from zero to a specified limit, then wraps. |
| Hidden inputs | `Limit`: int32 |
| Outputs | `Output`: int32 |

# Entry Detection

**Symbol 8.123:**

| | |
|---|---|
| SCADE name | **EntryDetect** |
| Package | smlk |
| Function | Detects entry into a range based on two Boolean inputs: one input indicates whether the signal is outside the border and the other one whether the signal is at the border. |
| Inputs | `Outside`: bool and `OnTheBorder`: bool |
| Outputs | `BorderCross`: bool |

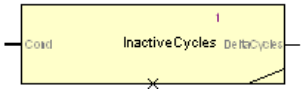
# Identity

**Symbol 8.124:**

| | |
|---|---|
| SCADE name | **Idt** |
| Package | smlk |
| Function | Passes signal through (Output=Input) |
| Inputs | `Idt_In`: 'T where 'T is numeric |
| Outputs | `Idt_Out`: 'T |
| KCG pragma | Operator expansion |

# Inactive Cycles

**Symbol 8.125:**




| | |
|---|---|
| SCADE name | **InactiveCycles** |
| Package | smlk |
| Function | Counter that increments an integer output in steps equal to a hidden integer value when Boolean input is `false`, else it returns a constant equal to its hidden value. |
| Inputs | `Cond`: bool |
| Hidden inputs | `NbCycles`: 'T where 'T is integer |
| Outputs | `DeltaCycles`: 'T where 'T is integer |

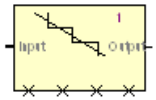
# Inactive Time

**Symbol 8.126:**

| | |
|---|---|
| SCADE name | **InactiveTime** |
| Package | smlk |
| Function | Counter that increments a real /float output in steps equal to the hidden value when the Boolean input is `false`, else it returns a constant equal to its hidden value. |
| Inputs | `Cond`: bool |
| Hidden inputs | `DTInit`: 'T where 'T is numeric |
| Outputs | `DeltaT`: 'T where 'T is numeric |

# Quantizer

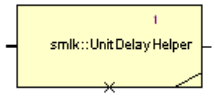
**Symbol 8.127:**




| | |
|---|---|
| SCADE name | **Quantizer** |
| Package | smlk |
| Function | Basic quantizer based on specified interval.<br>Discretizes the input at a specified interval and produces a stair-step output. |
| Inputs | `Input`: 'T where 'T is float |
| Hidden inputs | `Bits`: int32, `Umin` and `Umax`: 'T where 'T is float<br>`OutputNegativeValues`: bool |
| Outputs | `Output`: int32 |

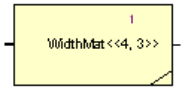
# Unit Delay (Helper)

**Symbol 8.128:**



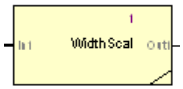

| | |
|---|---|
| SCADE name | **UnitDelayHelper** |
| Package | smlk |
| Function | Performs a unit delay based on init value. |
| Inputs | Input: 'T |
| Hidden inputs | `x0`: 'T |
| Outputs | Output: 'T |
| KCG pragma | Operator expansion |

# Width (Matrix)

**Symbol 8.129:** 


| | |
|---|---|
| SCADE name | **WidthMat** |
| Package | smlk |
| Function | Outputs the number of elements from the input matrix. |
| Size parameters | N1 and N2 |
| Inputs | `In1`: 'T ^N1^N2 where 'T numeric |
| Outputs | `Out1`: int32 |
| KCG pragma | Operator expansion |
| Comment | Simulink block Width mapped to operators WidthScal, WidthVect, or WidthMat depending on the kind of block input. |

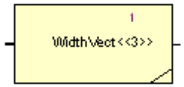
# Width (Scalar)

**Symbol 8.130:** 


| | |
|---|---|
| SCADE name | **WidthScal** |
| Package | smlk |
| Function | Outputs the number of elements from the input scalar, i.e. value 1. |
| Inputs | `In1`: 'T where 'T numeric |
| Outputs | `Out1`: int32 |
| KCG pragma | Operator expansion |
| Comment | Simulink block Width mapped to operators WidthScal, WidthVect, or WidthMat depending on the kind of block input. |

# Width (Vector)

**Symbol 8.131:**




| | |
|---|---|
| SCADE name | **WidthVect** |
| Package | smlk |
| Function | Outputs the number of elements from the input vector. |
| Size parameters | N |
| Inputs | `In1`: 'T ^N where 'T numeric |
| Outputs | `Out1`: int32 |
| KCG pragma | Operator expansion |
| Comment | Simulink block Width mapped to operators WidthScal, WidthVect, or WidthMat depending on the kind of block input. |

# 9 / Library libverif

This library contains operators designed to help expressing verification properties. Such operators are expressed as regular SCADE nodes. The timing diagram of each operator is provided when it applies.

Access the description of library operators in alphabetical order:


"After Nth Tick"
"Always After First Condition"
"At Least N Ticks"
"Has Never Been True"
"Implies"
"Implies Within N Ticks"


## After Nth Tick

**Symbol 9.1:**

| | |
|---|---|
| SCADE name | **AfterNthTick** |
| Function | Used to express that the output equals the input, except for the first N cycles during which the output is `true` (*i.e.*, input on the first N cycles is ignored). |
| Size parameters | N specifies the number of cycles |
| Inputs | `Input1`: bool |
| Outputs | `Output1`: bool |
| Comment | Check the operator timing diagram below. |

| **Timing Diagram** | **Cycle** | **1** | **2** | **...** | **10** | **11** | **12** |
|---|---|---|---|---|---|---|---|
| | Output1 (N=10 ticks) | true | true | true | true | Input1 | Input1 |

# Always After First Condition

**Symbol 9.2:**

| | |
|---|---|
| SCADE name | **AlwaysAfterFirstCond** |
| Function | Used to express that the output equals the input once a condition is `true`. Before that and since the initial cycle, the output is `true`. |
| Inputs | `Input1`: bool and `Cond`: bool |
| Outputs | `Output1`: bool |
| Comment | Check the operator timing diagram below. |

| **Timing Diagram** | **Cycle** | **1** | **2** | **3** | **4** | **5** | **6** |
|---|---|---|---|---|---|---|---|
| | Cond | false | false | true | * | * | * |
| | Output1 | true | true | Input1 | Input1 | Input1 | Input1 |

# At Least N Ticks

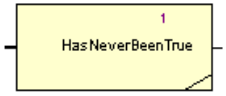
**Symbol 9.3:**

| | |
|---|---|
| SCADE name | **AtLeastNTicks** |
| Function | Used to express that the output equals the input as soon as the input is `true` for N cycles. Before this $N^{th}$ cycle, the output is `false`. |
| Inputs | `Input1`: bool |
| Hidden inputs | `N`: int32 |
| Outputs | `Output1`: bool |
| Comment | Check the operator timing diagram below. |

| **Timing Diagram** | **Cycle (N)** | **1** | **2** | **...** | **5** | **6** | **...** |
|---|---|---|---|---|---|---|---|
| | Input1 | true | true | true | true | * | * |
| | Output1 | false | false | false | Input1 | Input1 | Input1 |

# Has Never Been True

**Symbol 9.4:**




| | |
|---|---|
| SCADE name | **HasNeverBeenTrue** |
| Function | Used to express that the output becomes `false` as soon as its input becomes `true` for the first time. After this cycle, the output remains `false`. |
| Inputs | `Input1`: bool |
| Outputs | `Output1`: bool |
| KCG pragma | Operator expansion |
| Comment | Check the operator timing diagram below. |

| **Timing Diagram** | **Cycle** | **1** | **2** | **3** | **4** | **5** | **...** |
|---|---|---|---|---|---|---|---|
| | Input1 | false | false | true | true | * | * |
| | Output1 | true | true | false | false | * | * |

# Implies

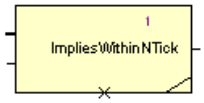
**Symbol 9.5:**

| | |
|---|---|
| SCADE name | **Implies** |
| Function | Implements the "implies" logical operator. "If A is `true`, then B is `true`": Not(A) or B. |
| Inputs | A: bool and B: bool |
| Outputs | C: bool |
| KCG pragma | Operator expansion |
| Comment | Check the operator timing diagram below. |

| | A | B | C |
|---|---|---|---|
| **Truth Table** | false | false | true |
| | false | true | true |
| | true | false | false |
| | true | true | true |

# Implies Within N Ticks

**Symbol 9.6:**




| | |
|---|---|
| SCADE name | **ImpliesWithinNTick** |
| Function | Output equal to "Input1 implies Input2" as soon as Input1 is true for "N" ticks. |
| Inputs | `Input1`: bool and `Input2`: bool |
| Hidden inputs | N: int32 |
| Outputs | `Output1`: bool |

| | **Cycle (N)** | **1** | **2** | **3** | **4** | **5** | **6** | **...** |
|---|---|---|---|---|---|---|---|---|
| **Timing Diagram** | N=3 | * | * | * | * | * | | * |
| | Input1 | false | false | true | true | true | true | false |
| | Input2 | * | * | * | * | false | true | * |
| | Output1 | true | true | true | true | false | true | true |

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